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Life Cycle Inventories of Wood as Fuel and Construction Material

Data v2.0 (2007)



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Life Cycle Inventories of Wood as Fuel and Construction Material

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Comments on this version 2.0

This part corresponds to the original report (Werner et al. 2003) that has been updated with the changes reported in (Frischknecht et al. 2006). Direct changes in the unit processes in this report are only made for the charcoal dataset and for the wood chopper dataset.

Part I

Life Cycle Inventories of Wood as Fuel and Construction Material

Data v2.0 (2007)

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Summary

In this project, more than 100 different products from wood have been inventoried. The aim of the work was to generate generic life cycle inventory data for wood as material and as fuel.

Forests are natural systems and production systems for wood in one. This complicates the LCI of wood for several reasons:

- Allocation of environmental burdens could be made to wood and other functions of the forest (not done in ecoinvent)
- It takes more then 100 years (or at least several decades) for a tree to grow. With this time frame, many assumptions about the past and the future have to be made.
- Different natural parameters (climate, wildlife,...) can influence the growth and the properties of wood.
- Different species of trees show different properties.
- Forest management strategies have a great influence on the LCI of wood.

For wood used as fuel, but also for material use the water content of the wood is very important because it changes the density and the heating value in wide ranges. To overcome this problem, the functional unit in this study is mostly given as m^3 , since the volume remains more or less constant with changing water content. The water content of each product is either given in the product name or in its meta information. This information is needed for inventorying transports of the products or to calculate the heating value.

Many processes in the wood chain produce relatively big amounts of by-products. To account for this correctly, these processes are modelled as multi output processes in ecoinvent. Thus part of the environmental burdens and of the biomass and the energy and carbon stored can be allocated to the by-product.

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1 Introduction

This report describes the processes and their inventories for the production of wood and wood based materials as they are modelled in ecoinvent data v 2.0.

Most of the relevant information to the processes is compiled in the figures depicting the input and output flows to the processes together with information about their use in the “real world” and about their representation in ecoinvent. This figure shows also the sources and the uncertainty information to the individual values.

2 Material and market characteristics of wood

This chapter is mainly based on (Werner 2002) and (Werner et al. 2002).

2.1 Wood as a material

Wood is the world's most important renewable material and regenerative fuel ((Bowyer 1995); Eldag 1980: p. 1105); Frühwald et al. 2001); Schulz 1993); Sutton 1993)). Wood has been utilised as material and fuel since prehistoric times. Therefore, it is deeply rooted in the entire history of civilisation and mankind. Today, wood is utilised in many forms, for example, sawn wood, particleboard, plywood, fibreboard and veneer. Its main applications are construction material for housing and civil engineering, furniture, railway ties (sleepers), poles and reinforcement for mining. Even today wood is still one of the leading construction materials.

Wood is also the most important raw material for the pulp and paper industry and plays a significant role in the packaging sector.

Wood can be used as solid wood, particles, fibres or even on a molecular basis. The material characteristics and the twofold nature of wood as material and fuel allows for an almost 100% utilisation of this resource.

Wood is anisotropic and inhomogeneous; its properties vary from tree to tree as well as within the same log. Anisotropy is reduced in wood panels and engineered wood products by suitable technologies. Despite the fact that wood is the oldest material utilised by man, numerous new developments have taken place in recent years, which have broadened the application of wood and wood products. Especially noteworthy are new impregnation techniques and combinations with other materials such as plastics and metals. The main advantages of wood lie in its relatively low price, favourable strength-to-weight ratio, its vapour permeability, its appealing, individual appearance, and in the ease in which it is worked with proven techniques ((Schulz 1972): p. 1416).

Wood as biogenic material is subject to biological and photochemical degradation. As a consequence, certain (outdoor) applications require (regular) constructive and/or chemical wood protection.

Its hygroscopic nature leads to changes in dimension, heating value and density ("swelling and shrinking") depending on the moisture content of the environment. This can lead to deformation and crack if wood is not applied in a carefully considered way.

Reuse & recycling of wood as material must always be considered as down cycling (except reuse in a very strict sense): diameters of wood pieces and fibre length decrease while "unwanted" contaminants (*e.g.* from wood treatment or coating) increase with each processing step ((Willeitner & Bucki 1994)). The more often wood is reprocessed, the more restricted are its potential applications. Original properties can only be restored with the investment of (non-renewable) energy and material ((Fraanje 1997), p 22f).

2.2 Some characteristics of wood industry

Wood as a naturally grown material, irregular in shape and structure; is not, a priori, suited for industrial processing or use in technical applications. As a consequence, the management and processing of wood generates a variety of co-products throughout the wood processing chain from its cultivation in managed forests, its extraction, sawing, processing to intermediate and finished products, on to its recycling, incineration or final disposal. Co-products generated are, *e.g.* residues from thinning, bark, sawdust, shavings, chips and fibres, side-cuts, wood waste and waste of intermediate products from wood and wood-based industries ((Jungmeier et al. 2002)).

If co-products cannot be used internally, e.g. for the generation of thermal energy for wood drying, they can be sold to other industrial sectors. This has in fact generated a considerable network of material flows between different wood processing industries that use co-products from other sectors as material input or fuel (e.g. chips from saw mills are used in the particleboard industry).

Even after the use phase of wood products, the material characteristics of wood (theoretically) still allow for a variety of options for the reuse of wood as material or energy carrier. Similar considerations on the generation and use of co-products throughout the material chain are also valid for the pulp and paper chain. In contrast to the wood processing industry, the (material) recycling of waste paper is widely established.

2.3 Sustainable Wood Flow Management

2.3.1 Forestry

The term "Nachhaltigkeit" (sustainability) is historically closely linked to forestry and is by no means an invention of nowadays. Conceptually, it was developed in the early 18th century, when population growth and industrialisation had increased pressure on the resource wood to an extent threatening further supply. It was Hans Carl Carlowitz who claimed in his *Sylvicultura Oeconomica. Die Naturmässige Anweisung zur Wilden Baum-Zucht*, published 1713, (...) *dass man mit dem Holtz pfleglich umgehe, and wie eine sothane Conservation und Anbau des Holtzes anzustellen, dass es eine continuirliche, beständige und nachhaltende Nutzung gebe* (Carlowitz (1713), quoted after (Grober 1999)).

The concept of sustainable forest management entered German and Swiss forest legislation at the turn of 19th century, with the main focus of maintaining forested area (for details see (Schuler 2000)).

Today, sustainable management of renewable resources is defined in a much broader sense. One of the most widely accepted definitions has been set up 1993 by the Ministerial Conference for the Conservation of Forests in Europe in Helsinki: *The stewardship and use of forests and forest lands in a way, and at a rate, that maintains their biodiversity, productivity, regeneration capacity, vitality and their potential to fulfil, now and in the future, relevant ecological, economic and social functions, at local, national and global levels, and that does not cause damage to other ecosystems* ((Anonymous 1998)).

In Europe, the local potential of wood as a resource is currently not exploited to its full extent ((Anonymous 1999)). In Switzerland, the average standing volume of wood per hectare has to be considered high, with a still increasing tendency. Moreover, forested area in Switzerland has been increasing constantly over the last fifty years. As a consequence of current under-exploitation, Swiss forests contain a disproportionally high share of elderly trees ((Anonymous 2000)).

The use of wood could be increased by an average of 20% to 40% in Switzerland even after defining large forest areas as protected nature reserves ((Anonymous 2000, p. 12); Mauch et al. 1995, p. 65ff)).

Mainly imports of raw material for paper industry could be substituted ((Hofer et al. 2001); Mauch et al. 1995)).

Quantitative criteria for sustainable forest management are generally met in Switzerland and the legal framework is considered appropriate. Nevertheless, qualitative criteria (e.g. the over-aged mountainous forests), the forest ecosystem health and vitality affected by air pollution, and a high degree of governmental subsidies of forestry activities are major concerns in the light of sustainable forest management in Switzerland ((Brassel & Brändli 1999); (Patosari 2000, p. 480ff)). On the other hand, native species cover more than 99% of the forest area.

Scant attention has been paid to another aspect: the closing of the cycle of nutrients that are withdrawn from forests by the harvesting of trees mainly with bark and branches. First experiences have been gained in the processing of wood-based ashes for their use as fertiliser. In this respect, the forest industry activity is beginning to participate in the recycling of nutrients as base cations in the ashes are re-

turned to the forests. In the future, this effort should be increased in order to complete the natural-industrial nutrient cycle ((Korhonen et al. 2001); Winckler & Pflug 1980, p. 152)).

An increased use of domestic wood in Switzerland would not only respect the environmental dimension of sustainable development but would also foster its social and economic dimensions by providing income in marginalized regions.

A discussion on the use of wood from tropical forests, the world-wide deforestation and the loss of biodiversity related to it is beyond the scope of this study.

2.3.2 Use of wood

Based on the concept of sustainability, a reduced use of virgin natural resources and a relative shift from non-renewable to renewable resources has to be fostered. An expanded use of renewable materials due to replacement of non-renewable resources, however, leads, *ceteris paribus*, to an unacceptable pressure on land resources ((Müller 1996)). Therefore, the efficiency of resource utilisation needs to be increased in order to reduce or even better avoid this pressure ((Fraanje 1997, p. 22); Schulz 1972, p. 1416)).

However, the claim for dematerialisation—the fulfilment of a need using less material and energy input—does not make an exception for wood. But—contrary to all other important materials, such as steel, aluminium or concrete—the resource efficiency of wood should only be increased by a factor of 2—instead of a factor of 3 to 8—in order to reach sustainable consumption levels in Europe ((Mauch et al. 1995), p. 65). Basically, this requires a more effective and efficient use of wood including optimised process technology and products with longer service life and an aptitude for repair, material recycling, and finally incineration with energy recovery ((Lafleur & Fraanje 1997, p. 26f); Werner 1999, p. 74)).

Fraanje (1997) states that wood is often not used according to the "appropriate application"-principle of resource cascading (see (Sirkin & ten Houten 1994)). *Appropriate application means that the resource is applied on the basis of its (typical) properties; at the highest quality level it is possible. In this view one should not make pulp out of tree directly, but first profit by specific qualities of massive wood (like strength). When making paper directly from vegetable resources, one should rather use flax or hemp, instead of wood. Appropriate application has reference to the whole* ((Fraanje 1997, p. 24)).

Use of primary wood in energy production is not economically feasible under current market conditions in Switzerland, owing to the high cost of labour and the infrastructure for fuel preparation ((Hofer et al. 2001, p. 19)). Hence, only the use of wood as material may generate enough revenue to cover current forest exploitation costs.

2.3.3 Post-consumer wood

Handling post-consumer wood is situated between waste treatment/disposal and exploitation of a secondary material resource for material or energy purposes ((Anonymous 1995); Anonymous 1996); Voss 1998)).

Reuse & recycling of post-consumer wood brings both advantages and disadvantages. Among the advantages are ((Marutzky 1997, p. 2204); Roeffael 1997)):

- Reintegration of waste into the socio-economic system;
- Careful treatment of resources and amplification of the resource basis;
- Less occupation of space in landfills;
- Lower thermal energy consumption and VOC-emissions from drying processes if used (dry) wood is recycled as material;
- Substitution of fossil fuels (CO₂-neutrality);

- Higher energy efficiency due to lower moisture content compared to "fresh" wood if used for energy purposes;
- Destruction of harmful (organic) chemicals or their immobilisation and export from ecosphere (in the case of inorganic chemicals) if used for energy purposes in cement kilns.

Among the disadvantages are:

- Dispersion of pollutants if recycled as material (see below);
- Generation of harmful emissions if used for energy purposes;
- Higher requirements of logistics and transports than primary material;
- Displacement of other wooden raw materials such as residues from thinning or sawmills leading to lower prices for products from primary processes with already low profits or even to subsidised sources;
- Even less economically viable wood extraction and a correspondingly reduced potential substitution of fossil fuels.

Environmentally relevant problems specifically related to the incineration of post-consumer wood are ((Nussbaumer 1994)):

- High ash content and low ash melting temperature;
- Fuel nitrogen leading to NO_x emissions;
- Sub-optimal combustion processes leading to emissions of NO_x, CO and particles;
- Chlorine in the fuel leading to HCl or even to PCDD/F emissions;
- Emissions of heavy metals (Pb, Zn);
- Post-consumer wood can be contaminated with stones, sand, metals, etc.

For both the material and energy uses of post-consumer wood, the potential content of harmful substances from chemical wood conservation is a major limitation. Most agents applied (biocides) are designed for a long persistence. They remain in the wood even after the service life of the product is reached and can thus considerably limit the use and disposal options of the post-consumer wood. Among the substances used for their biocidal character are mercury, zinc and arsenic compounds, creosote or pentachlorophenol (PCP). Some of them (mercury, arsenic, PCP, chloro-organic compounds) are prohibited today in most Western European countries. Still, they are present in current post-consumer wood fractions.

Processes have been developed to remove organic wood preservatives (creosote) from post-consumer wood in order to allow its use as secondary material in the particleboard industry. However, such processes have not been put into industrial practice ((Kühne & Schwarz 1997)).

Additionally to chemical wood preservatives, adhesives, toxic pigments of coatings, and mineral contaminants are other substances limiting further utilisation of post-consumer wood.

Up to now, no labelling scheme has been put into practice (although it is currently being discussed). Visual distinction of untreated and treated post-consumer wood, in the latter case differentiating between organic and inorganic active groups, is feasible only to a very limited extent. In practice, it is limited to certain homogenous fractions such as utility poles or railway sleepers ((Peek 1998)). Large-scale analytic techniques for the sorting of post-consumer wood according to its contamination are currently being investigated in pilot plants ((Peylo 1998a); Peylo 1998b); Weis et al. 1999)).

Limits to the recycling of post-consumer wood are not only set by the possible content of harmful substances but also by the unavoidable material degradation during reprocessing. Degradation during reprocessing includes, *e.g.* a reduction in dimensions of wooden parts, an unavoidable amount of sawdust produced during reprocessing, the more cubical dimensions of reprocessed secondary chips for particleboard production (if applying dry separation processes), or reduced fibre length of reprocessed fibreboards or paper, etc.

The above situation requires three basic strategies to extend the utilisation potential of post-consumer wood as a secondary resource:

- Labelling of wood products with declaration of used additives;
- Utilisation of homogenous materials concerning the type of wood or of the wood-based intermediate products used, and with regard to the treatment with substances that limit further utilisation;
- Simple structure of wood products to facilitate disassembly and the recuperation of homogenous fractions free of substances limiting further (thermal) utilisation.

2.3.4 Wood in the international discussion on climate change

Forestry and wood products hold a particular position in the current political discussion on climate change and on the implementation of the "Kyoto-Protocol" ((Anonymous 1997)), for a summary of the current discussion, see ((Fischlin 2000)).

On one hand, the destruction of large forest areas taking place mainly in countries of the Southern Hemisphere is a considerable source of greenhouse gas emissions. On the other hand, afforestation, reforestation and forest conservation are considered valuable sinks for mitigation and storage of CO₂ ((Schärer & Jakobi 2000); Various 2000); Wegener et al. 1994)).

Due to its carbon-neutrality¹, wood is given an important role for its carbon storing potential in applications with long service life ((Skog & Nicholson 1998)) as well as for its substitution potential of non-renewable (fossil) fuels and of material whose extraction and processing heavily relies on non-renewable energy carriers ((Sikkema & Nabuurs 1995); (Matthews et al. 1996); (Gielen 1995)).

2.4 Legal framework for a sustainable utilisation of post-consumer wood

Since the beginning of the debate on environmental issues in the 80-ies, the utilisation of post-consumer wood has been subject to increasing control and regulatory works. Corresponding to the threefold nature of post-consumer wood as waste², secondary material, and fuel, a variety of ordinances³ based on the Federal Law on Environmental Protection (USG) and some international conventions⁴ apply to the utilisation of post-consumer wood (for details, see (Hofer 2000b)).

According to the Waste Concept of Switzerland ((Anonymous 1976)) the following four strategies have to be followed for the use of wood in descending order of importance:

- Reducing waste at the source;
- Minimising pollutants in processes and in products;
- Reducing waste by improved recovery operations;
- Improving domestic treatment facilities for non-recoverable wastes.

In principle, material utilisation of waste has priority over thermal utilisation. Given the current conditions, this does not hold for post-consumer wood for two reasons:

1 Atmospheric carbon is fixed as cellulose and lignin in wood by photosynthesis and released during natural rotting or technical incineration processes.

2 in the sense of the Federal Law on Environmental Protection (USG) Art. 7, para. 6

3 e.g., the Technical Ordinance on Waste (TVA), the Ordinance Relating to Environmentally Harmful Materials (StoV), the Ordinance on Air Pollution Control (LRV)

4 The Basler Übereinkommen über die Kontrolle der grenzüberschreitenden Verbringung gefährlicher Abfälle und ihrer Entsorgung, SR 0.814.05 ("Basel Convention"), the OECD-resolution C(92)39/FINAL on the control of frontier passing traffic with wastes destined for recovery operations

1. Post-consumer wood fractions can be contaminated with heavy metals or organic substances. Instead of concentrating and disposing of these contaminants, they are more likely to be distributed and reintegrated into anthroposphere. This leaves unsolved the problematic nature of contaminated post-consumer wood and may even accentuate problems in the future. The dilution of contaminated waste is forbidden under the Technical Ordinance on Waste (TVA, Art. 10);
2. For economic reasons, the potential of European forests as fuel suppliers is not exploited; currently only the revenue of material arising from material uses justifies the cost of harvesting wood. Recycling of post-consumer wood reduces primary extraction and thus the amount of wood available for incineration and substitution of fossil fuels. As a consequence, the substitution potential of wood concerning fossil fuels remains largely not exploited.

Hofer 2000a) proposes four basic strategies for the management of post-consumer wood:

3. Post-consumer wood generated in Switzerland will be supplied to thermal utilisation if feasible under sustainable conditions;
4. Material utilisation within Switzerland and abroad will take place under environmental requirements that are equivalent to the thermal utilisation of post-consumer wood in Switzerland;
5. Disposal/treatment of post-consumer wood will comply with legal regulations;
6. The quality of post-consumer wood will be improved on a mid- to long-term basis ((Hofer 2000a), p. 19; translated from German by F. Werner).

2.5 Rules for a sustainable utilisation of wood

Summarising the above chapters, the following principles for a sustainable use of wood and particularly post-consumer wood can be stated:

1. Wood production in forestry respecting qualitative and quantitative criteria for sustainable forest management;
2. Efficient and effective processing and use of wood as material in wood industry;
3. Avoidance of chemical wood protection where possible (e.g. by constructive means) and careful selection of additives like overlays, adhesives, coatings etc. (see also point 5);
4. Manufacturing of products that can easily be disassembled thus providing single-material fractions for easy recycling and incineration in appropriately equipped plants;
5. Keeping up the level of the incineration potential for the substitution of non-regenerative fossil fuels;
6. Recovery of the ashes from incineration of post-consumer wood to return them to the forest ecosystem as fertiliser in an ecologically acceptable way;
7. Incineration of post-consumer wood and maximisation of the substitution of fossil fuels;
8. Maximisation of the amount of wood stored in long-term applications such as buildings.

These principles should be incorporated properly into LCA of wood products.

3 Examples of ecoinvent processes

This chapter means to give an example of how the processes developed in this part of the ecoinvent project are connected. The grey fields in Fig. 3.1 and Fig. 3.2 represent multi-output processes for which no cumulative results can be obtained from the database.

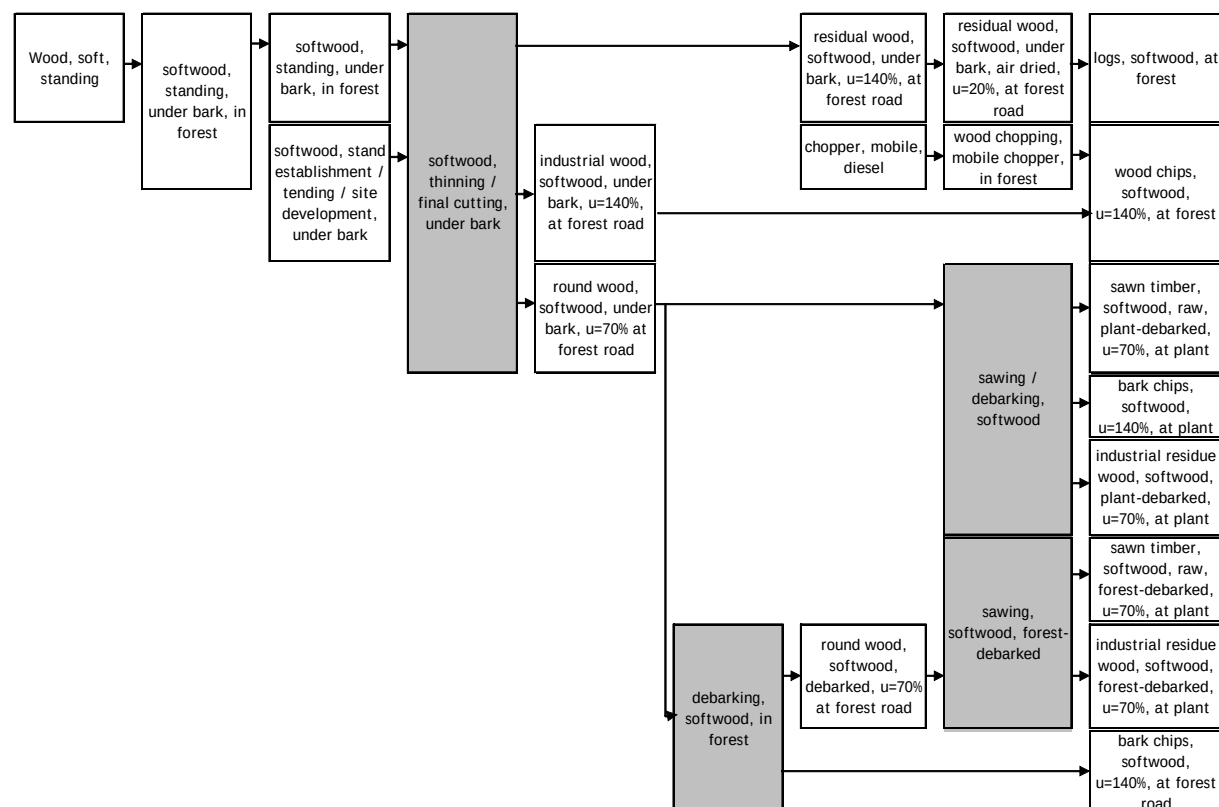


Fig. 3.1 Overview of softwood production

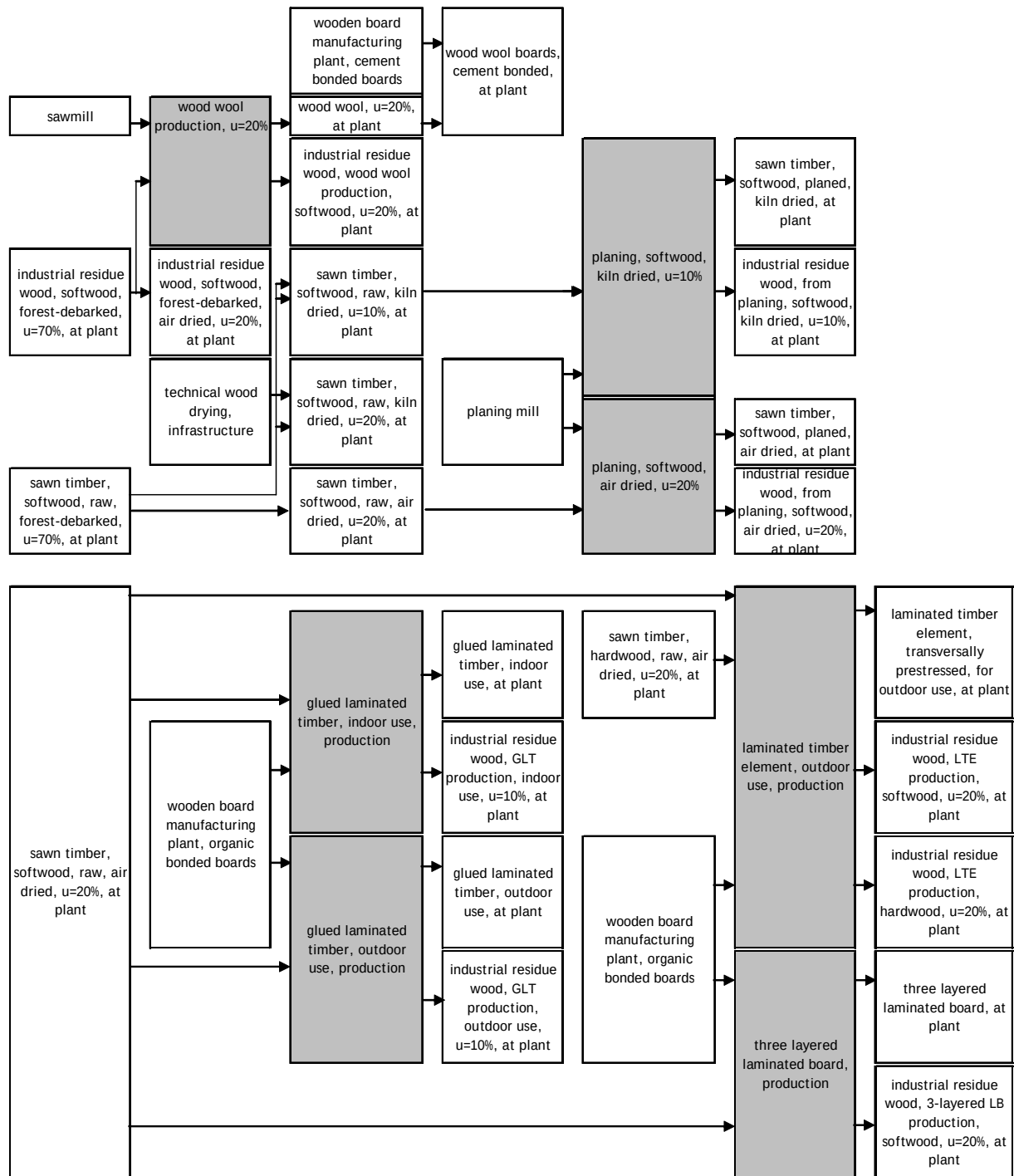
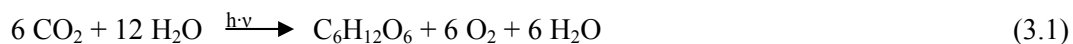


Fig. 3.2 Overview of softwood production cont.

4 Natural wood production

This chapter is mainly based on (Nimz et al. 1997); (Miller 1999a) and (Miller 1999b).

Trees compose their biomass from carbon dioxide (CO₂), water (H₂O) and several nutrients using solar energy. Simplified, the process of photosynthesis can be described approximately as:



The glucose molecule (C₆H₁₂O₆) stands for the biomass of the tree, which is composed of 6 molecules of CO₂ and 12 molecules of H₂O. The theoretical energy stored as glucose adds up to 2'872 kJ/mol or around 16 MJ per kg oven-dried biomass (u=0%).

In reality, dry wood is primarily composed of cellulose, lignin, hemi-celluloses, and minor amounts (usually 5% to 10% (w/w) of extraneous materials. Cellulose, the major component, constitutes approximately 50% (w/w) of wood substance by weight. It is a high-molecular-weight linear polymer consisting of chains of 1 to more than 4 glucose monomers. During growth of the tree, the cellulose molecules are arranged into ordered strands called fibrils, which in turn are organised into the larger structural elements that make up the cell wall of wood fibres. Lignin constitutes 23% to 33% (w/w) of the wood substance in softwoods and 16% to 25% (w/w) in hardwood. Although lignin occurs in wood throughout the cell wall, it is concentrated toward the outside of the cells and between cells. Lignin is often called the cementing agent that binds individual cells together. The hemi-celluloses are associated with cellulose and are branches of low-molecular-weight polymers composed of different kinds of pentose and hexose sugar monomers. Their relative amounts of these sugars vary markedly with species. Unlike the major constituents of wood, extraneous materials are non-structural components. Both organic and inorganic extraneous materials are found in wood. The organic components include tannins and other polyphenolics, colouring matter, essential oils, fats, resins, waxes, gum starch, and simple metabolic intermediates. These components may constitute roughly 5% to 30% (w/w) of the wood substance, depending on such factors as species, growth conditions, and time of year when the tree is cut. The inorganic components such as calcium, potassium, and magnesium generally constitute 0.2% to 1.0% (w/w) of the wood substances.

Trees are divided into two broad classes, usually referred to as hardwoods and softwoods. These names can be confusing since some softwoods are actually harder than some hardwoods, and vice-versa. Botanically, hardwoods are Angiosperms; the seeds are enclosed in the ovary of the flower. Anatomically, hardwoods are porous; i.e., they contain vessel elements. A vessel element is a wood cell with open ends; when vessel elements are set one above another, they form a continuous tube (vessel), which serves as a conduit for transporting water or sap in the tree. Typically, hardwoods are plants with broad leaves that, with few exceptions in the temperate region, lose their leaves in autumn or winter. Most imported tropical woods are hardwoods. Botanically, softwoods are Gymnosperms or conifers; the seeds are naked (not enclosed in the ovary of the flower). Anatomically, softwoods are nonporous and do not contain vessels. Softwoods are usually cone-bearing plants with needles or scale-like evergreen leaves. Some softwood, such as larches (*larix* sp.), lose their needles during autumn or winter.

Softwood timber and plywood are used in construction for forms, scaffolding, framing, sheeting, flooring, moulding, panelling, cabinets, poles and piles, and many other building components. Softwoods may also be found in the form of shingles, sashes, doors, and other millwork, in addition to some rough products such as timber and round posts. Hardwoods are used in construction for flooring, architectural woodwork, interior woodwork, and panelling. Most hardwood timber and dimension stocks are remanufactured into furniture, flooring, pallets, containers, dunnage, and blocking.

In Switzerland, the commercially most important of about 40 hardwood species and the commercially most important of about 12 softwood species are summarised in Tab. 4.1:

Tab. 4.1 Important commercially used softwood and hardwood species in Switzerland

Hardwood		Softwood	
Beech (<i>Fagus silvatica.</i>)		European spruce (<i>Picea abies</i>)	
European oak (<i>Quercus robur</i>)		Silver fir (<i>Abies alba</i>)	
European ash (<i>Fraxinus excelsior</i>)		Scots pine (<i>Pinus silvatica.</i>)	
Sweet chestnut (<i>Castanea sativa</i>)		European larch (<i>Larix decidua</i>)	
		Swiss pine (<i>Pinus cembra</i>)	

Beech constitutes 60% of total hardwood and spruce 66% of total softwood in Swiss forests. (BfS/BUWAL 2000). Therefore beech and spruce are chosen to represent hardwood and softwood in this project.

Growth of natural wood, i.e. the physiological processes of trees, is normally not covered in LCA. An exception is usually the consideration of the CO₂-uptake during photosynthesis and the embodiment of solar energy (de Feyter 1995). In this project the embodiment of solar energy is defined to equal the calorific gross value of the wood. The carbon dioxide uptake is calculated based on the carbon content of wood which for this project is 49.4% of dry wood matter.

4.1 Natural wood production in ecoinvent database

The datasets for hardwood and softwood standing in forest are used to account for the land on which trees grow and for the CO₂ assimilated by the trees. They refer to the living tree before harvesting.

The land use for wood production and forest processes is calculated according to information and assumptions given in Tab. 4.2. The total forest area is allocated to the wood production despite the fact that forests have many other functions e.g. as room for recreational activities or as protection against avalanches or floods.

The land occupied by forest roads is not considered in the natural wood production but in the forest processes. Again, uses other than for establishing, tending and harvesting of wood are not considered.

Tab. 4.2 Land use for wood production and forest processes

Nr.		Hardwood	Softwood	Source	refers to	comment
1	Yield [m^3 / ha]	7.84E+02	1.34E+03	Schweinle 2001	total area including forest roads	including bark (hardwood 12%; softwood 10% (v/v) bark in total yield)
2	Time from planting trees to final harvesting [a]	1.50E+02	1.20E+02	Schweinle 2001		
3	Forest road (length) [m / ha]	5.40E+01	5.40E+01	Schweinle 2001	total area including forest roads	
4	Forest road width [m]	2.00E+00	2.00E+00	estimate		
5	Forest road area [m^2/m^2]	1.08E-02	1.08E-02	calculated (3, 4)	total area including forest roads	
6	Yield [m^3/m^2]	7.84E-02	1.34E-01	calculated (1)	total area including forest roads	
7	Yield [m^3/m^2]	7.92E-02	1.35E-01	calculated (5, 6)	forest area excluding forest roads	
8	Land use forest [m^2/m^3]	1.26E+01	7.40E+00	calculated (7)	total yield (including bark)	without forest roads
9	Land use forest roads [m^2/m^3]	1.38E-01	8.08E-02	calculated (5,1)	total yield (including bark)	
10	Time for which land is used as forest (after transformation to forest) [a]	1.00E+03	1.00E+03	estimate		
11	Land transformation forest [m^2/m^3]	1.89E+00	8.88E-01	calculated (10, 12)	total yield (including bark)	
12	Land occupation forest [$\text{m}^2\text{a}/\text{m}^3$]	1.89E+03	8.88E+02	calculated (2, 8)	total yield (including bark)	
13	Land transformation forest road [m^2/m^3]	2.07E-02	9.70E-03	calculated (10, 14)	total yield (including bark)	
14	Land occupation forest road [$\text{m}^2\text{a}/\text{m}^3$]	2.07E+01	9.70E+00	calculated (2, 9)	total yield (including bark)	

The background information to the processes are given in Tab. 4.3, flows are shown in:

- hardwood, standing, including bark, in forest Fig. 4.1
- softwood, standing, including bark, in forest Fig. 4.2

4.1.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007).

Data for CO_2 -assimilation and for the energy embodied in the biomass are calculated from the carbon and energy content. For those, the uncertainty comes from different values for different wood species.

Land use data can vary considerably depending on how intensive the forest is used. Since the data used are valid for one specific species and region, the uncertainty is rather high.

Tab. 4.3 Ecoinvent meta information to the datasets for "hardwood, standing, under bark, in forest" and "softwood, standing, under bark, in forest"

Name	hardwood, standing, under bark, in forest	softwood, standing, under bark, in forest
Location	RER	RER
Infrastructure Process	0	0
Unit	m3	m3
Data Set Version	2.0	2.0
Included Processes	Includes the CO ₂ absorption from air and the land use for the trees to grow. Forest road area is not included in the land use	Includes the CO ₂ absorption from air and the land use for the trees to grow. Forest road area is not included in the land use
Amount	1	1
Local Name	Laubholz, stehend, ohne Rinde, im Wald	Nadelholz, stehend, ohne Rinde, im Wald
Synonyms	Beech//Buche	Spruce//Fichte
General Comment to reference function	The volume does not include the bark. CO ₂ assimilation is based on 49.4% carbon in the wood including the bark (12%). Biomass energy equals gross calorific value including bark.	The volume does not include the bark. CO ₂ assimilation is based on 49.4% carbon in the wood including the bark (10%). Biomass energy equals gross calorific value including bark.
Start Date	2001	2001
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Land use data for Germany used for central Europe	Land use data for Germany used for central Europe
Technology text	Natural growth. Relates to the forest area and CO ₂ from air needed to produce beech. It is assumed, that other hardwood species need similar time and area to grow.	Natural growth. Relates to the forest area and CO ₂ from air needed to produce spruce. It is assumed, that other softwood species need similar time and area to grow.
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumptions that different species are growing at the same rate as beech and that Germany is representative for central Europe are considered in the uncertainties.	The assumptions that different species are growing at the same rate as spruce and that Germany is representative for central Europe are considered in the uncertainties.

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Beech	→ hardwood, standing, under bark, in forest		including bark (12%)	resource	biotic			Wood, hard, standing	1.12E+00	m3	calculated	1	1.30	(4,1,1,1,3,1,12)
Solar energy			equals to gross calorific value of hardwood	resource	biotic			Energy, gross calorific value, in biomass	1.43E+04	MJ	calculated	1	1.30	(4,1,1,1,3,1,12)
CO2 regenerativ			Dried matter = 650 kg/m3, C content of dried matter = 49.4% excluding forest roads: Schweinle 2001: 700 m3/ha (incl. bark and forest roads); time from planting to final harvesting: 150 a; assumption for total time of use as forest: 150 a	resource	in air			Carbon dioxide, in air	1.32E+03	kg	calculated	1	1.30	(4,1,1,1,3,1,12)
Transformation, from forest, extensive				resource	land			Transformation, from forest, extensive	1.41E+01	m2	calculated from Schweinle 2001	1	2.10	(2,3,1,3,3,5,8)
Transformation, to forest, intensive				resource	land			Transformation, to forest, intensive, normal	1.41E+01	m2	calculated from Schweinle 2001	1	2.10	(2,3,1,3,3,5,8)
Land occupation				resource	land			Occupation, forest, intensive, normal	2.12E+03	m2a	calculated from Schweinle 2001	1	1.63	(2,3,1,3,3,5,7)
		→ Beech in forest	under bark (Inputs refer to 1.12 m3 oven-dried beech (u=0%) including bark).	wooden materials	extraction	No	RER	hardwood, standing, under bark, in forest	1.00E+00	m3				

Fig. 4.1 Flows for "hardwood, standing, under bark, in forest" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Spruce	→ softwood, standing, under bark, in forest		including bark (10%)	resource	biotic			Wood, soft, standing	1.10E+00	m3	calculated	1	1.30	(4,1,1,1,3,1,12)
Solar energy			equals to gross calorific value of softwood	resource	biotic			Energy, gross calorific value, in biomass	1.01E+04	MJ	calculated	1	1.30	(4,1,1,1,3,1,12)
CO2 regenerativ			Dried matter = 450 kg/m3, C content of dried matter = 49.4% excluding forest roads: Schweinle 2001: 1215 m3/ha (incl. bark and forest roads); time from planting to final harvesting: 120 a; assumption for total time of use as forest: 120 a	resource	in air			Carbon dioxide, in air	8.97E+02	kg	calculated	1	1.30	(4,1,1,1,3,1,12)
Transformation, from forest, extensive				resource	land			Transformation, from forest, extensive	8.14E+00	m2	calculated from Schweinle 2001	1	2.10	(2,3,1,3,3,5,8)
Transformation, to forest, intensive				resource	land			Transformation, to forest, intensive, normal	8.14E+00	m2	calculated from Schweinle 2001	1	2.10	(2,3,1,3,3,5,8)
Land occupation				resource	land			Occupation, forest, intensive, normal	9.77E+02	m2a	calculated from Schweinle 2001	1	1.63	(2,3,1,3,3,5,7)
		→ Spruce in forest	under bark (Inputs refer to 1.1 m3 oven-dried spruce (u=0%) including bark).	wooden materials	extraction	No	RER	softwood, standing, under bark, in forest	1.00E+00	m3				

Fig. 4.2 Flows for "softwood, standing, under bark, in forest" and its representation in the ecoinvent database

5 Forest processes

Technical wood production covers tree nurseries, tending, thinning, cutting, and hauling of the useable wood to the nearest forest road. Technical wood production has to be differentiated from natural wood production, *i.e.* the physiological processes of tree growth (see chapter 4)¹.

Fig. 5.1 gives an overview of the processes for technical wood production. It is based on an extensive analysis of forestry processes ((Schweinle 2000); Schweinle 2001)). These data sources are used, as they refer to the only extensive LCA-study on forestry processes covering the four economically most important species for Central Europe: spruce, fir, beech and oak. The definition of the process steps, its inputs and outputs are pre-set by the given structure of the data source.

The process steps defined are:

Tree nursing: the cultivation of tree seedlings from seeds in tree nurseries (German: Setzlingszucht);

Stand establishment: the planting of the trees in the foreseen grounds and additional protective measures against damages caused by forest fauna (German: Bestandesgründung). Planting is commonly made with planting machines mounted on forestry tractors;

Tending: Tending of forest plantations, *i.e.* of artificially established stands or crops, during the establishment stage. Tending mainly consists in grass cleaning with a hand-held motor grass mower (German: Kulturpflege);

Young-growth tending: The tending carried out in pre-thicket and thicket stands (both naturally and artificially regenerated) in order to encourage the even development of the trees and to improve the quality (German: Jungwuchspflege). Similar equipment is used as for tending;

Cleaning: Improvement felling during the thicket stage aiming at regulating species composition, stocking level and stand stability, usually without utilisation of the felled tree (German: Läuterung). Selective cleaning is usually made with small chainsaws whereas systematic cleaning is normally made with special equipment mounted on forestry tractors;

Thinning: Several subsequent fellings made in an immature crop or stand in order primarily to accelerate diameter increment but also to salvage potential mortality and to improve the average form of the remaining trees. **Thinning** vary in technique, in degree and in frequency of application. Motor-manual thinning is made with chainsaws; mechanised thinning is made with diesel-powered harvesters. Harvested trees are hauled with tractors to the nearest forest road.

For commercialisation, trees are sorted into several classes and sub-classes such as industrial wood (for the use in paper or wood fibre industry), fuel wood, and round wood, depending on diameter, branchiness, length, stature, and qualitative damages;

Cutting: Several subsequent cuts made in the mature stand as final harvesting. Again, industrial wood and round wood are harvested, hauled and sorted into several classes. Cutting is usually made with big chainsaws for felling and with small chainsaws for disbranching whereas hauling is normally made with forestry tractors (or horses).

Debarking: The main function of debarking is the removal of bark, bast, and dirt. Manual debarking is rare, because it is expensive, slow, and inefficient. Mechanical debarking is usual; debarking with cutting tools can be movable or stationary. As a tendency, softwood is debarked in the forest with movable debarkers while hardwood is commonly debarked at the sawmill with stationary ones. The amount of bark, based on the total volume of the stem with bark is about 10 to 17% for pine, for spruce 8 to 15%, for birch 7 to 15%, for beech 6 to 10% ((Kollmann et al. 1975); (Sell 1997)).

¹ This study follows the distinction made, e.g. by Fallscheer F. C. (1997), Frühwald A. et al. (1996), Jungmeier G. et al. (2002), Schweinle J. (2000), Wegener G. et al. (1994).

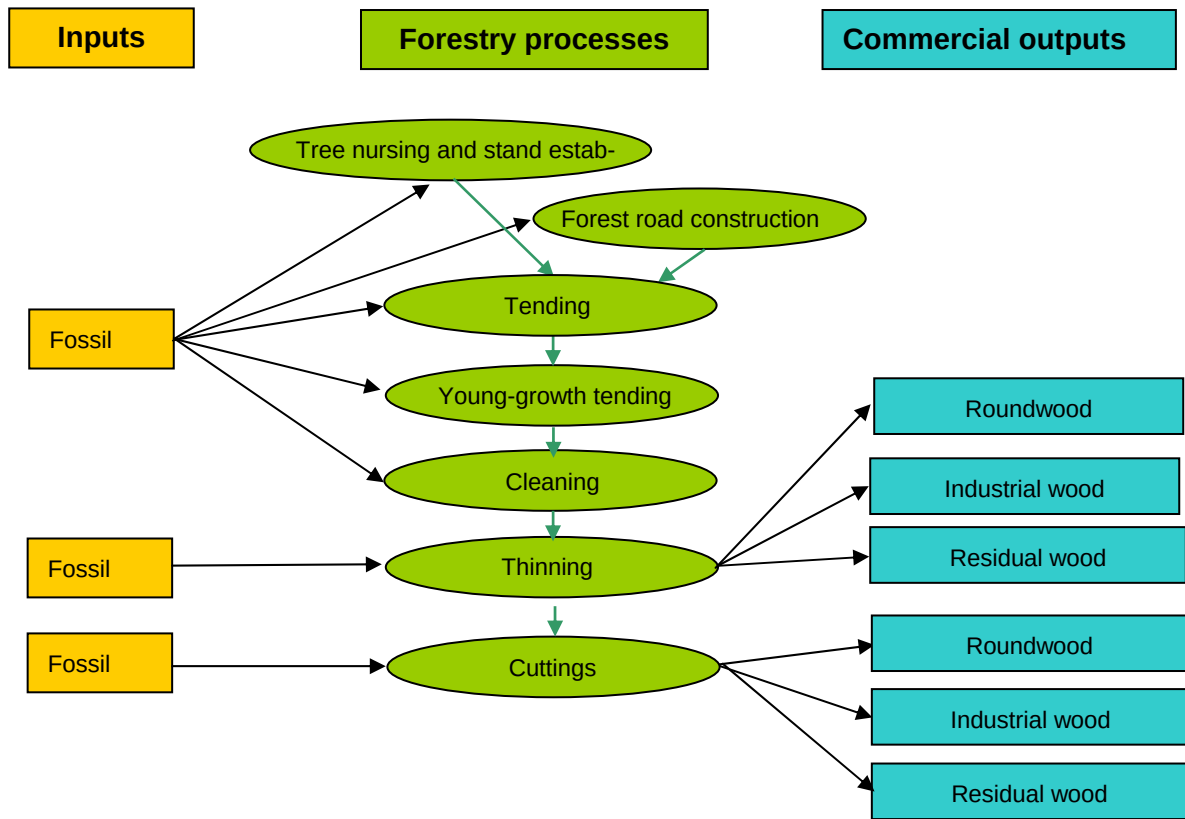


Fig. 5.1 Forestry processes, their inputs and commercial output related to the production of beech wood for railway sleepers (from (Werner et al. 2002); modified)

Several processes are not considered in this example, such as soil liming, the application of pesticides, debarking in the forest, and (wet timber) storage. Furthermore, the environmental consequences of soil compression due to the use of heavy machines are not considered for a lack of data.

Several fractions of round wood, industrial wood and residual wood result from forestry. Due to a lack of more specific data, the following simplifying assumptions are made for performing the calculations:

- Forest management aiming at the II. yield class (intense thinning) is applied in the beech stand ((Schober 1967));
- Forest management aiming at the I. yield class (intense thinning) is applied in the spruce stand ((Wiedemann 1936/42));
- Three fractions result both from thinning and cutting, round wood, industrial wood, and residual wood;
- The respective fractions of industrial wood, residual wood and round wood from thinning and cutting are equivalent;

5.1 Forest processes in ecoinvent database

Providing wood at forest road requires the usage of chainsaws and transport vehicles. The type of transport vehicle used depends very much on the type of wood harvested, its sizes as well as on spe-

cific site conditions (steepness, accessibility, etc.). Transport vehicles include agricultural tractors, harvesters with trailers, forwarders, horses, aerial ropeways, or helicopters.

The following modules cover forestry actions commonly applied in the Swiss Midland using chain-saw, tractor, and trailer.

For round wood, the tree is cut and disbranched with a chainsaw, hauled with a chain pulled by the tractor and moved by the same tractor to the forest road.

Industrial wood and residual wood is harvested with chainsaws and cut into suitable pieces. Tractor-powered or manual hauling is common. The wood is then loaded onto a tractor-driven trailer and forwarded to the forest road.

Infrastructure requirements of the chainsaw and the tractor are included in the respective modules.

The trailer assumed has a loading capacity of 8 t, a net mass of 2 t, a live time of 15 years and a yearly service time of 80 h (see (Nemecek et al. 2007)). Thus 1.667 kg of the trailer has to be inventoried per hour of its use.

The land use for forest roads included in the "stand establishment / tending / site development" - processes is calculated according to Tab. 4.2. The gravel needed for maintaining the forest roads is estimated assuming that new gravel is distributed over the forest road area in a 2 cm thick layer every five years.

The allocation of the burdens to the different outputs of the thinning/final cutting processes is proposed according to the economic return of the products. Inputs that can be clearly attributed to specific products (e.g. trailer is used only for the transport of industrial and residual wood but not for round wood) are allocated to them exclusively.

The economic allocation can't respect the mass and energy balance of the products. Therefore, correcting modules are defined that can add or subtract the CO₂ uptake and the resource (wood and solar energy) consumption from nature. They are used as shown in the example in Fig. 5.2:

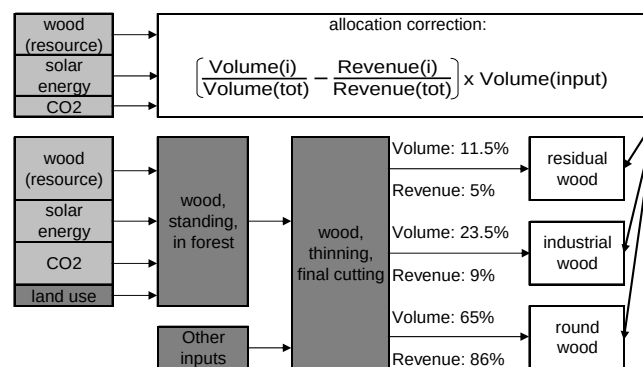


Fig. 5.2 Example for use of allocation correction modules

The background information to the allocation correction processes are given in Tab. 5.1, flows are shown in:

- hardwood, allocation correction, x
- softwood, allocation correction, x

Fig. 5.3

Fig. 5.4

Tab. 5.1 Ecoinvent meta information to the datasets for allocation correction

Name	hardwood, allocation correction, x	softwood,allocation correction, x
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Includes the wood resource, the CO ₂ absorption from air and the energy in biomass	Includes the wood resource, the CO ₂ absorption from air and the energy in biomass
Amount	1	1
Local Name	Laubholz, Allokationskorrektur, 1	Nadelholz, Allokationskorrektur, 1
Synonyms		
General Comment to reference function	This module is used to allocate the proper mass, energy and CO ₂ uptake from nature for the products for which economic allocation is made	This module is used to allocate the proper mass, energy and CO ₂ uptake from nature for the products for which economic allocation is made
Start Date	2002	2002
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geograhcy text		
Technology text	none	none
Representativeness [%]		
Production Volume	none	none
Sampling Procedure	none	none
Extrapolations	none	none
Uncertainty Adjustments	none	none

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Beech	hardwood, allocation correction, x			resource	biotic			Wood, hard, standing	1.00E+00	m3	calculated	1	1.00	calculated correction term
Solar energy			equals to gross calorific value of hardwood	resource	biotic			Energy, gross calorific value, in biomass	1.27E+04	MJ	calculated	1	1.00	calculated correction term
CO2 regenerativ			Dried matter =650 kg/m3, C content of dried matter = 49.4%	resource	in air			Carbon dioxide, in air	1.18E+03	kg	calculated	1	1.00	calculated correction term
		Hardwood, allocation correction	used to correct the mass and energy balance where economic allocation is made	wooden materials	extraction	No	RER	hardwood, allocation correction, x	1.00E+00	m3				

Fig. 5.3 Flows for "hardwood, allocation correction, x" (x = 1-3) and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Spruce	softwood, allocation correction, x			resource	biotic			Wood, soft, standing	1.00E+00	m3	calculated	1	1.00	calculated correction term
Solar energy			equals to gross calorific value of softwood	resource	biotic			Energy, gross calorific value, in biomass	9.18E+03	MJ	calculated	1	1.00	calculated correction term
CO2 regenerativ			Dried matter =450 kg/m3, C content of dried matter = 49.4%	resource	in air			Carbon dioxide, in air	8.15E+02	kg	calculated	1	1.00	calculated correction term
		softwood, allocation correction	used to correct the mass and energy balance where economic allocation is made	wooden materials	extraction	No	RER	softwood, allocation correction, x	1.00E+00	m3				

Fig. 5.4 Flows for "softwood, allocation correction, x" (x = 1-3) and its representation in the ecoinvent database

The background information to the actual forest processes are given in Tab. 5.2, flows are shown in:

- hardwood, stand establishment / tending / site development, under bark Fig. 5.5
- softwood, stand establishment / tending / site development, under bark Fig. 5.6
- hardwood, thinning/final cutting, under bark
- Outputs:
 - round wood, hardwood, under bark, $u=70\%$, at forest road
 - residual wood, hardwood, under bark, $u=80\%$, at forest road
 - industrial wood, hardwood, under bark, $u=80\%$, at forest road Fig. 5.7
- Softwood, thinning/final cutting, under bark
- Outputs:
 - round wood, softwood, under bark, $u=70\%$ at forest road
 - residual wood, softwood, under bark, $u=140\%$, at forest road
 - industrial wood, softwood, under bark, $u=140\%$, at forest road Fig. 5.8
- Debarking, softwood, in forest
- Outputs:
 - round wood, softwood, debarked, $u=70\%$ at forest road
 - bark chips, softwood, $u=140\%$, at forest road Fig. 5.9

The different water content in softwood and hardwood is due to the different anatomy of the wood species (see chapter 2.1).

5.1.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Data quality for the type of forest management that relates to the data is good. Data uncertainty is adjusted to cover also wood from short-rotation areas for energy wood production.

Tab. 5.2 Ecoinvent meta information for forest processes

Name	hardwood, stand establishment / tending / site development, under bark	softwood, stand establishment / tending / site development, under bark
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Includes motormanual processes for stand establishment, tending and site development for beech and the land and material (including transports) use for the forest roads.	Includes motormanual processes for stand establishment, tending and site development for spruce and the land and material (including transports) use for the forest roads.
Amount	1	1
Local Name	Laubholz, Bestandesgründung / Kulturpflege / Läuterung, Festkubikmeter	Nadelholz, Bestandesgründung / Kulturpflege / Läuterung, Festkubikmeter
Synonyms	Beech//Buche	Spruce//Fichte
General Comment to reference function	The volume refers to the wood not including the bark.	The volume refers to the wood not including the bark
Start Date	1996	1996
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Germany used for central Europe	Data for Germany used for central Europe
Technology text	Considers only one type of forest interventions (Motormanual processes for "Buche, II. Ertragsklasse, starke Durchforstung")	Considers only one type of forest interventions (Motormanual processes for "Fichte, I. Ertragsklasse, starke Durchforstung")
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that Germany is representative for central Europe is considered in the uncertainties.	The assumption that Germany is representative for central Europe is considered in the uncertainties.

Tab. 5.2 Ecoinvent meta information for forest processes (cont.)

Name	hardwood, thinning / final cutting, under bark	softwood, thinning / final cutting, under bark	debarking, softwood, in forest
Location	RER	RER	RER
Infrastructure Process	0	0	0
Unit	m ³	m ³	m ³
Data Set Version	2.0	2.0	2.0
Included Processes	Includes motormanual processes for thinning and final cutting as well as transport of the products to the nearest forest road.	Includes motormanual processes for thinning and final cutting as well as transport of the products to the nearest forest road.	Tractor driven debarking; allocation based on proceeds
Amount	1	1	1
Local Name	Laubholz, Durchforstung / Endnutzung, Festkubikmeter	Nadelholz, Durchforstung / Endnutzung, Festkubikmeter	Entrindung, Nadelholz, im Wald
Synonyms	Beech//Buche	Spruce//Fichte	Spruce//Fichte
General Comment to reference function	The volume refers to the wood not including the bark. The multi-output process "hardwood, thinning/final cutting, under bark" delivers the three coproducts "round wood, hardwood, under bark, u=70%, at forest road", "residual wood, hardwood, under bark, u=80%, at forest road" and "industrial wood, hardwood, under bark, u=80%, at forest road". Allocation is based on the over-all proceeds of the site.	The volume refers to the wood not including the bark. The multi-output process "softwood, thinning/final cutting, under bark" delivers the three coproducts "round wood, softwood, under bark, u=70%, at forest road", "residual wood, softwood, under bark, u=140%, at forest road" and "industrial wood, softwood, under bark, u=140%, at forest road". Allocation is based on the over-all proceeds of the site.	The volume refers to the wood not including the bark. The multi-output process "debarking, softwood, in forest" delivers the two coproducts "round wood, softwood, debarked, u=70% at forest road" and "bark chips, softwood, u=140%, at forest road". Allocation is based on the over-all proceeds of the process.
Start Date	1996	1996	1994
End Date	2002	2002	2001
Data Valid For Entire Period	1	1	1
Geography text	Data for Germany used for central Europe	Data for Germany used for central Europe	Data for Germany used for central Europe
Technology text	Considers only one type of forest interventions (Motormanual processes for "Buche, II. Ertragsklasse, starke Durchforstung")	Considers only one type of forest interventions (Motormanual processes for "Fichte, I. Ertragsklasse, starke Durchforstung")	Refers to a tractor driven process.
Representativeness [%]			
Production Volume	unknown	unknown	unknown
Sampling Procedure	Literature	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that Germany is representative for central Europe is considered in the uncertainties.	The assumption that Germany is representative for central Europe is considered in the uncertainties.	The assumption that Germany is representative for central Europe is considered in the uncertainties.

General Flow information				Representation in ecoinvent									Uncertainty information				
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Diesel burned in building machine	→	hardwood, stand establishment / tending / site development, under bark			I. yield class, intense thinning	construction processes	machinery	No	GLO	diesel, burned in building machine	5.75E+00	MJ	Schweinle 2000; Werner 2002	1	2.01	(1,3,2,3,1,1,5)	
Chainsaw carburator without catalytic converter	→					construction processes	machinery	No	RER	power sawing, without catalytic converter	5.30E-02	h	Schweinle 2000; Werner 2002	1	2.01	(1,3,2,3,1,1,5)	
Land transformation from forest	→				Schweinle 2001: 54 lfm / ha forest. Assumption: average width 2 m.	resource	land			Transformation, from forest, extensive	1.38E-01	m2	calculated from Schweinle 2001	1	2.06	(2,3,1,3,1,5,8)	
Land transformation to forest roads	→					resource	land			Transformation, to traffic area, road embankment	1.38E-01	m2	calculated from Schweinle 2001	1	2.06	(2,3,1,3,1,5,8)	
Land occupation of forest roads	→					resource	land			Occupation, traffic area, road embankment	2.07E+01	m2a	calculated from Schweinle 2001	1	1.57	(2,3,1,3,1,5,7)	
Gravel for forest roads	→					Road maintenance; no new forest roads built in Switzerland	construction materials	additives	No	CH	gravel, crushed, at mine	1.32E+02	kg	estimated	1	1.12	(1,4,2,3,1,1,4)
Transport lorry 16 t	→					Gravel: 50 km, by lorry 16t	transport systems	road	No	RER	transport, lorry 3.5-16t, fleet average	6.62E+00	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
				→	Stand establishment / tending / site development beech	under bark	wooden materials	extraction	No	RER	hardwood, stand establishment / tending / site development, under bark	1.00E+00	m3				

Fig. 5.5 Flows for "hardwood, stand establishment / tending / site development, under bark" and its representation in the ecoinvent database

5 Forest processes

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Diesel burned in building machine	softwood, stand establishment / tending / site development, under bark		II. yield class, intense thinning	construction processes	machinery	No	GLO	diesel, burned in building machine	2.06E+00	MJ	Schweinle 2000; Werner 2002	1	2.01	(1,3,2,3,1,1,5)
Chainsaw carburetor without catalytic converter				construction processes	machinery	No	RER	power sawing, without catalytic converter	2.01E-02	h	Schweinle 2000; Werner 2002	1	2.01	(1,3,2,3,1,1,5)
Land transformation from forest				resource	land			Transformation, from forest, extensive	8.08E-02	m2	calculated from Schweinle 2001	1	2.06	(2,3,1,3,1,5,8)
Land transformation to forest roads				resource	land			Transformation, to traffic area, road embankment	8.08E-02	m2	calculated from Schweinle 2001	1	2.06	(2,3,1,3,1,5,8)
Land occupation as forest roads				resource	land			Occupation, traffic area, road embankment	9.70E+00	m2a	calculated from Schweinle 2001	1	1.57	(2,3,1,3,1,5,7)
Gravel for forest roads			Road maintenance; no new forest roads built in Switzerland	construction materials	additives	No	CH	gravel, crushed, at mine	6.21E+01	kg	estimated	1	1.12	(1,4,2,3,1,1,4)
Transport lorry 16 t			Gravel: 50 km, by lorry 16t	transport systems	road	No	RER	transport, lorry 3.5-16t, fleet average	3.10E+00	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
		Stand establishment / tending / site development spruce	under bark	wooden materials	extraction	No	RER	softwood, stand establishment / tending / site development, under bark	1.00E+00	m3				

Fig. 5.6 Flows for "softwood, stand establishment / tending / site development, under bark" and its representation in the ecoinvent database

5 Forest processes

General Flow information						Representation in ecoinvent									Uncertainty information		
Input	Process Name	Output	Allocation	source allocation	Remarks	Category	Sub category	Infra struc ture	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Beech in forest	hardwood, thinning / final cutting, under bark				under bark (--> 1.12 m3 wood including bark)	wooden materials	extraction	No	RER	hardwood, standing, under bark, in forest	1.00E+00	m3	Frühwald et al. 1996	1	1.07	(1,2,2,3,1,1,11)	
Stand establishment / tending / site development beech					under bark	wooden materials	extraction	No	RER	hardwood, stand establishment / tending / site development, under bark	1.00E+00	m3	Schweinle 2000; Werner 2002	1	1.08	(1,3,2,3,1,1,3)	
Diesel burned in building machine						construction processes	machinery	No	GLO	diesel, burned in building machine	4.54E+01	MJ	Schweinle 2000; Werner 2002	1	2.01	(1,3,2,3,1,1,5)	
Chainsaw carburator without catalytic converter						construction processes	machinery	No	RER	power sawing, without catalytic converter	1.48E-01	h	Schweinle 2000; Werner 2002	1	2.01	(1,3,2,3,1,1,5)	
Transport						workers and equipment --> neglected											
Trailer				to residual wood: 0.33; to industrial wood 0.67	allocation only to residual and industrial wood	Assumption: trailer is in use for 15 minutes per unit process	agricultural means of production	machinery	Yes	CH	trailer, production	4.17E-01	kg	estimated	1	2.05	(4,n.A.,n.A.,n.A.,n.A.,5)
Hardwood, allocation correction				100% to residual wood		adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	1.09E-01	m3	calculated	1	1.00	calculated correction term
Hardwood, allocation correction				100% to industrial wood			wooden materials	extraction	No	RER	hardwood, allocation correction, 2	2.39E-01	m3	calculated	1	1.00	calculated correction term
Hardwood, allocation correction				100% to roundwood			wooden materials	extraction	No	RER	hardwood, allocation correction, 3	-3.47E-01	m3	calculated	1	1.00	calculated correction term
			Residual wood beech, at forest road (u=80%)	Price recommendation (year 2000): 32 CHF/m3; revenue: 5 CHF; revenue process: 83 CHF; alloc. factor: 0.06	www.wvs.ch Richtpreise für Energieholzpreise 2000/2001	under bark	wooden materials	extraction	No	RER	residual wood, hardwood, under bark, u=80%, at forest road	1.57E-01	m3	Source moisture content: Frühwald et al. 1996; yield: Schweinle 2000, Schweinle 2001; BFS/BUWAL 2000			
			Industrial wood beech, at forest road (u=80%)	Weighted average (year 2000): 44 CHF/t (u=0%) = 29 CHF/m3; revenue: 10 CHF; revenue process: 83 CHF; alloc. factor: 0.12	after BUWAL/BFS, Tab. 7.1	under bark	wooden materials	extraction	No	RER	industrial wood, hardwood, under bark, u=80%, at forest road	3.33E-01	m3	Source moisture content: Frühwald et al. 1996; yield: Schweinle 2000, Schweinle 2001; BFS/BUWAL 2000			
			Roundwood beech, at forest road (u=70%)	Weighted average (year 2000): 133 CHF/m3; revenue: 68 CHF; revenue process: 83 CHF; alloc. factor: 0.82	after BUWAL/BFS, Tab. 7.2	under bark	wooden materials	extraction	No	RER	round wood, hardwood, under bark, u=70%, at forest road	5.10E-01	m3	Source moisture content: Frühwald et al. 1996; yield: Schweinle 2000, Schweinle 2001; RES/RIWAL 2000			

Fig. 5.7 Flows for "hardwood, thinning/final cutting, under bark" and its representation in the ecoinvent database

5 Forest processes

General Flow information						Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Spruce in forest	softwood, thinning / final cutting, under bark				under bark (-> 1.1 m3 wood including bark)	wooden materials	extraction	No	RER	softwood, standing, under bark, in forest	1.00E+00	m3	Frühwald et al. 1996	1	1.07	(1,2,2,3,1,1,11)
Stand establishment / tending / site development spruce					under bark	wooden materials	extraction	No	RER	softwood, stand establishment / tending / site development, under bark	1.00E+00	m3	Schweinle 2000; Werner 2002	1	1.08	(1,3,2,3,1,1,3)
Diesel burned in building machine						construction processes	machinery	No	GLO	diesel, burned in building machine	7.72E+01	MJ	Schweinle 2000; Werner 2002	1	2.01	(1,3,2,3,1,1,5)
Chainsaw carburator without catalytic converter						construction processes	machinery	No	RER	power sawing, without catalytic converter	2.51E-01	h	Schweinle 2000; Werner 2002	1	2.01	(1,3,2,3,1,1,5)
Transport					workers and equipment -> neglected											
Trailer					to residual wood: 0.36; to industrial wood 0.64	agricultural means of production	machinery	Yes	CH	trailer, production	4.17E-01	kg	estimated	1	2.05	(4,nA,nA,nA,nA,5)
Softwood, allocation correction					100% to residual wood	wooden materials	extraction	No	RER	softwood, allocation correction, 1	7.15E-02	m3	calculated	1	1.00	calculated correction term
Softwood, allocation correction					100% to industrial wood	wooden materials	extraction	No	RER	softwood, allocation correction, 2	1.60E-01	m3	calculated	1	1.00	calculated correction term
Softwood, allocation correction					100% to roundwood	wooden materials	extraction	No	RER	softwood, allocation correction, 3	-2.31E-01	m3	calculated	1	1.00	calculated correction term
		Residual wood spruce, at forest road (u=140%)		Price recommendation (year 2000): 28 CHF/m3; revenue: 3 CHF; revenue process: 57 CHF; alloc. factor: 0.05	www.wvs.ch Richtpreise für Energieholzpreise 2000/2001	wooden materials	extraction	No	RER	residual wood, softwood, under bark, u=140%, at forest road	1.15E-01	m3	Source moisture content: Frühwald et al. 1996; yield: Schweinle 2000, Schweinle 2001; BFS/BUWAL 2000			
		Industrial wood spruce, at forest road (u=140%)		Weighted average (year 2000): 44 CHF/t (u=0%) = 20 CHF/m3; revenue: 5 CHF; revenue process: 57 CHF; alloc. factor: 0.09	after BUWAL/BFS, Tab. 7.1	wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=140%, at forest road	2.35E-01	m3	Source moisture content: Frühwald et al. 1996; yield: Schweinle 2000, Schweinle 2001; BFS/BUWAL 2000			
		Roundwood spruce, at forest road (u=70%)		Weighted average (year 2000): 75 CHF/m3 (-7 CHF for bark; estimated); revenue: 49 CHF; revenue process: 57 CHF; alloc. factor: 0.86	after BUWAL/BFS, Tab. 7.2	wooden materials	extraction	No	RER	round wood, softwood, under bark, u=70% at forest road	6.50E-01	m3	Source moisture content: Frühwald et al. 1996; yield: Schweinle 2000, Schweinle 2001; BFS/BUWAL 2000			

Fig. 5.8 Flows for "softwood, thinning/final cutting, under bark" and its representation in the ecoinvent database

5 Forest processes

General Flow information						Representation in ecoinvent									Uncertainty information		
Input	Process Name	Output	Allocation	source allocation	Remarks	Category	Sub category	Infra struc ture	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Roundwood spruce, at forest road (u=70%) →	de-barking, softwood, in forest				under bark	wooden materials	extraction	No	RER	round wood, softwood, under bark, u=70% at forest road	1.00E+00	m3	calculated	1	1.51	(1,3,5,3,1,1,4)	
Diesel burned in building machine →						construction processes	machinery	No	GLO	diesel, burned in building machine	2.84E+01	MJ	Wegener et al. 1994	1	2.02	(2,3,3,3,1,3,5)	
Lubricating oil →						chemicals	organics	No	RER	lubricating oil, at plant	2.52E-02	kg	Wegener et al. 1994	1	1.15	(2,3,3,3,1,3,4)	
Softwood, allocation correction →		100% to bark chips			adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	softwood, allocation correction, 1	8.68E-02	m3	calculated	1	1.00	calculated correction term	
Softwood, allocation correction →		100% to roundwood				wooden materials	extraction	No	RER	softwood, allocation correction, 2	-8.68E-02	m3	calculated	1	1.00	calculated correction term	
		Bark chips spruce, at forest road (u=140%) →	10 CHF/m3 (estimated); revenue: 1 CHF; revenue process: 83.- CHF; alloc. factor: 0.012			residues remaining in the forest	wooden materials	extraction	No	RER	bark chips, softwood, u=140%, at forest road	1.00E-01	m3	Frühwald et al. 1996			
		Roundwood spruce debarked, at forest road (u=70%) →	Weighted average (year 2000): 82 CHF/m3; revenue: 82 CHF; revenue process: 83.- CHF; alloc. factor: 0.988	after BUWAL/BFS, Tab. 7.1			wooden materials	extraction	No	RER	round wood, softwood, debarked, u=70% at forest road	1.00E+00	m3	calculated			
		Dust →				neglegible because of high water content of wood											

Fig. 5.9 Flows for "de-barking, softwood, in forest" and its representation in the ecoinvent database

6 „Direct“ forest products

This chapter describes the production of split wood (logs) generally used as fuel wood and wood chips used as fuel or primary material for industrial purposes such as particle- or fibreboard or paper. Split wood is normally made of residual wood; chips are mainly made of industrial wood. Of course, also residual wood can be chipped.

The advantage of chips as raw material lies in its higher transport density compared to industrial wood.

6.1 Direct forest products in ecoinvent database

6.1.1 Split wood

For the production of split wood, residual wood (branches, small trees from tending or thinning) are cut into one meter long log pieces, loaded onto a tractor-driven trailer, and transported to the forest road. The logs are split along the grain with a mobile, tractor-powered cleaving machine; the resulting split wood is piled along the forest road for natural drying; small-diameter logs and branches are piled without splitting.

The background information to the processes are given in Tab. 6.1, flows are shown in:

- logs, hardwood, at forest Fig. 6.1
- logs, softwood, at forest Fig. 6.2
- logs, mixed, at forest Fig. 6.3

6.1.2 Wood chips

This chapter deals with wood chips directly from forest. For wood chips from industrial residual wood see the following chapters (especially chapter 11).

For the production of wood chips, industrial wood at forest road is fed into a mobile chopper. The water content of the chips is derived by the assumption that the industrial wood is chopped immediately after harvesting. If it is chopped only a few days later the water content could be considerably lower (in dry weather) due to evaporation over the leaves.

The background information to the processes are given in Tab. 6.2, flows are shown in:

- wood chips, hardwood, u=80%, at forest Fig. 6.4
- wood chips, softwood, u=140%, at forest Fig. 6.5
- wood chips, mixed, u=120%, at forest Fig. 6.6

6.1.3 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

The energy demand for splitting wood and for wood chips can vary considerably depending on many parameters. Thus this figure is rather uncertain. Material input is derived from physical relations thus being certain.

Tab. 6.1 Ecoinvent meta information for the split wood processes

Name	logs, hardwood, at forest	logs, softwood, at forest	logs, mixed, at forest
Location	RER	RER	RER
Infrastructure Process	0	0	0
Unit	m ³	m ³	m ³
Data Set Version	2.0	2.0	2.0
Included Processes	Includes only the splitting of residual wood with a diesel powered splitting machine	Includes only the splitting of residual wood with a diesel powered splitting machine	Includes only the splitting of the average mix of 72% residual softwood and 28% residual hardwood (reflecting Swiss average wood consumption) with a diesel powered splitting machine
Amount	1	1	1
Local Name	Stückholz, Laubholz, ab Wald	Stückholz, Nadelholz, ab Wald	Stückholz, Mix, ab Wald
Synonyms	Buche//Beech//Brennholz	Fichte//Spruce//Brennholz	Brennholz
General Comment to reference function	The volume refers to the wood including the bark. The logs have a water content (u) of 20%, a mass of 780 kg/m ³ and a lower heating value of 15.0 MJ/kg.	The volume refers to the wood including the bark. The logs have a water content (u) of 20%, a mass of 540 kg/m ³ and a lower heating value of 15.7 MJ/kg.	The volume refers to the wood including the bark. The logs have a water content (u) of 20%, a mass of 607 kg/m ³ and a lower heating value of 15.5 MJ/kg.
Start Date	1996	1996	1996
End Date	1996	1996	2000
Data Valid For Entire Period	1	1	1
Other Period Text			
Geography text	Data for Switzerland used for central Europe	Data for Switzerland used for central Europe	Data for Switzerland used for central Europe
Technology text	Motormanual production with diesel powered splitting machine	Motormanual production with diesel powered splitting machine	Motormanual production with diesel powered splitting machine
Representativeness [%]			
Production Volume	unknown	unknown	unknown
Sampling Procedure	Literature	Literature	Literature
Extrapolations	see Geography	see Geography	see Geography
Uncertainty Adjustments	The assumption that Switzerland is representative for central Europe is considered in the uncertainties.	The assumption that Switzerland is representative for central Europe is considered in the uncertainties.	The assumption that Switzerland is representative for central Europe is considered in the uncertainties.

Tab. 6.2 Ecoinvent meta information for the wood chips in forest processes

Name	wood chips, hardwood, u=80%, at forest	wood chips, softwood, u=140%, at forest	wood chips, mixed, u=120%, at forest
Location	RER	RER	RER
Infrastructure Process	0	0	0
Unit	m ³	m ³	m ³
Data Set Version	2.0	2.0	2.0
Included Processes	Includes the chopping of residual hardwood with a mobile chopper in the forest. Also includes the driving of the mobile chopper to and within the forest	Includes the chopping of residual softwood with a mobile chopper in the forest. Also includes the driving of the mobile chopper to and within the forest	Includes the chopping of the average mix of 72% residual softwood and 28% residual hardwood (reflecting Swiss average wood consumption) with a mobile chopper in the forest. Also includes the driving of the mobile chopper to and within the forest
Amount	1	1	1
Local Name	Holzschnitzel, Laubholz, u=80%, ab Wald	Holzschnitzel, Nadelholz, u=140%, ab Wald	Holzschnitzel, Mix, u=120%, ab Wald
Synonyms	Buche//Beech//Häcksel	Fichte//Spruce//Häcksel	Häcksel
General Comment to reference function	The volume refers to the bulked volume: dried matter content = 239 kg/m ³ dried matter (u=0%), lower heating value = 4048 MJ/m ³ . Wood is unstored/undried and not stemming from short-rotation areas for energy wood production!	The volume refers to the bulked volume: dried matter content = 169 kg/m ³ dried matter (u=0%), lower heating value = 3001 MJ/m ³ . Wood is unstored/undried and not stemming from short-rotation areas for energy wood production!	The volume refers to the bulked volume: dried matter content = 188.6 kg/m ³ dried matter (u=0%), lower heating value = 3294 MJ/m ³ . Wood is unstored/undried and not stemming from short-rotation areas for energy wood production!
Start Date	1996	1996	1996
End Date	1998	1998	2002
Data Valid For Entire Period	1	1	1
Other Period Text			
Geography text	Data for Austria used for central Europe	Data for Austria used for central Europe	Data for Austria (process) and Switzerland (mix) used for central Europe
Technology text	Chopping with a large mobile chopper. Wood is unstored/undried and not stemming from short-rotation areas for energy wood production!	Chopping with a large mobile chopper. Wood is unstored/undried and not stemming from short-rotation areas for energy wood production!	Chopping with a large mobile chopper. Wood is unstored/undried and not stemming from short-rotation areas for energy wood production!
Representativeness [%]			
Production Volume	unknown	unknown	unknown
Sampling Procedure	Literature	Literature	Literature
Extrapolations	see Geography	see Geography	see Geography
Uncertainty Adjustments	The assumption that Austria is representative for central Europe is considered in the uncertainties.	The assumption that Austria is representative for central Europe is considered in the uncertainties.	The assumption that Austria / Switzerland are representative for central Europe is considered in the uncertainties.

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Residual wood beech, naturally dried, at forest road (u=20%)	logs, hardwood, at forest		under bark	wooden materials	extraction	No	RER	residual wood, hardwood, under bark, air dried, u=20%, at forest road	8.93E-01	m3	Frischknecht et al. 1995	1	1.35	(3,3,3,3,5,3)
Diesel burned in building machine			7 l diesel fuel/MWH at 0.19 MWH/stere corresponding to 0.7 m3 wood (MWH: Man Working Hour)	construction processes	machinery	No	GLO	diesel, burned in building machine	6.83E+01	MJ	Frischknecht et al. 1996	1	2.12	(3,3,3,3,5,5)
		Split wood, hardwood, 1 m, at forest	Including bark	wood energy	fuels	No	RER	logs, hardwood, at forest	1.00E+00	m3	Frischknecht et al. 1996			

Fig. 6.1 Flows for "logs, hardwood, at forest" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Residual wood spruce, naturally dried, at forest road (u=20%)	logs, softwood, at forest		under bark	wooden materials	extraction	No	RER	residual wood, softwood, under bark, air dried, u=20%, at forest road	9.09E-01	m3	Frischknecht et al. 1996	1	1.35	(3,3,3,3,5,3)
Diesel burned in building machine			7 l diesel fuel/MWH at 0.19 MWH/stere corresponding to 0.7 m3 wood (MWH: Man Working Hour)	construction processes	machinery	No	GLO	diesel, burned in building machine	6.83E+01	MJ	Frischknecht et al. 1996	1	2.12	(3,3,3,3,5,5)
		Split wood, softwood, 1 m, at forest	Including bark	wood energy	fuels	No	RER	logs, softwood, at forest	1.00E+00	m3	Frischknecht et al. 1996			

Fig. 6.2 Flows for "logs, softwood, at forest" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
Split wood, softwood, 1 m, at forest	logs, mixed, at forest			wood energy	fuels	No	RER	logs, softwood, at forest	7.20E-01	m3	BFS/BUWAL 2000	1	1.35	(3,3,3,3,3,5,3)
Split wood, hardwood, 1 m, at forest				wood energy	fuels	No	RER	logs, hardwood, at forest	2.80E-01	m3	BFS/BUWAL 2000	1	1.35	(3,3,3,3,3,5,3)
		Split wood, mix, 1 m, at storehouse	Including bark	wood energy	fuels	No	RER	logs, mixed, at forest	1.00E+00	m3				

Fig. 6.3 Flows for "logs, mixed, at forest" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
Industrial wood beech, at forest road (u=80%)	wood chips, hardwood, u=80%, at forest		volume under bark	wooden materials	extraction	No	RER	industrial wood, hardwood, under bark, u=80%, at forest road	3.29E-01	m3	calculated	1	1.35	(3,3,3,3,3,5,3)
Wet wood chopping with mobile chopper in forest			includes chopping and driving	wooden materials	extraction	No	RER	wood chopping, mobile chopper, in forest	2.39E+02	kg	Bergmair 1996 according to Anonymous 1998 (GEMIS)	1	2.12	(3,3,3,3,3,5,5)
			bulked volume: dried matter content = 239 kg/m3 dried matter (u=0%), lower heating value = 4048 MJ/m3. Wood is unstored/undried and not stemming from short-rotation areas for energy wood production!	wood energy	fuels	No	RER	wood chips, hardwood, u=80%, at forest	1.00E+00	m3				

Fig. 6.4 Flows for "wood chips, hardwood, u=80%, at forest " and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	STDv 95%	General Comment
Industrial wood spruce, at forest road (u=140%)	wood chips, softwood, u=140%, at forest	Wood chips softwood, at forest road (u=140%)	volume under bark	wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=140%, at forest road	3.41E-01	m3	calculated	1	1.35	(3,3,3,3,3,5,3)
Wet wood chopping with mobile chopper in forest			includes chopping and driving	wooden materials	extraction	No	RER	wood chopping, mobile chopper, in forest	1.69E+02	kg	Bergmair 1996 according to Anonymous 1998 (GEMIS)	1	2.12	(3,3,3,3,3,5,5)
			bulk volume: dried matter content = 169 kg/m3 dried matter (u=0%), lower heating value = 3001 MJ/m3. Wood is unstored/undried and not stemming from short-rotation areas for energy wood production!	wood energy	fuels	No	RER	wood chips, softwood, u=140%, at forest	1.00E+00	m3	Frischknecht et al. 1996			

Fig. 6.5 Flows for "wood chips, softwood, u=140%, at forest " and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	STDv 95%	General Comment
Wood chips softwood, at forest road (u=140%)	wood chips, mixed, u=120%, at forest	Wood chips mix, at forest road (u=120%)	bulk volume: dried matter content = 169 kg/m3 dried matter (u=0%)	wood energy	fuels	No	RER	wood chips, softwood, u=140%, at forest	7.20E-01	m3	Frischknecht et al. 1996, BFS/BUWAL 2000	1	1.35	(3,3,3,3,3,5,3)
Wood chips hardwood, at forest road (u=80%)			bulk volume: dried matter content = 239 kg/m3 dried matter (u=0%)	wood energy	fuels	No	RER	wood chips, hardwood, u=80%, at forest	2.80E-01	m3	Frischknecht et al. 1996, BFS/BUWAL 2000	1	2.12	(3,3,3,3,3,5,5)
			bulk volume: dried matter content = 188.6 kg/m3 dried matter (u=0%), lower heating value = 3294 MJ/m3. Wood is unstored/undried and not stemming from short-rotation areas for energy wood production!	wood energy	fuels	No	RER	wood chips, mixed, u=120%, at forest	1.00E+00	m3				

Fig. 6.6 Flows for "wood chips, mixed, u=120%, at forest " and its representation in the ecoinvent database

7 Sawmill processes

This chapter is mainly based on (Schniewind 1989) and (Kollmann et al. 1975).

Wood machining is the process of manufacturing wood products such as lumber, veneer and furniture parts. The objective of wood machining is to produce a desired shape and dimension with requisite accuracy and surface quality in the most economical way. Machining processing in the manufacture of wood products may be classified as follows: sawing; peeling, and slicing; planing, moulding and shaping; turning and slicing; sanding, and non-traditional machining processes such as cutting with laser beams, high-velocity liquid jets and vibrating cutters.

Sawing is the most important and most frequent cutting process. Sawing machines are classified according to the basic machine design, that is, sash gang saws (reciprocating, multiple blade frame saws), circular saws, band saws and chain saws. Saws are designated rip saws if they are designed to cut along the grain, as bucking or trim saws if they are designed to cut across the grain, as well as at a certain angle to the grain (e.g. miter saws). Sawing machines are further classified according to their use. For example, a bucking saw is used for cutting logs to length, a head rig or head saw for primary log breakdown, a resaw for resawing cants into boards, an edger for edging boards, a trimmer for cutting boards to length, and a scroll saw for general-purpose sawing in furniture plants.

Saw blades are made of cold-rolled, hardened and tempered steel. High carbon content, nickel-alloyed saw steel is used in most cases. Other saw steel alloys may contain manganese, chromium and vanadium. Circular saws typically range from 1 to 5 mm in thickness and from 100 to 1800 mm in diameter. The thickness of band-saw and gang-saw blades may range from 0.4 to 2.1 mm. Band-saw width ranges from 6 to 50 mm for the narrow band saws used in furniture manufacturing, and from 60 to 360 mm for saws used in timber manufacturing. Band-saw thickness and width depend on the saw wheel diameter and width. As a rule, the saw blade thickness should not exceed 0.1% of the wheel diameter, and the band-saw width should not be greater than wheel width plus gullet depth and an additional 5 mm. The typical gang saws are 2 mm in thickness and approximately 175 mm in width.

Saws vary considerably with regard to tooth and gullet design. The primary design considerations include tooth strength and gullet loading capacity, the function of the gullet being sawdust removal. Other important factors are tooth wear and noise generation.

The effective stiffness and stability of saws can be increased by introducing radial slots, by prestressing or tensioning, by cutting guiding systems, by on-line cooling near the cutting edge, and by heating near the centre. Radial slots in circular saws reduce compression hoop stresses at the saw periphery due to temperature gradients, introduce asymmetry into the saw-blade design and consequently reduce transverse vibration and noise. The application of various guiding systems in conjunction with the use of floating-collar or splined-arbour saws, which can float on the arbour, is a common and particularly effective method used for stability control of thin-kerf circular saws. Conventional guide systems are either contacting or non-contacting displacement-limiting devices normally placed near the cutting edge. Regardless of the operating conditions, the stress level in the saw blade should be kept constant. This can be achieved, for example, by on-line thermal tensioning and/or feed rate control.

Sawn timber is a product family including many different products (Fig. 7.1). The energy consumption and the output ratio between sawn timber and industrial residue wood depend on the specific products. The figures used in this project reflect the average production of sawn timber in Germany in 1984. There is no information available about the split in different products.

The average capacity of small sawmills lies around 5'000 m³/a (under bark), middle-sized plants reach 10'000 m³/a, and the largest Swiss company produces around 200'000 m³/a. In the European context, even the largest Swiss company is relatively small; large plants in Europe produce up to 1'000'000 m³/a ((Anonymous 2002)).

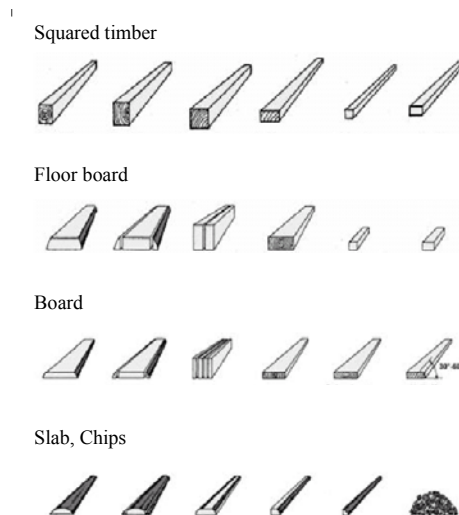


Fig. 7.1 Different products of the sawn timber family

7.1 Sawn timber in ecoinvent database

The first step of sawn timber production is the debarking of wood. If not made in the forest (see chap. 5), stationary debarking devices are used. Several principles are applied for debarking: with cutting tools (manual debarking with mechanised hand-tools, small-peelers with milling or cutter-heads and cutter-discs, gloating cutter-head peelers), by friction forces between wood (drum-peelers), by friction of tools against wood (rotor-peelers), and hydraulic debarking (in connection with friction forces, hydraulic ring-barkers). Energy consumption, wood losses, labour forces required, log diameter and capacity per hour vary considerably for these debarking processes (for details, see (Kollmann et al. 1975), p. 344f, p. 575ff).

The allocation of the burdens to the different outputs is proposed according to the economic return of the products. None of the inputs can be clearly attributed to one specific product.

The economic allocation can't respect the mass and energy balance of the products. Therefore, correcting modules are defined that can add or subtract the CO₂ uptake and the resource (wood and solar energy) consumption from nature. They are used as shown in the example in Fig. 5.2:

The background information to the processes are given in Tab. 7.1, flows are shown in:

- sawing / debarking, hardwood, at plant
 - Outputs:
 - sawn timber, hardwood, raw, plant-debarked, u=70%, at plant
 - industrial residue wood, hardwood, including bark, u=70%, at plant
- sawing / debarking, softwood, at plant
 - Outputs
 - sawn timber, softwood, raw, plant-debarked, u=70%, at plant
 - industrial residue wood, softwood, plant-debarked, u=70%, at plant
 - bark chips, softwood, u=140%, at plant
- sawing, softwood, forest-debarked, at plant
 - Outputs:
 - sawn timber, softwood, raw, forest-debarked, u=70%, at plant
 - industrial residue wood, softwood, forest-debarked, u=70%, at plant

Fig. 7.2

Fig. 7.3

Fig. 7.4

Tab. 7.1 Ecoinvent meta information for the sawing processes

Name	sawing / debarking, hardwood	sawing / debarking, softwood	sawing, softwood, forest-debarked
Location	RER	RER	RER
Infrastructure Process	0	0	0
Unit	m ³	m ³	m ³
Data Set Version	2.0	2.0	2.0
Included Processes	Includes transports from forest, sawing, and debarking at sawmill.	Includes transports from forest, sawing, and debarking at sawmill.	Includes transports of debarked round wood from forest, sawing a sawmill
Amount	1	1	1
Local Name	Einschneiden / entrinden, Laubholz	Einschneiden / entrinden, Nadelholz	Einschneiden, Nadelholz, waldentrindet
Synonyms	Beech//Buche	Spruce//Fichte	Spruce//Fichte
General Comment to reference function	The volume refers to the wood not including the bark. The multi-output process "sawing / debarking, hardwood, at plant" delivers the two coproducts "sawn timber, hardwood, raw, plant-debarked, u=70%, at plant" and "industrial residue wood, hardwood, including bark, u=70%, at plant". Allocation is based on the over-all proceeds of the process.	The volume refers to the wood not including the bark. The multi-output process "sawing / debarking, softwood, at plant" delivers the three coproducts "sawn timber, softwood, raw, plant-debarked, u=70%, at plant", "industrial residue wood, softwood, plant-debarked, u=70%, at plant" and "bark chips, softwood, u=140%, at plant". Allocation is based on the over-all proceeds of the process.	The volume refers to the wood not including the bark. The multi-output process "sawing, softwood, forest-debarked, at plant" delivers the two coproducts "sawn timber, softwood, raw, forest-debarked, u=70%, at plant" and "industrial residue wood, softwood, forest-debarked, u=70%, at plant". Allocation is based on the over-all proceeds of the process.
Start Date	1986	1986	1986
End Date	1996	1996	1996
Data Valid For Entire Period	1	1	1
Other Period Text			
Geography text	Data for Germany used for central Europe	Data for Germany used for central Europe	Data for Germany used for central Europe
Technology text	Technology mix used in Germany 1986	Technology mix used in Germany 1986	Technology mix used in Germany 1986
Representativeness [%]			
Production Volume	unknown	unknown	unknown
Sampling Procedure	Literature	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that Germany is representative for central Europe is considered in the uncertainties.	The assumption that Germany is representative for central Europe is considered in the uncertainties.	The assumption that Germany is representative for central Europe is considered in the uncertainties.

7 Sawmill processes

General Flow information						Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Roundwood, beech, at forest road (u=70%)	sawing / debarking, hardwood				under bark	wooden materials	extraction	No	RER	round wood, hardwood, under bark, u=70%, at forest road	1.59E+00	m3	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Electricity medium voltage - at grid UCTE					Debarking (at plant): 5.4 kWh, main cut: 17.2 kWh, conversion: 5.6 kWh, chopper: 2.6 kWh, suction: 3.0 kWh	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.38E+01	kWh	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Sawmill					infrastructure without land for air drying	wooden materials	extraction	Yes	RER	sawmill	4.00E-07	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
Transport					40 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	7.03E+01	tkm	Hurst 1996	1	2.01	(2,3,2,3,1,1,5)
Hardwood, allocation correction		100% to sawn timber			adds / subtracts the amount of CO2 uptake, resource consumption and embodied energy	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	-7.27E-01	m3	calculated	1	1.00	calculated correction term
Hardwood, allocation correction		100% to industrial residual wood			that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	hardwood, allocation correction, 2	7.27E-01	m3	calculated	1	1.00	calculated correction term
		Sawn timber beech rough, at sawmill (u=70%)		for furniture = 680 CHF/m3; revenue: 680 CHF; revenue process: 703 CHF; alloc. factor: 0.97	estimation after Werner 2002, based on Gautschi 2001 (personal communication)	wooden materials	extraction	No	RER	sawn timber, hardwood, raw, plant-debarked, u=70%, at plant	1.00E+00	m3				
		Industrial residual wood beech incl. bark, at sawmill (u=70%)		0.047 CHF/kg = 30 CHF/m3; revenue: 23 CHF; revenue process: 703 CHF; alloc. factor: 0.03	estimation after Werner 2002, based on Gautschi 2001	wooden materials	extraction	No	RER	industrial residue wood, hardwood, including bark, u=70%, at plant	7.81E-01	m3	calculated			
		Dust			neglegible because of high water content of wood											
		Waste heat into air				air	low population density			Heat, waste	1.22E+02	MJ	calculated	1	1.51	(1,3,5,3,1,1,13)

Fig. 7.2 Flows for "sawing / debarking, hardwood, at plant" and its representation in the ecoinvent database

7 Sawmill processes

General Flow information						Representation in ecoinvent									Uncertainty information				
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra struc ture	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Roundwood spruce, at forest road (u=70%)	→	sawing / debarking, softwood					under bark	wooden materials	extraction	No	RER	round wood, softwood, under bark, u=70% at forest road	1.71E+00	m3	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)	
Electricity medium voltage - at grid UCTE	→					Debarking (at plant): 4.5 kWh, main cut: 15.8 kWh, conversion: 5.4 kWh, chopper: 4.5 kWh, suction: 2.5 kWh in kWh/m3 sawn timber	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.27E+01	kWh	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)		
Sawmill	→					infrastructure without land for air drying	wooden materials	extraction	Yes	RER	sawmill	4.00E-07	unit	estimated	1	3.36	(4,5,2,3,4,5,9)		
Transport	→					40 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	5.97E+01	tkm	Hurst 1996	1	2.01	(2,3,2,3,1,1,5)		
Softwood, allocation correction	→				100% to sawn timber			adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-8.06E-01	m3	calculated	1	1.00	calculated correction term
Softwood, allocation correction	→				100% to bark chips				wooden materials	extraction	No	RER	softwood, allocation correction, 2	1.71E-01	m3	calculated	1	1.00	calculated correction term
Softwood, allocation correction	→				100% to industrial residual wood				wooden materials	extraction	No	RER	softwood, allocation correction, 3	6.35E-01	m3	calculated	1	1.00	calculated correction term
	→			Sawn timber spruce rough, at sawmill (debarked at plant) (u=70%)	400 CHF/m3 (weighted average for construction wood); revenue: 400 CHF; revenue process: 419 CHF; alloc. factor: 0.96	after BUWAL/BFS, Tab. 7.2	Module is not used further. For calculations within Ecoinvent 2000 the process "Sawn timber spruce rough, at sawmill (debarked in forest) (u=70%) is used instead.	wooden materials	extraction	No	RER	sawn timber, softwood, raw, plant-debarked, u=70%, at plant	1.00E+00	m3	Ressel 1986				
	→			Bark chips spruce, at sawmill (u=140%)	10 CHF/m3 (estimated); revenue: 2 CHF; revenue process: 419 CHF; alloc. factor: 0		Module is not used further. For calculations within Ecoinvent 2000 the process "Bark chips spruce, at forest road (u=140%) is used instead.	wooden materials	extraction	No	RER	bark chips, softwood, u=140%, at plant	1.71E-01	m3	Frühwald et al. 1996				
	→			Industrial residual wood spruce, at sawmill (debarked at plant) (u=70%)	0.054 CHF/kg (u=0%) = 24 CHF/m3; revenue: 17 CHF; revenue process: 419 CHF; alloc. factor: 0.04	estimation after Werner 2002, based on Gautschi 2001	Module is not used further. For calculations within Ecoinvent 2000 the process "Industrial residual wood timber spruce, at sawmill (debarked in forest) (u=70%) is used instead.	wooden materials	extraction	No	RER	industrial residue wood, softwood, plant-debarked, u=70%, at plant	7.10E-01	m3	Ressel 1986				
	→			Dust			neglegible because of high water content of wood												
	→			Waste heat into air					air	low population density			Heat, waste	1.18E+02	MJ	calculated	1	1.51	(1,3,5,3,1,1,13)

Fig. 7.3 Flows for "sawing / debarking, softwood, at plant" and its representation in the ecoinvent database

7 Sawmill processes

General Flow information						Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Roundwood spruce debarked, at forest road (u=70%)	sawing, softwood, forest-debarked					wooden materials	extraction	No	RER	round wood, softwood, debarked, u=70% at forest road	1.71E+00	m3	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Electricity medium voltage - at grid UCTE					Debarking in forest , main cut: 15.8 kWh, conversion: 5.4 kWh, chopper: 4.5 kWh, suction: 2.5 kWh; in kWh/m3 sawn timber	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.27E+01	kWh	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Sawmill					infrastructure without land for air drying	wooden materials	extraction	Yes	RER	sawmill	4.00E-07	unit		1	3.36	(4,5,2,3,4,5,9)
Transport					40 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	5.23E+01	tkm	Hurst 1996	1	2.01	(2,3,2,3,1,1,5)
Softwood, allocation correction		100% to sawn timber			adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-6.42E-01	m3	calculated	1	1.00	calculated correction term
Softwood, allocation correction		100% to industrial residual wood			that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	softwood, allocation correction, 2	6.42E-01	m3	calculated	1	1.00	calculated correction term
		Sawn timber spruce rough, at sawmill (debarked in forest) (u=70%)	400 CHF/m3 (weighted average for construction wood); revenue: 400 CHF; revenue process: 417 CHF; alloc. factor: 0.96	after BUWAL/BFS, Tab. 7.2		wooden materials	extraction	No	RER	sawn timber, softwood, raw, forest-debarked, u=70%, at plant	1.00E+00	m3	Ressel 1986			
		Industrial residual wood spruce, at sawmill (debarked in forest) (u=70%)	0.054 CHF/kg (u=0%) = 24 CHF/m3; revenue: 17 CHF; revenue process: 417 CHF; alloc. factor: 0.04	estimation after Werner 2002, based on Gautschi 2001		wooden materials	extraction	No	RER	industrial residue wood, softwood, forest-debarked, u=70%, at plant	7.10E-01	m3	Ressel 1986			
		Dust			neglegible because of high water content of wood											
		Waste heat into air				air	low population density			Heat, waste	1.18E+02	MJ		1	1.51	(1,3,5,3,1,1,13)

Fig. 7.4 Flows for "sawing, softwood, forest-debarked, at plant" and its representation in the ecoinvent database

7.1.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

The flows for sawing can vary considerably depending on many variables. Thus these figures are rather uncertain. It is, however, not possible to give general data of higher certainty for processes of such diversity. Data quality is not sufficient to allow for comparing specific products (e.g. products of different thickness).

8 Timber drying

This chapter is mainly based on (Simpson 1999).

In the living tree, wood contains large quantities of water. The moisture remaining in the wood after drying tends to come to equilibrium with the relative humidity of the surrounding air. Correct drying, handling, and storage of wood will minimise moisture content changes that might occur after drying when the wood is in service and such changes are undesirable. If moisture content is controlled within reasonable limits by such methods, major problems from dimensional changes can usually be avoided.

Wood in service is virtually always undergoing at least slight changes in moisture content. Changes in response to daily humidity variation are small and usually without consequences. Changes that occur as a result of seasonal variation, although gradual, tend to be of more concern. Protective coatings can retard dimensional changes in wood but do not prevent them. In general, no significant dimensional changes will occur if wood is fabricated or installed at a moisture content corresponding to the average atmospheric conditions to which it will be exposed. When incompletely dried material is used in construction, some minor dimensional changes can be tolerated if the proper design is used.

The key to successful and efficient drying is controlling the drying process. Timely application of optimum or at least adequate temperature, relative humidity, and air circulation conditions is critical. Uncontrolled drying leads to drying defects that can adversely affect the serviceability and economics of the product. The usual strategy is to dry as fast as the particular species, thickness, and end-product requirements allow without damaging the wood. Slower drying can be uneconomical as well as introduce the risk of stain.

Softwood timber (and hardwood timber, although not commonly used) intended for framing in construction is usually targeted for drying to an average moisture content of 15% (w/w), not exceeding 19%. Softwood timber for many other uses is dried to moisture content as low as 7% to 9% for furniture, cabinets, and millwork. Hardwood timber for furniture, cabinets, and millwork is usually dried to 6% to 8% moisture content.

Timber drying is usually accomplished by some combination of air drying, accelerated air drying or pre-drying, and kiln drying. Wood species, timber thickness, economics, and end use are often the main factors determining the details of the drying process.

Below 30% moisture content, species-specific volume changes of wood are associated with the drying process. A shrinkage ratio of 9% (v/v) for spruce and 12% (v/v) for beech are assumed in (Frühwald et al. 1996).

Natural drying

The main purpose of natural (or air) drying of timber is to evaporate as much of the water as possible before transport, end use or transfer to a dry kiln. Air drying occurs during storage at the margin of forest road or at the sawmill plant.

Air drying is usually extended until wood moisture contents decrease to 20% to 25%, then timber is transferred to a dry kiln if final drying to lower moisture content is required. Sometimes, depending on individual mill's scheduling, air drying may be cut short at higher moisture content before the wood is sent to the dry kiln. Air drying saves energy costs and reduces required dry kiln capacity.

Limitations of air drying are generally associated with uncontrolled drying. The drying rate is very slow during the cold winter months. During other seasons, hot, dry winds may increase degradation and volume losses as a result of severe surface checking and end splitting. Warm, humid periods with little air movement may promote the growth of fungal stains, as well as aggravate chemical stains. Another limitation of air drying is the high cost of stocking a large inventory of high value timber for extended periods. Air drying time to 20% to 25% moisture content varies widely, depending on species, thickness, location, and the time of year the timber is stacked. As examples of extreme situations, for 25-mm thick timber 15 to 30 days are necessary for some of the low density species, such as pine,

spruce, red alder, and soft maple, stacked in favourable times of the year, whereas 200 to 300 days are required for slow drying species, such as sinker hemlock and pine, oak, and birch, in northern locations and stacked at unfavourable times of the year.

Accelerated air drying and pre-drying

The limitations of air drying have led to increased use of technology that reduces drying time and introduces some control into drying from green to 20% to 25% moisture content. Accelerated air drying involves the use of fans to force air through timber piles in a shed. This protects the timber from the elements and improves air circulation compared with natural air drying. Some additional heat is sometimes used to reduce relative humidity and slightly increase temperature. Pre-dryers take this acceleration and control a step further by controlling temperature and relative humidity and by providing forced air circulation in a completely enclosed compartment. Typical conditions in a pre-dryer are 27 C° to 38C° and 65% to 85% relative humidity.

Technical drying

Technical drying is required for wood to be used in most products, especially for indoor uses. With technical drying, warping, splitting, checking, and other harmful effects resulting from uncontrolled drying are largely eliminated. Dried timber has many advantages over green timber:

- Reduced weight thus reduced shipping and handling costs.
- Proper drying confines shrinking and swelling of wood products in its use to manageable margins under all but extreme conditions.
- Most of wood's mechanical, it's electrical and thermal insulating properties are increased.
- Properly dried timber can be cut to precise dimensions and machined more easily and efficiently.
- Dried wood can be fitted and fastened together more securely with nails, screws, bolts, and adhesives.
- Paint, varnish, and other finishes are more effectively applied and maintained. (Wood must be relatively dry before it can be glued or treated with decay-preventing and fire-retardant chemicals.)

Kiln drying

In kiln drying, higher temperatures and faster air circulation are used to increase the drying rate considerably. Specific kiln schedules have been developed to control temperature and relative humidity in accordance with the moisture content and internal stress within the wood, thus minimising defects caused by shrinkage. Most dry kilns are thermally insulated compartments designed for a batch process. Temperature and relative humidity are kept as uniform as possible throughout the kiln and can be controlled over a wide range. Temperature and relative humidity are change as the wood dries-out based on a schedule that takes into account the moisture content and/or the drying rate of the timber. All dry kilns use forced-air circulation, with air moving through the timber perpendicular to the length of the timber and parallel to the spacers (stickers) that separate each layer of timber in a stack.

In general, three types of kilns are commonly used. One is the wagon-loaded type, where timber is stacked on kiln wagons that are rolled in and out of the kiln on tracks. The majority of softwood timber is dried in this type of kiln. Another major type of kiln is the package-loaded kiln, where individual stacks of timber are fork-lifted into place in the kiln. This type of kiln is commonly used for drying hardwood timber. These kilns are most commonly heated with steam. Another common type of kiln, usually package loaded, is the dehumidification kiln. Instead of venting humid air, water is removed by condensation on cold dehumidifier coils.

Technical drying can cause drying defects. Most drying defects or problems that develop in wood products during drying can be classified as fracture or distortion, warp, or as chemical or fungal discoloration. Defects in any one of theses categories are caused by an interaction of wood properties with processing factors. Wood shrinkage is mainly responsible for wood ruptures and distortion of shape. Cell structure and chemical composition of wood contribute to defects associated with uneven

moisture content, undesirable colour, and undesirable surface texture. Drying temperature is the most important processing factor because it can be responsible for defects in each category.

8.1 Timber drying in ecoinvent database

8.1.1 Natural drying

The space needed for natural drying residual wood in forests is not considered in the respective datasets because it is already accounted for as land occupied by forests.

The space needed for natural drying of wood at the sawmill is calculated according to the estimates given in Tab. 8.1.

Tab. 8.1 Calculation of land use for natural drying of wood

	Hardwood	Softwood	Source	refers to / comment
Height of staples [m]	5.00E+00	5.00E+00	estimate	
Height usage factor [-]	6.00E-01	7.00E-01	estimate	Fraction of the height that is wood (rest is air)
Area usage fraction [-]	7.00E-01	7.00E-01	estimate	Fraction of the area that is occupied by the staples (rest is empty)
Time for drying [a]	2.00E+00	2.00E+00	estimate	
Time for which land is used as drying place for wood [a]	1.00E+02	1.00E+02	estimate	
Land use for air drying [m ² / m ³]	4.76E-01	4.08E-01	calculated	wet wood volume
Shrinkage [-]	1.20E-01	9.00E-02	Frühwald 2000	wet wood volume
Land use for air drying [m ² / m ³]	4.25E-01	3.74E-01	calculated	dry wood volume
Land transformation for air drying [m ² / m ³]	8.50E-03	7.49E-03	calculated	dry wood volume
Land occupation for air drying [m ² a / m ³]	8.50E-01	7.49E-01	calculated	dry wood volume

The background information to the processes are given in Tab. 8.2, flows are shown in:

- residual wood, hardwood, under bark, air dried, u=20%, at forest road Fig. 8.1
- residual wood, softwood, under bark, air dried, u=20%, at forest road Fig. 8.2
- sawn timber, hardwood, raw, air dried, u=20%, at plant Fig. 8.3
- sawn timber, softwood, raw, air dried, u=20%, at plant Fig. 8.4
- industrial residue wood, hardwood, including bark, air dried, u=20%, at plant Fig. 8.5
- industrial residue wood, softwood, forest-debarked, air dried, u=20%, at plant Fig. 8.6

8.1.2 Data quality

The land use is estimated based on physical relations, thus relevant figures may be considered of good overall quality.

Tab. 8.2 Ecoinvent meta information for the air drying processes

Name	residual wood, hardwood, under bark, air dried, u=20%, at forest road	residual wood, softwood, under bark, air dried, u=20%, at forest road
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Natural drying in the forest. Land use of forest is already considered with the "hardwood, standing, in forest" process. Thus it is not again included here. Air emissions released from the wood are not accounted for because the same emissions would occur if the wood would not be used.	Natural drying in the forest. Land use of forest is already considered with the "softwood, standing, in forest" process. Thus it is not again included here. Air emissions released from the wood are not accounted for because the same emissions would occur if the wood would not be used.
Amount	1	1
Local Name	Restholz, Laubholz, Festkubikmeter, luftgetrocknet, u=20%, ab Waldstrasse	Restholz, Nadelholz, Festkubikmeter, luftgetrocknet, u=20%, ab Waldstrasse
Synonyms	Beech//Buche	Fichte//Spruce
General Comment to reference function	The volume refers to the wood not including the bark.	The volume refers to the wood not including the bark.
Start Date	1996	1996
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	not specific	not specific
Technology text	Naturally dried on staples for 2 years. Shrinkage for beech	Naturally dried on staples for 2 years. Shrinkage for spruce
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Technology	see Technology
Uncertainty Adjustments	The assumption that the volume change of beech is the same as for other hardwood is considered in the uncertainties.	The assumption that the volume change of spruce is the same as for other softwood is considered in the uncertainties.

Tab. 8.2 Ecoinvent meta information for the air drying processes (cont.)

Name	sawn timber, hardwood, raw, air dried, u=20%, at plant	sawn timber, softwood, raw, air dried, u=20%, at plant
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Natural drying at the sawmill. Land use is accounted for. Air emissions released from the wood are not accounted for because the same emissions would occur if the wood would not be used.	Natural drying at the sawmill. Land use is accounted for. Air emissions released from the wood are not accounted for because the same emissions would occur if the wood would not be used.
Amount	1	1
Local Name	Schnittholz, Laubholz, sägerau, luftgetrocknet, u=20%, ab Werk	Schnittholz, Nadelholz, sägerau, luftgetrocknet, u=20%, ab Werk
Synonyms	Beech//Buche	Fichte//Spruce
General Comment to reference function	The volume refers to the wood not including the bark.	The volume refers to the wood not including the bark.
Start Date	1996	1996
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	not specific	not specific
Technology text	Naturally dried on 5 m high staples for 2 years. Shrinkage for beech	Naturally dried on 5 m high staples for 2 years. Shrinkage for spruce
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Technology	see Technology
Uncertainty Adjustments	The assumption that the volume change of beech is the same as for other hardwood is considered in the uncertainties.	The assumption that the volume change of spruce is the same as for other softwood is considered in the uncertainties.

Tab. 8.2 Ecoinvent meta information for the air drying processes (cont.)

Name	industrial residue wood, hardwood, including bark, air dried, u=20%, at plant	industrial residue wood, softwood, forest-debarked, air dried, u=20%, at plant
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Natural drying at the sawmill. Land use is accounted for. Air emissions released from the wood are not accounted for because the same emissions would occur if the wood would not be used.	Natural drying at the sawmill. Land use is accounted for. Air emissions released from the wood are not accounted for because the same emissions would occur if the wood would not be used.
Amount	1	1
Local Name	Industrierestholz, Laubholz, inklusive Rinde, luftgetrocknet, u=20%, ab Werk	Industrierestholz, Nadelholz, waldentrindet, luftgetrocknet, u=20%, ab Werk
Synonyms	Beech//Buche	Fichte//Spruce
General Comment to reference function	The volume refers to the wood not including the bark.	The volume refers to the wood not including the bark.
Start Date	1996	1996
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	not specific	not specific
Technology text	Naturally dried on 5 m high staples for 2 years. Shrinkage for beech	Naturally dried on 5 m high staples for 2 years. Shrinkage for spruce
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Technology	see Technology
Uncertainty Adjustments	The assumption that the volume change of beech is the same as for other hardwood is considered in the uncertainties.	The assumption that the volume change of spruce is the same as for other softwood is considered in the uncertainties.

General Flow information					Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Residual wood beech, at forest road (u=80%)	→	residual wood, hardwood, under bark, air dried, u=20%, at forest road			Shrinkage ratio: 12 Vol%	wooden materials	extraction	No	RER	residual wood, hardwood, under bark, u=80%, at forest road	1.14E+00	m3	Frühwald et al. 1996	1	1.21	(1,1,1,1,3,1,3)
Shrinkage correction	→				to maintain CO2 and biomass balance	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	-1.52E-01	m3	calculated	1	1.00	calculated
Land transformation / occupation	→				included in forest											
				Residual wood beech, naturally dried, at forest road (u=20%)		wooden materials	extraction	No	RER	residual wood, hardwood, under bark, air dried, u=20%, at forest road	1.00E+00	m3				

Fig. 8.1 Flows for "residual wood, hardwood, under bark, air dried, u=20%, at forest road" and its representation in the ecoinvent database

General Flow information					Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Residual wood spruce, at forest road (u=140%)	→	residual wood, softwood, under bark, air dried, u=20%, at forest road			Shrinkage ratio: 9 Vol%	wooden materials	extraction	No	RER	residual wood, softwood, under bark, u=140%, at forest road	1.10E+00	m3	Frühwald et al. 1996	1	1.21	(1,1,1,1,3,1,3)
Shrinkage correction	→				to maintain CO2 and biomass balance	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-1.09E-01	m3	calculated	1	1.00	calculated
Land transformation / occupation	→				included in forest											
				Residual wood spruce, naturally dried, at forest road (u=20%)		wooden materials	extraction	No	RER	residual wood, softwood, under bark, air dried, u=20%, at forest road	1.00E+00	m3				

Fig. 8.2 Flows for "residual wood, softwood, under bark, air dried, u=20%, at forest road" and its representation in the ecoinvent database

8 Timber drying

General Flow information					Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Sawn timber beech rough, at sawmill (u=70%)	→	sawn timber, hardwood, raw, air dried, u=20%, at plant			Shrinkage ratio: 12 Vol%	wooden materials	extraction	No	RER	sawn timber, hardwood, raw, plant-debarked, u=70%, at plant	1.14E+00	m3	Frühwald et al. 1996	1	1.21	(1,1,1,1,3,1,3)
Shrinkage correction	→				to maintain CO2 and biomass balance	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	-1.36E-01	m3	calculated	1	1.00	calculated
Land transformation from former state	→				Assumption: dired in staples of 5 m hight for 2 years. 60% of hight is wood, 70% of area is staples. Area is used for 100 a for wood drying	resource	land			Transformation, from unknown	8.50E-03	m2	estimated	1	2.19	(4,5,1,3,3,5,8)
Land transformation to air drying space	→					resource	land			Transformation, to industrial area, vegetation	8.50E-03	m2	estimated	1	2.19	(4,5,1,3,3,5,8)
Land occupation as air drying space	→					resource	land			Occupation, industrial area, vegetation	8.50E-01	m2a	estimated	1	1.73	(4,5,1,3,3,5,7)
				Sawn timber beech rough, naturally dried, at sawmill (u=20%)		wooden materials	extraction	No	RER	sawn timber, hardwood, raw, air dried, u=20%, at plant	1.00E+00	m3				

Fig. 8.3 Flows for "sawn timber, hardwood, raw, air dried, u=20%, at plant" and its representation in the ecoinvent database

General Flow information					Representation in ecoinvent								Uncertainty information			
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Sawn timber spruce rough, at sawmill (debarked in forest) (u=70%)	→	sawn timber, softwood, raw, air dried, u=20%, at plant			Shrinkage ratio: 9 Vol%	wooden materials	extraction	No	RER	sawn timber, softwood, raw, forest-debarked, u=70%, at plant	1.10E+00	m3	Frühwald et al. 1996	1	1.21	(1,1,1,1,3,1,3)
Shrinkage correction	→				to maintain CO2 and biomass balance	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-9.90E-02	m3	calculated	1	1.00	
Land transformation from former state	→				Assumption: dired in staples of 5 m hight for 2 years. 70% of hight is wood, 70% of area is staples. Area is used for 100 a for wood drying	resource	land			Transformation, from unknown	7.49E-03	m2	estimated	1	2.19	(4,5,1,3,3,5,8)
Land transformation to air drying space	→					resource	land			Transformation, to industrial area, vegetation	7.49E-03	m2	estimated	1	2.19	(4,5,1,3,3,5,8)
Land occupation as air drying space	→					resource	land			Occupation, industrial area, vegetation	7.49E-01	m2a	estimated	1	1.73	(4,5,1,3,3,5,7)
				Sawn timber spruce rough, naturally dried, at sawmill (u=20%)			wooden materials	extraction	No	RER	sawn timber, softwood, raw, air dried, u=20%, at plant	1.00E+00	m3			

Fig. 8.4 Flows for "sawn timber, softwood, raw, air dried, u=20%, at plant" and its representation in the ecoinvent database

8 Timber drying

General Flow information					Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
industrial residual wood beech incl. bark, at sawmill (u=70%)	→	industrial residue wood, hardwood, including bark, air dried, u=20%, at plant			Shrinkage ratio: 12 Vol%	wooden materials	extraction	No	RER	industrial residue wood, hardwood, including bark, u=70%, at plant	1.14E+00	m3	Frühwald et al. 1996	1	1.21	(1,1,1,1,3,1,3)
Shrinkage correction	→				to maintain CO2 and biomass balance	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	-1.36E-01	m3	calculated	1	1.00	
Land transformation from former state	→				assumed to be the same as for sawn timber	resource	land			Transformation, from unknown	8.50E-03	m2	estimated	1	2.37	(4,5,1,3,4,5,8)
Land transformation to air drying space	→					resource	land			Transformation, to industrial area, vegetation	8.50E-03	m2	estimated	1	2.37	(4,5,1,3,4,5,8)
Land occupation as air drying space	→					resource	land			Occupation, industrial area, vegetation	8.50E-01	m2a	estimated	1	1.92	(4,5,1,3,4,5,7)
				Industrial residual wood beech incl. bark, naturally dried, at sawmill (u=20%)		wooden materials	extraction	No	RER	industrial residue wood, hardwood, including bark, air dried, u=20%, at plant	1.00E+00	m3				

Fig. 8.5 Flows for "industrial residue wood, hardwood, including bark, air dried, u=20%, at plant" and its representation in the ecoinvent database

General Flow information					Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Industrial residual wood spruce, at sawmill (u=70%)	→	industrial residue wood, softwood, forest-debarked, air dried, u=20%, at plant			Shrinkage ratio: 9 Vol%	wooden materials	extraction	No	RER	industrial residue wood, softwood, forest-debarked, u=70%, at plant	1.10E+00	m3	Frühwald et al. 1996	1	1.21	(1,1,1,1,3,1,3)
Shrinkage correction	→				to maintain CO2 and biomass balance	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-9.90E-02	m3	calculated	1	1.00	
Land transformation from former state	→				assumed to be the same as for sawn timber	resource	land			Transformation, from unknown	7.49E-03	m2	estimated	1	2.37	(4,5,1,3,4,5,8)
Land transformation to air drying space	→					resource	land			Transformation, to industrial area, vegetation	7.49E-03	m2	estimated	1	2.37	(4,5,1,3,4,5,8)
Land occupation as air drying space	→					resource	land			Occupation, industrial area, vegetation	7.49E-01	m2a	estimated	1	1.92	(4,5,1,3,4,5,7)
				Industrial residual wood spruce, naturally dried, at sawmill (u=20%)		wooden materials	extraction	No	RER	industrial residue wood, softwood, forest-debarked, air dried, u=20%, at plant	1.00E+00	m3				

Fig. 8.6 Flows for "industrial residue wood, softwood, forest-debarked, air dried, u=20%, at plant" and its representation in the ecoinvent database

8.1.3 Kiln drying

Drying time and energy consumption depend on the shape and dimension of the timber to be dried. There is no information available about the mix of shapes and dimensions that is represented by these data.

The background information to the processes are given in Tab. 8.3, flows are shown in:

- | | | |
|---|---|-----------|
| - | sawn timber, hardwood, raw, air / kiln dried, u=10%, at plant | Fig. 8.7 |
| - | sawn timber, hardwood, raw, kiln dried, u=10%, at plant | Fig. 8.8 |
| - | sawn timber, softwood, raw, kiln dried, u=20%, at plant | Fig. 8.9 |
| - | sawn timber, softwood, raw, kiln dried, u=10%, at plant | Fig. 8.10 |

8.1.4 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Data for the total energy consumption of the processes are good despite their age. Little is known about the fuels commonly used to supply this energy today. The overall data quality can be considered medium. Data quality is not sufficient to allow for comparing specific products (e.g. boards to beams etc.).

Tab. 8.3 Ecoinvent meta information for the kiln drying processes

Name	sawn timber, hardwood, raw, air / kiln dried, u=10%, at plant	sawn timber, hardwood, raw, kiln dried, u=10%, at plant
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Includes kiln drying of air dried wood (u=20%) down to u=10%, and shrinkage. Air emissions released from the wood are not accounted for because the same emissions would occur if the wood would not be used. Kiln is assumed to be located on the sawmill site. No transports are considered.	Includes kiln drying of wet wood (u=70%) down to u=10%, and shrinkage. Air emissions released from the wood are not accounted for because the same emissions would occur if the wood would not be used. Kiln is assumed to be located on the sawmill site. No transports are considered.
Amount	1	1
Local Name	Schnittholz, Laubholz, sägerau, luft-/kammergetrocknet, u=10%, ab Werk	Schnittholz, Laubholz, sägerau, kammergetrocknet, u=10%, ab Werk
Synonyms	Buche//Beech	Buche//Beech
General Comment to reference function	Drying time (in kiln): 10 days	Drying time: 30 days
Start Date	1986	1986
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Germany used for central Europe	Data for Germany used for central Europe
Technology text	Technology mix used in Germany (1986)	Technology mix used in Germany (1986)
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that the German technology mix is representative for central Europe is considered in the uncertainties.	The assumption that the German technology mix is representative for central Europe is considered in the uncertainties.

Tab. 8.3 Ecoinvent meta information for the kiln drying processes (cont.)

Name	sawn timber, softwood, raw, kiln dried, u=20%, at plant	sawn timber, softwood, raw, kiln dried, u=10%, at plant
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Includes kiln drying of wet wood (u=70%) down to u=20%, and shrinkage. Air emissions released from the wood are not accounted for because the same emissions would occur if the wood would not be used. Kiln is assumed to be located on the sawmill site. No transports are considered.	Includes kiln drying of wet wood (u=70%) down to u=10%, and shrinkage. Air emissions released from the wood are not accounted for because the same emissions would occur if the wood would not be used. Kiln is assumed to be located on the sawmill site. No transports are considered.
Amount	1	1
Local Name	Schnittholz, Nadelholz, sägerau, kammergetrocknet, u=20%, ab Werk	Schnittholz, Nadelholz, sägerau, kammergetrocknet, u=10%, ab Werk
Synonyms	Fichte//Spruce	Fichte//Spruce
General Comment to reference function	Drying time: 3 days	Drying time: 9 days
Start Date	1986	1986
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Germany used for central Europe	Data for Germany used for central Europe
Technology text	Technology mix used in Germany (1986)	Technology mix used in Germany (1986)
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that the German technology mix is representative for central Europe is considered in the uncertainties.	The assumption that the German technology mix is representative for central Europe is considered in the uncertainties.

General Flow information					Representation in ecoinvent									Uncertainty information			
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Sawn timber beech rough, naturally dried, at sawmill (u=20%)	→	sawn timber, hardwood, raw, air / kiln dried, u=10%, at plant				wooden materials	extraction	No	RER	sawn timber, hardwood, raw, air dried, u=20%, at plant	1.00E+00	m3	calculated	1	1.51	(1,3,5,3,1,1,4)	
Heat from wood furnace, fueled with industrial residual wood	→				Estimation assuming that the energy used for drying wood from u=20% to u= 10% is about a third of the total enegy consumption	wood energy	heating systems	No	CH	wood chips, from industry, hardwood, burned in furnace 300kW	4.32E+02	MJ	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)	
Electricity	→					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.10E+01	kWh	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)	
Kiln	→				including hall and land use	wooden materials	extraction	Yes	RER	technical wood drying, infrastructure	2.03E-05	unit	estimated	1	3.36	(4,5,2,3,4,5,9)	
			→	Sawn timber beech rough, naturally/technically dried, at sawmill (u=10%)		wooden materials	extraction	No	RER	sawn timber, hardwood, raw, air / kiln dried, u=10%, at plant	1.00E+00	m3					
				Emissions from wood to air	not balanced as these emissions would also result if the wood would be left in the forest .												
			→	Waste heat into air		air	unspecified			Heat, waste	3.96E+01	MJ	calculated	1	1.51	(1,3,5,3,1,1,13)	

Fig. 8.7 Flows for "sawn timber, hardwood, raw, air / kiln dried, u=10%, at plant" and its representation in the ecoinvent database

8 Timber drying

General Flow information					Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Sawn timber beech rough, at sawmill (u=70%)	→	sawn timber, hardwood, raw, kiln dried, u=10%, at plant			Shrinkage ratio: 12%	wooden materials	extraction	No	RER	sawn timber, hardwood, raw, plant-debarked, u=70%, at plant	1.14E+00	m3	Shrinkage ratio: Frühwald et al. 1996	1	1.51	(1,3,5,3,1,1,4)
Shrinkage correction	→				to maintain CO2 and biomass balance	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	-1.36E-01	m3	calculated	1	1.00	
Heat from wood furnace, fueled with industrial residual wood	→					wood energy	heating systems	No	CH	wood chips, from industry, hardwood, burned in furnace 300kW	1.30E+03	MJ	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Electricity	→					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.30E+01	kWh	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Kiln	→				including hall and land use	wooden materials	extraction	Yes	RER	technical wood drying, infrastructure	6.09E-05	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
				Sawn timber beech rough, technically dried, at sawmill (u=10%)		wooden materials	extraction	No	RER	sawn timber, hardwood, raw, kiln dried, u=10%, at plant	1.00E+00	m3				
				Emissions from wood to air	not balanced as these emissions would also result if the wood would be left in the forest.											
			Waste heat into air		air	unspecified				Heat, waste	1.19E+02	MJ	calculated	1	1.51	(1,3,5,3,1,1,13)

Fig. 8.8 Flows for "sawn timber, hardwood, raw, kiln dried, u=10%, at plant" and its representation in the ecoinvent database

8 Timber drying

General Flow information					Representation in ecoinvent									Uncertainty information			
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Sawn timber spruce rough, at sawmill (u=70%)	→	sawn timber, softwood, raw, kiln dried, u=20%, at plant			Shrinkage ratio: 9%	wooden materials	extraction	No	RER	sawn timber, softwood, raw, forest-debarked, u=70%, at plant	1.10E+00	m3	Shrinkage ratio: Frühwald et al. 1996	1	1.51	(1,3,5,3,1,1,3)	
Shrinkage correction	→				to maintain CO2 and biomass balance	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-9.90E-02	m3	calculated	1	1.00		
Heat from wood furnace, fueled with industrial residual wood	→				2/3 of energy for drying to 10%	wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 300kW	6.38E+02	MJ	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)	
Electricity	→				2/3 of energy for drying to 10%	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.67E+01	kWh	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)	
Kiln	→				including hall and land use	wooden materials	extraction	Yes	RER	technical wood drying, infrastructure	6.09E-06	unit	estimated	1	3.36	(4,5,2,3,4,5,9)	
				Sawn timber spruce rough, technically dried, at sawmill (u=20%)			wooden materials	extraction	No	RER	sawn timber, softwood, raw, kiln dried, u=20%, at plant	1.00E+00	m3				
				Emissions from wood to air		not balanced as these emissions would also result if the wood would be left in the forest .											
				Waste heat into air			air	unspecified			Heat, waste	5.99E+01	MJ	calculated	1	1.51	(1,3,5,3,1,1,13)

Fig. 8.9 Flows for "sawn timber, softwood, raw, kiln dried, u=20%, at plant" and its representation in the ecoinvent database

8 Timber drying

General Flow information					Representation in ecoinvent								Uncertainty information			
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Sawn timber spruce rough, at sawmill (u=70%)	→	sawn timber, softwood, raw, kiln dried, u=10%, at plant			Shrinkage ratio: 9%	wooden materials	extraction	No	RER	sawn timber, softwood, raw, forest-debarked, u=70%, at plant	1.10E+00	m3	Shrinkage ratio: Frühwald et al. 1996	1	1.51	(1,3,5,3,1,1,3)
Shrinkage correction	→				to maintain CO2 and biomass balance	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-9.90E-02	m3	calculated	1	1.00	
Heat from wood furnace, fueled with industrial residual wood	→					wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 300kW	9.59E+02	MJ	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Electricity	→					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.50E+01	kWh	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Kiln	→				including hall and land use	wooden materials	extraction	Yes	RER	technical wood drying, infrastructure	1.83E-05	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
				Sawn timber spruce rough, technically dried, at sawmill (u=10%)		wooden materials	extraction	No	RER	sawn timber, softwood, raw, kiln dried, u=10%, at plant	1.00E+00	m3				
				Emissions from wood to air	not balanced as these emissions would also result if the wood would be left in the forest.											
				Waste heat into air		air	unspecified				Heat, waste	9.00E+01	MJ	calculated	1	1.51

Fig. 8.10 Flows for "sawn timber, softwood, raw, kiln dried, u=10%, at plant" and its representation in the ecoinvent database

9 Planing mill

This chapter is mainly based on (Schniewind 1989).

Planing refers to the peripheral milling of wood. Its purpose is to smooth one or more surfaces of the work piece and at the same time bring the work piece to some predetermined dimension. The machinery used for planing operations include: a) surfacers designed to smooth one or two sides of the work piece and reduce it to a predetermined thickness, and b) planers and matchers designed as double surfacers equipped with two opposed profile side heads that can simultaneously machine the two edges to the desired pattern or profile.

In planing, it is of primary importance to adjust the operation conditions and knife geometry so that machining defects are maintained at a satisfactorily low level. The most commonly encountered defects are torn or fuzzy grain, raised and loosened grain, and chip marks. These defects are caused by improper cutting angles, chip thickness which is too large, dull knives, low density species and often the presence of reaction wood.

Usually only dried wood is planed.

9.1 Planing in ecoinvent database

The allocation of the burdens to the different outputs is proposed according to the economic return of the products. None of the inputs can be clearly attributed to one specific product.

The economic allocation can't respect the mass and energy balance of the products. Therefore, correcting modules are defined that can add or subtract the CO₂ uptake and the resource (wood and solar energy) consumption from nature. They are used as shown in the example in Fig. 5.2:

The background information to the processes are given in Tab. 9.1, flows are shown in:

- planing, hardwood, air / kiln dried, u=10%
Outputs:
 - sawn timber, hardwood, planed, air / kiln dried, u=10%, at plant
 - industrial residue wood, from planing, hardwood, air/kiln dried, u=10%, at plant Fig. 9.1
- planing, hardwood, kiln dried, u=10%
Outputs:
 - sawn timber, hardwood, planed, kiln dried, u=10%, at plant
 - industrial residue wood, from planing, hardwood, kiln dried, u=10%, at plant Fig. 9.2
- planing, softwood, air dried, u=20%
Outputs:
 - sawn timber, softwood, planed, air dried, at plant
 - industrial residue wood, from planing, softwood, air dried, u=20%, at plant Fig. 9.3
- planing, softwood, kiln dried, u=10%
Outputs:
 - sawn timber, softwood, planed, kiln dried, at plant
 - industrial residue wood, from planing, softwood, kiln dried, u=10%, at plant Fig. 9.4

9.1.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Maintenance of the equipment is neglected for a lack of data. It would, however, not play an important role. Energy consumption data are of good quality.

Tab. 9.1 Ecoinvent meta information for the planing processes

Name	planing, hardwood, air / kiln dried, u=10%	planing, hardwood, kiln dried, u=10%
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Includes planing process. planing mill is assumed to be located on the sawmill site. No transports are considered. Dust emissions are neglected for a lack of data.	Includes planing process. planing mill is assumed to be located on the sawmill site. No transports are considered. Dust emissions are neglected for a lack of data.
Amount	1	1
Local Name	Hobeln, Laubholz, luft-/kammergetrocknet, u=10%	Hobeln, Laubholz, kammergetrocknet, u=10%
Synonyms	Buche//Beech	Buche//Beech
General Comment to reference function	The multi-output process "planing, hardwood, air / kiln dried, u=10%" delivers the two coproducts "sawn timber, hardwood, planed, air / kiln dried, u=10%, at plant" and "industrial residue wood, from planing, hard, air/kiln dried, u=10%, at plant". Allocation is based on the over-all proceeds of the process.	The multi-output process "planing, hardwood, kiln dried, u=10%" delivers the two coproducts "sawn timber, hardwood, planed, kiln dried, u=10%, at plant" and "industrial residue wood, from planing, hardwood, kiln dried, u=10%, at plant". Allocation is based on the over-all proceeds of the process.
Start Date	1986	1986
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Germany used for central Europe	Data for Germany used for central Europe
Technology text	Technology mix used in Germany (1986)	Technology mix used in Germany (1986)
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that the German technology mix is representative for central Europe is considered in the uncertainties.	The assumption that the German technology mix is representative for central Europe is considered in the uncertainties.

Tab. 9.1 Ecoinvent meta information for the planing processes (cont.)

Name	planing, softwood, air dried, u=20%	planing, softwood, kiln dried, u=10%
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Includes planing process. planing mill is assumed to be located on the sawmill site. No transports are considered. Dust emissions are neglected for a lack of data.	Includes planing process. planing mill is assumed to be located on the sawmill site. No transports are considered. Dust emissions are neglected for a lack of data.
Amount	1	1
Local Name	Hobeln, Nadelholz, luftgetrocknet, u=20%	Hobeln, Nadelholz, kammergetrocknet, u=10%
Synonyms	Fichte//Spruce	Fichte//Spruce
General Comment to reference function	The multi-output process "planing, softwood, air dried, u=20%" delivers the two coproducts "sawn timber, softwood, planed, air dried, at plant" and "industrial residue wood, from planing, softwood, air dried, u=20%, at plant". Allocation is based on the over-all proceeds of the process.	The multi-output process "planing, softwood, kiln dried, u=10%" delivers the two coproducts "sawn timber, softwood, planed, kiln dried, at plant" and "industrial residue wood, from planing, softwood, kiln dried, u=10%, at plant". Allocation is based on the over-all proceeds of the process.
Start Date	1986	1986
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Germany used for central Europe	Data for Germany used for central Europe
Technology text	Technology mix used in Germany (1986)	Technology mix used in Germany (1986)
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that the German technology mix is representative for central Europe is considered in the uncertainties.	The assumption that the German technology mix is representative for central Europe is considered in the uncertainties.

General Flow information							Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Sawn timber beech rough, naturally/technically dried, at sawmill (u=10%)	→	planing, hardwood, air / kiln dried, u=10%					technical drying of naturally dried wood (u=20% --> u=10%)	wooden materials	extraction	No	RER	sawn timber, hardwood, raw, air / kiln dried, u=10%, at plant	1.15E+00	m3	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Electricity	→						suction: 3.0 kWh, planing: 28.1 kWh	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.11E+01	kWh	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Planning mill	→							wooden materials	extraction	Yes	RER	planing mill	8.00E-07	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
Hardwood, allocation correction	→			100% to sawn timber planed			adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	-1.39E-01	m3	calculated	1	1.00	calculated correction term
Hardwood, allocation correction	→			100% to industrial residual wood				wooden materials	extraction	No	RER	hardwood, allocation correction, 2	1.39E-01	m3	calculated	1	1.00	calculated correction term
			→	Sawn timber beech planed, naturally/technically dried, at planing mill (u=10%)	500 CHF/m3; revenue: 500 CHF; revenue process: 507 CHF; alloc. factor: 0.99	estimation based on Gautschi 2001 (personal communication)	yield 80%	wooden materials	extraction	No	RER	sawn timber, hardwood, planed, air / kiln dried, u=10%, at plant	1.00E+00	m3	Ressel 1986			
			→	Industrial residual wood beech planed, naturally/technically dried, at planing mill (u=10%)	47 CHF/m3; revenue: 7 CHF; revenue process: 507 CHF; alloc. factor: 0.01	estimation after Werner 2002, based on Gautschi 2001 (value increased by 1.61 based on wood with u=140% due to lower water content (after Vhe Nr. 407 (2002))		wooden materials	extraction	No	RER	industrial residue wood, from planing, hard, air/kiln dried, u=10%, at plant	1.50E-01	m3	Ressel 1986			
			→	Dust			no data available											
			→	Waste heat into air					air	low population density			Heat, waste	1.12E+02	MJ	calculated	1	1.51

Fig. 9.1 Flows for "planing, hardwood, air / kiln dried, u=10%" and its representation in the ecoinvent database

9 Planing mill

General Flow information							Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Sawn timber beech rough, technically dried, at sawmill (u=10%)	→	planing, hardwood, kiln dried, u=10%						wooden materials	extraction	No	RER	sawn timber, hardwood, raw, kiln dried, u=10%, at plant	1.15E+00	m3	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Electricity	→						suction: 3.0 kWh, planing: 28.1 kWh	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.11E+01	kWh	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Planing mill	→							wooden materials	extraction	Yes	RER	planing mill	8.00E-07	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
Hardwood, allocation correction	→			100% to sawn timber planed			adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	-1.39E-01	m3	calculated	1	1.00	calculated correction term
Hardwood, allocation correction	→			100% to industrial residual wood				wooden materials	extraction	No	RER	hardwood, allocation correction, 2	1.39E-01	m3	calculated	1	1.00	calculated correction term
	→			Sawn timber beech planed, technically dried, at planing mill (u=10%)	500 CHF/m3; revenue: 500 CHF; revenue process: 507 CHF; alloc. factor: 0.99	estimation based on Gautschi 2001 (personal communication)	yield 80%	wooden materials	extraction	No	RER	sawn timber, hardwood, planed, kiln dried, u=10%, at plant	1.00E+00	m3	Ressel 1986			
	→			Industrial residual wood beech planed, technically dried, at planing mill (u=10%)	47 CHF/m3; revenue: 7 CHF; revenue process: 507 CHF; alloc. factor: 0.01	estimation after Werner 2002, based on Gautschi 2001 (value increased by 1.61 based on wood with u=140% due to lower water content (after Vhe Nr. 407 (2002))		wooden materials	extraction	No	RER	industrial residue wood, from planing, hardwood, kiln dried, u=10%, at plant	1.50E-01	m3	Ressel 1986			
	→			Dust			no data available											
	→			Waste heat into air				air	low population density				Heat, waste	1.12E+02	MJ	calculated	1	1.51

Fig. 9.2 Flows for "planing, hardwood, kiln dried, u=10%" and its representation in the ecoinvent database

General Flow information							Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Sawn timber spruce rough, naturally dried, at sawmill (u=20%)	→	planing, softwood, air dried, u=20%						wooden materials	extraction	No	RER	sawn timber, softwood, raw, air dried, u=20%, at plant	1.15E+00	m3	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Electricity	→						suction: 3.0 kWh, planing: 28.1 kWh	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.11E+01	kWh	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Planing mill	→							wooden materials	extraction	Yes	RER	planing mill	8.00E-07	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
Softwood, allocation correction	→			100% to sawn timber planed			adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-1.39E-01	m3	calculated	1	1.00	calculated correction term
Softwood, allocation correction	→			100% to industrial residual wood				wooden materials	extraction	No	RER	softwood, allocation correction, 2	1.39E-01	m3	calculated	1	1.00	calculated correction term
	→			Sawn timber spruce planed, naturally dried, at planing mill (u=20%)	450 CHF/m3; revenue: 450 CHF; revenue process: 456 CHF; alloc. factor: 0.99	estimation based on Gautschi 2001 (personal communication)	yield 80%	wooden materials	extraction	No	RER	sawn timber, softwood, planed, air dried, at plant	1.00E+00	m3	Ressel 1986			
	→			Industrial residual wood spruce, naturally dried, at planing mill (u=20%)	39 CHF/m3; revenue: 6 CHF; revenue process: 456 CHF; alloc. factor: 0.01	estimation after Werner 2002, based on Gautschi 2001 (value increased by 1.61 based on wood with u=140% due to lower water content (after Vhe Nr. 407 (2002))		wooden materials	extraction	No	RER	industrial residue wood, from planing, softwood, air dried, u=20%, at plant	1.50E-01	m3	Ressel 1986			
	→			Dust			no data available											
	→			Waste heat into air				air	low population density				Heat, waste	1.12E+02	MJ	calculated	1	1.51

Fig. 9.3 Flows for "planing, softwood, air dried, u=20%" and its representation in the ecoinvent database

9 Planing mill

General Flow information							Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Sawn timber spruce rough, technically dried, at sawmill (u=10%)	→	planing, softwood, kiln dried, u=10%						wooden materials	extraction	No	RER	sawn timber, softwood, raw, kiln dried, u=10%, at plant	1.15E+00	m3	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)
Electricity	→					suction: 3.0 kWh, planing: 28.1 kWh	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.11E+01	kWh	Ressel 1986	1	1.51	(1,3,5,3,1,1,4)	
Planing mill	→						wooden materials	extraction	Yes	RER	planing mill	8.00E-07	unit	estimated	1	3.36	(4,5,2,3,4,5,9)	
Softwood, allocation correction	→			100% to sawn timber planed			adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-1.39E-01	m3	calculated	1	1.00	calculated correction term
Softwood, allocation correction	→			100% to industrial residual wood				wooden materials	extraction	No	RER	softwood, allocation correction, 2	1.39E-01	m3	calculated	1	1.00	calculated correction term
	→			Sawn timber spruce planed, technically dried, at planing mill (u=10%)	500 CHF/m3; revenue: 500 CHF; revenue process: 506 CHF; alloc. factor: 0.99	estimation based on Gautschi 2001 (personal communication)	yield 80%	wooden materials	extraction	No	RER	sawn timber, softwood, planed, kiln dried, at plant	1.00E+00	m3	Ressel 1986			
	→			Industrial residual wood spruce, technically dried, at planing mill (u=10%)	39 CHF/m3; revenue: 6 CHF; revenue process: 506 CHF; alloc. factor: 0.01	estimation after Werner 2002, based on Gautschi 2001 (value increased by 1.61 based on wood with u=140% due to lower water content (after Vhe Nr. 407 (2002))		wooden materials	extraction	No	RER	industrial residue wood, from planing, softwood, kiln dried, u=10%, at plant	1.50E-01	m3	Ressel 1986			
	→			Dust			no data available											
	→			Waste heat into air				air	low population density				Heat, waste	1.12E+02	MJ	calculated	1	1.51

Fig. 9.4 Flows for "planing, softwood, kiln dried, u=10%" and its representation in the ecoinvent database

10 Wood boards based on round wood

This chapter is mainly based on (Nimz et al. 1997) and (Kollmann et al. 1975).

The primary aim of the development of wood-based construction materials is at eliminating or reducing undesirable properties as much as possible. At the same time, the favourable characteristics of wood should be preserved or even enhanced. Conversion of solid wood into wood-based materials always follows the same principle. Wood is cut into small pieces that are combined with a suitable binder. Cutting into smaller pieces involves sawing, peeling, chopping, or defibration. Boards, veneers, particles of varying dimensions, and fibres are thus produced. As the size of the piece decreases, the properties of the wood-based panel or construction material differ more and more from those of natural wood. The wood-based material becomes more homogeneous, and the possibilities for size enlargement in two or three dimensions and moulding increase. However, at a comparable density, the strength decreases and the typical characteristics of wood are lost. This is exemplified by the changing properties in the series sawn timber, plywood, particle board, and medium-density fibreboard (MDF). (Products describes in the following sections.)

10.1 Plywood

Plywood is made out of glued veneers. Veneers are thin sheets of wood obtained by sawing, cutting, or peeling. They are among the longest-known wood-based construction materials. Veneers are produced from cordwood or from specially divided blocks by using peeling and cutting machines or, less commonly, by sawing. Many types of wood - e.g., poplar (*Populus* sp.), birch (*Betula* sp.), or Douglas fir (*Pseudotsuga* sp.) - can be peeled or cut either fresh or saturated with moisture through storage in water. Other types of wood - e.g., beech (*Fagus silvatica*), oak (*Quercus* sp.), ash (*Fraxinus* sp.), walnut (*Juglans* sp.), or various tropical high-density deciduous woods - must first be softened by steaming or boiling to allow the wood to be cut more easily. After peeling, the veneers have a moisture content of 30-110% (based on oven dry weight). This must therefore be reduced by drying to 4-12% for further processing. Special veneer dryers are used through which hot air is blown at 80-200 °C.

By combining veneers of different types of wood, densities, numbers of layers (plies), and different core materials, a large number of materials can be produced with widely differing properties and areas of use. Plywood can be classified according to different criteria into: veneer, core, and composite plywood.

10.1.1 Types of Plywood

Veneer plywood

Veneer plywood normally consists of at least three layers of wood glued to one another crosswise. The barrier technique (orientation of the grain of adjacent layers at an angle, usually 90°) requires the middle layer to form a plane of symmetry with regard to type of wood, grain direction, and thickness of individual layers. By forming a barrier, the contraction in width of layers with longitudinal grain direction depends extensively on the mechanical deformation of transverse layers. At the same time, the strength and deformation properties in the longitudinal and transverse directions of the plies are aligned with one another, thus decreasing the anisotropy. Plywood boards are thus considerably more dimensionally stable and can be subjected to loads more uniformly than solid wood boards. The number of veneers is usually uneven, and the arrangements are symmetrical. Three, five, and seven-fold boards, etc., are produced normally.

Wood Core plywood (block board)

In wood core plywood, the core consists of battens or strips, usually of light coniferous wood that has been cut from a block of boards or veneers that have been glued together and then blocked on both

sides with veneers. Depending on the core, two types of sandwich board exist: *battenboard* and *lamin-board*. Wood core plywood is used for high-grade joinery.

Composite plywood

Two types of composite plywood exist. In the first type the core consists of a material, usually of low density, that is planked on both sides with veneers. Practically any material can be used for the core (e.g., extruded particle board, paper honeycomb, and for special applications, polyurethane foam). These types of boards are used, for example, for doors and partitions. In the reverse case, the core can be made of plywood and the covering of metal sheeting or glass-fibre-reinforced plastic. A multitude of combinations is possible. Composite materials with special properties, such as high strength, good insulating capacity, and high abrasion resistance, can be designed for particular applications (e.g., containers).

Moulded plywood is formed when the plywood is moulded by using appropriately shaped presses. It is used, for example, for chair backs and seats, furniture, wall panelling, and girder webs.

The natural adhesives that were initially used, such as protein adhesives (gluten, casein, soybean, and blood albumin adhesives), have disadvantages like low resistance to climate variations, sensitivity to attack by micro-organisms, long pressing times, and sometimes highly complex processing. Since the 1930s they have been increasingly replaced by synthetic adhesives, and these are nowadays used almost exclusively. They include urea-formaldehyde (UF), melamine-formaldehyde (MF), and phenol formaldehyde (PF) resins, as well as MUF or MUPF mixed condensates. The use envisaged and the cost determine the choice of adhesive.

10.1.2 Resins in Plywood

Urea-formaldehyde (UF) resins

The most important adhesives for gluing wood are urea-formaldehyde resins. The use of UF resins to glue plywood was first described in 1929. Production of these wood adhesives has since developed enormously. Urea-formaldehyde resins are used for adhesion of wood materials used in the interior and have limited resistance to moisture. They can be readily adapted to operational requirements through the nature and dosage of curing aids and additives. Ammonium salts (e.g. NH_4Cl) are generally used as curing aids. They lower the pH by reacting with formaldehyde. The rate of curing increases with increasing temperature and decreasing pH. The viscosity of the adhesive liquor is adapted to processing conditions by the addition of water, depending on the content of extenders and fillers. The pressing temperature is 80-120 °C.

Melamine-formaldehyde (MF) resins

Melamine-formaldehyde resins undergo a curing reaction analogous to that of UF resins. MF resins are characterised by their high water resistance and colourless glue lines. For cost reasons, however, pure MF resins are rarely employed in plywood production. Instead, MUF or MUPF mixed condensates are used when high weather resistance is required. Depending on the melamine content and the type of wood, adhesions are obtained for semi-exterior or exterior applications.

Phenol-formaldehyde (PF) resins

Phenol-formaldehyde resins are generally cured by heating and also by a combination of heating and curing aids. PF resins are less reactive than UF or MF resins. They therefore require higher pressing temperatures (115-140 °C). Curing of phenol-formaldehyde resins is more sensitive to wood moisture and to pressing conditions than curing with other adhesives. Their alkali content can have damaging effects, e.g. on subsequent surface treatment of veneered boards.

Extenders and fillers

In plywood production, extenders and fillers are used together with adhesives to avoid glue penetration during hot pressing.

Extenders are finely ground products of organic origin that are capable of swelling and gelatinising and thus act as adhesives themselves (e.g., starches and cereal and vegetable flours). They are generally used with UF resins at levels of 15-30% or higher (extender plus water, based on the adhesive resin) to reduce the brittleness of the glue line and resin consumption.

Fillers do not have adhesive properties. They are finely powdered materials (e.g. nut shell or olive stone flours) or minerals, such as kaolin or pulverised gypsum (lenzin) that can be swelled only slightly or not at all. Fillers are normally used with PF resins.

The addition of extenders and fillers can decrease the resistance of the adhesion to water and micro-organisms. Mineral fillers also increase the wear on tools.

10.1.3 Plywood production

The main operations involved in the production of veneer plywood are preparation, assembly, application of adhesive, stacking coated veneers (sometimes pre-pressing), hot pressing, conditioning, forming (sometimes sanding), and sometimes fault elimination.

The adhesive is generally applied by using two- or four-roll machines, by pouring using an adhesive curtain, or by melting an adhesive film. The coated veneers are stacked either manually or by machine. To make transportation easier and reduce stack height, coated veneer stacks are frequently pre-compressed in an unheated one-daylight press. Hot pressing occurs in one- or multi-daylight presses but usually with a smaller size. Pressing conditions (temperature, time, and pressure) depend on the type of wood, the strength of the plywood, and the type of adhesive. Pressures of 0.6-2.5 N/mm² and temperatures of 80-140 °C are usually applied. After gluing and sometimes climatisation, veneer plywood is sanded, cut to standard or ordered measurements, and graded according to quality.

Veneer plywood is normally produced in thickness' of 1-12 mm and as multiply boards with thickness' of 13-50 mm. Wood core plywood is available in thickness' of 10-60 mm. The density of veneer plywood depends on the type of wood and the plywood construction (500-800 kg/m³). Wood core plywood generally has a lower density (450-750 kg/m³).

10.1.4 Use of plywood

Depending on the type of adhesive and the wood, a distinction is made between plywood for exterior and interior use; further classification is made for the following applications:

- In dry areas, i.e. a normal indoor climate (interior)
- In moist areas, i.e. with protection against direct weathering also suitable for outdoor climates (semi-exterior)
- For outdoor use, suitable for long-term weathering (exterior)

Most of the plywood produced is used in building. Douglas fir and pine are used predominantly world-wide. Geographically, the consumption of plywood varies widely and depends mainly on local building traditions. Countries such as Canada and the United States are the largest plywood consumers world-wide, with 105 and 77 m³ per 1000 inhabitants, respectively. In Europe, the Netherlands is the leading consumer, with 28 m³ per 1000 inhabitants; in Germany, consumption is 12 m³, and in Italy 6 m³ per 1000 inhabitants.

Typical uses for veneer plywood are as construction plywood for wall, floor, and ceiling panels and also as panels for concrete shuttering. Industrial plywood is used to build railway coaches (e.g., for

high-speed trains). A large area of use is for containers, boxes, and packaging. Plywood with decorative covering veneers is frequently used in interior furnishings (e.g. wall panels or doors).

10.1.5 Plywood in ecoinvent database

The allocation of the burdens to the different outputs is proposed according to the economic return of the products. Because the amount and the value of the industrial residue wood are both small compared to plywood, the allocation factor for industrial residue wood is (practically) zero.

The economic allocation can't respect the mass and energy balance of the products. Therefore, correcting modules are defined that can add or subtract the CO₂ uptake and the resource (wood and solar energy) consumption from nature. They are used as shown in the example in Fig. 5.2:

The background information to the processes are given in Tab. 10.1, flows are shown in:

- plywood, indoor use, production
 Outputs:
 - plywood, indoor use, at plant
 - industrial residue wood, plywood prod., indoor use, hardwood, u=20%, at plant Fig. 10.1
- plywood, outdoor use, production
 Outputs:
 - plywood, outdoor use, at plant
 - industrial residue wood, plywood prod., outdoor use, hardwood, u=20%, at plant Fig. 10.2

10.1.6 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

These data are of medium quality. Material and energy input data are good, emissions are not comprehensive and based on weakly supported data.

Tab. 10.1 Ecoinvent meta information for the plywood production processes

Name	plywood, indoor use, production	plywood, outdoor use, production
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Includes the inputs to the production processes, transports of those inputs and the process emissions as far as they are available.	Includes the inputs to the production processes, transports of those inputs and the process emissions as far as they are available.
Amount	1	1
Local Name	Sperrholz, Innenanwendung, Produktion	Sperrholz, Aussenanwendung, Produktion
Synonyms		
General Comment to reference function	Different types of plywood are available depending on the number and height of the layers (type not specified). The multi-output process "plywood, indoor use, production" delivers the two coproducts "plywood, indoor use, at plant" and "industrial residue wood, plywood prod., indoor use, hardwood, u=20%, at plant". Allocation is based on the over-all proceeds of the process.	Different types of plywood are available depending on the number and height of the layers (type not specified). The multi-output process "plywood, outdoor use, production" delivers the two coproducts "plywood, outdoor use, at plant" and "industrial residue wood, plywood prod., outdoor use, hardwood, u=20%, at plant". Allocation is based on the over-all proceeds of the process.
Start Date	1996	1996
End Date	1996	1996
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Production data for one plant in Switzerland used for central Europe	Production data for one plant in Switzerland used for central Europe
Technology text	Medium sized enterprise technology in Switzerland (1996)	Medium sized enterprise technology in Switzerland (1996)
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that the technology is representative for central Europe is considered in the uncertainties.	The assumption that the technology is representative for central Europe is considered in the uncertainties.

10 Wood boards based on round wood

General Flow information							Representation in ecoinvent								Uncertainty information				
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Roundwood beech, at forest road (u=70%)	→	plywood, indoor use, production, inputs					volume including bark is 2.7*1.12 m3	wooden materials	extraction	No	RER	round wood, hardwood, under bark, u=70%, at forest road	2.70E+00	m3	Plant data Hess & Co. AG Döttingen	1	1.12	(1,4,2,3,1,1,3)	
Urea-formadehyde resin incl. hardener, at plant	→							paintings	production	No	RER	urea formaldehyde resin, at plant	8.32E+01	kg	Plant data Hess & Co. AG Döttingen	1	1.12	(1,4,2,3,1,1,3)	
Wood chips beech burned in 50kW furnace	→							wood energy	heating systems	No	CH	wood chips, from industry, hardwood, burned in furnace 50kW	8.11E+03	MJ	Plant data Hess & Co. AG Döttingen	1	1.12	(1,4,2,3,1,1,1)	
Chips	→						Since the fuel for the furnace is included in the input, the corresponding amount inventoried with the energy module is subtracted. Since the volume of the chips is bulked, it is higher then the difference of input and output volume.	wood energy	fuels	No	RER	wood chips, hardwood, from industry, u=40%, at plant	-1.93E+00	m3	calculated	1	1.00	calculated	
Diesel burned in building machine	→							construction processes	machinery	No	GLO	diesel, burned in building machine	3.20E+00	MJ	Plant data Hess & Co. AG Döttingen	1	2.01	(1,4,2,3,1,1,5)	
Electricity	→							electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.06E+02	kWh	Plant data Hess & Co. AG Döttingen	1	1.12	(1,4,2,3,1,1,2)	
Transport lorry	→						Chemicals: 100 km, wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	1.57E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport rail	→						Chemicals: 600 km, wood: 100 km	transport systems	train	No	RER	transport, freight, rail	3.48E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Process and cooling water	→							resource	in water			Water, cooling, unspecified natural origin	1.84E+00	m3	Plant data Hess & Co. AG Döttingen	1	1.12	(1,4,2,3,1,1,4)	
plant	→							wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	3.33E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)	
Hardwood, allocation correction	→				100% to plywood			adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation. Also adds the 12% bark	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	-1.31E+00	m3	calculated	1	1.00	calculated correction term
Hardwood, allocation correction	→				100% to industrial residual wood				wooden materials	extraction	No	RER	hardwood, allocation correction, 2	1.31E+00	m3	calculated	1	1.00	calculated correction term

Fig. 10.1 Flows for "plywood, indoor use, production" and its representation in the ecoinvent database

10 Wood boards based on round wood

General Flow information							Representation in ecoinvent								Uncertainty information			
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
plywood, indoor use, production, outputs	→			Plywood, indoor use, at plant	Industrial wood as result of veneer production (veneer beech around 7'000 CHF/m3 to 10'000 CHF/m3; allocation fact. 1	estimation based on price list from Heftli & Grob, Niederhasli		wooden materials	extraction	No	RER	plywood, indoor use, at plant	1.00E+00	m3				
	→			Industrial residual wood beech, at plywood-plant I (u=20%)	60 CHF/Stere; 84 CHF/m3 u.b.; alloc. factor: 0	estimation after Werner 2002, based on Gautschi 2001 (value increased by 1.61 based on wood with u=140% due to lower water content (after Vhe Nr. 407 (2002))		wooden materials	extraction	No	RER	industrial residue wood, plywood prod., indoor use, hardwood, u=20%, at plant	1.31E+00	m3	calculated by balance of dry wood mass			
	→			Waste heat into air				air	unspecified			Heat, waste	1.10E+03	MJ	calculated	1	1.12	(1,4,2,3,1,1,13)
	→			Formaldehyde into air				air	unspecified			Formaldehyde	8.32E-02	kg	Nimz 1997	1	2.81	(4,5,2,5,5,5,23)
	→			Wastewater				waste management	wastewater treatment	No	CH	treatment, plywood production effluent, to wastewater treatment, class 3	1.84E+00	m3	Plant data Hess & Co. AG Döttingen	1	1.52	(1,4,2,3,1,1,32)

Fig. 10.1 Flows for "plywood, indoor use, production" and its representation in the ecoinvent database (cont.)

10 Wood boards based on round wood

General Flow information							Representation in ecoinvent								Uncertainty information			
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Roundwood beech, at forest road (u=70%)	→	plywood, outdoor use, production, input					volume including bark is 2.7*1.12 m3	wooden materials	extraction	No	RER	round wood, hardwood, under bark, u=70%, at forest road	2.70E+00	m3	Plant data Hess & Co. AG Döttingen	1	1.12	(1,4,2,3,1,1,3)
melamine formaldehyde resin, at plant	→							paintings	production	No	RER	melamine formaldehyde resin, at plant	8.32E+01	kg	Plant data Hess & Co. AG Döttingen	1	1.12	(1,4,2,3,1,1,3)
Wood chips beech burned in 50kW furnace	→							wood energy	heating systems	No	CH	wood chips, from industry, hardwood, burned in furnace 50kW	8.11E+03	MJ	Plant data Hess & Co. AG Döttingen	1	1.12	(1,4,2,3,1,1,1)
Chips	→						Since the fuel for the furnace is included in the input, the corresponding amount inventoried with the energy module is subtracted. Since the volume of the chips is bulked, it is higher then the difference of input and output volume.	wood energy	fuels	No	RER	wood chips, hardwood, from industry, u=40%, at plant	-1.93E+00	m3	calculated	1	1.00	calculated
Diesel burned in building machine	→							construction processes	machinery	No	GLO	diesel, burned in building machine	3.20E+00	MJ	Plant data Hess & Co. AG Döttingen	1	2.01	(1,4,2,3,1,1,5)
Electricity	→							electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.06E+02	kWh	Plant data Hess & Co. AG Döttingen	1	1.12	(1,4,2,3,1,1,2)
Transport lorry	→						Chemicals: 100 km, wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	1.57E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Transport rail	→						Chemicals: 600 km, wood: 100 km	transport systems	train	No	RER	transport, freight, rail	3.48E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Process and cooling water	→							resource	in water			Water, cooling, unspecified natural origin	1.84E+00	m3	Plant data Hess & Co. AG Döttingen	1	1.12	(1,4,2,3,1,1,4)
plant	→							wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	3.33E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
Hardwood, allocation correction	→				100% to plywood		adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	-1.31E+00	m3	calculated	1	1.00	calculated correction term
Hardwood, allocation correction	→				100% to industrial residual wood			wooden materials	extraction	No	RER	hardwood, allocation correction, 2	1.31E+00	m3	calculated	1	1.00	calculated correction term

Fig. 10.2 Flows for "plywood, outdoor use, production" and its representation in the ecoinvent database

10 Wood boards based on round wood

General Flow information							Representation in ecoinvent							Uncertainty information		
Input	Process Name	Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
plywood, outdoor use, production, output		→ Plywood, for outdoor use, at plant	Industrial wood as result of veneer production (veneer beech around 7'000 CHF/m3 to 10'000 CHF/m3; allocation fact. 1	estimation based on price list from Heftli & Grob, Niederhasli		wooden materials	extraction	No	RER	plywood, outdoor use, at plant	1.00E+00	m3				
		→ Industrial residual wood beech, at plywood plant II (u=20%)	60 CHF/Stere; 84 CHF/m3 u.b.; alloc. factor: 0	estimation after Werner 2002, based on Gautschi 2001 (value increased by 1.61 based on wood with u=140% due to lower water content (after Vhe Nr. 407 (2002))		wooden materials	extraction	No	RER	industrial residue wood, plywood prod., outdoor use, hardwood, u=20%, at plant	1.31E+00	m3	calculated by balance of dry wood mass			
		→ Waste heat into air				air	unspecified			Heat, waste	1.10E+03	MJ	calculated	1	1.12	(1,4,2,3,1,1,13)
		→ Formaldehyde into air				air	unspecified			Formaldehyde	8.32E-02	kg	according to Uhlmann 1997	1	2.81	(4,5,2,5,5,5,23)
		→ Wastewater				waste management	wastewater treatment	No	CH	treatment, plywood production effluent, to wastewater treatment, class 3	1.84E+00	m3	Plant data Hess & Co. AG Döttingen	1	1.52	(1,4,2,3,1,1,32)

Fig. 10.2 Flows for "plywood, outdoor use, production" and its representation in the ecoinvent database (cont.)

10.2 Laminated timber elements

The geometry and cross sections of laminated timber beams can be varied widely and adapted to requirements of static and aesthetics. Lamination allows bent beams with differing profile shapes and spans of 60-100 m.

Laminated wood is clearly superior in performance to squared logs. By eliminating the natural wood faults through careful sorting of the boards, a more homogeneous construction material is formed with regard to appearance and strength.

The behaviour of wooden girders in the event of fire is especially favourable. Through the formation of a charcoal layer on the surface, the centre of the girder is protected from further rapid attack by the fire and is thus insulated. Therefore, the desired extent of fire resistance is determined by the dimensions of the load-bearing beams.

Glued laminated timber elements

The art of building with wood is many centuries old. Unfortunately, the quantity of trunk wood of sufficient dimensions available for making load-bearing building components with desired spans is continuously decreasing. This consideration gave the Weimar carpenter Otto Hetzer the idea in 1906 of using a completely new method. He cut the round wood into boards and glued the graded and dried lamellar boards together with adhesive to form new cross sections (glued laminated "glulam" girders, or laminated beams).

The actual breakthrough of the process for using glulam girders in building occurred in 1930 with the development of synthetic adhesive resins. Large-scale industrial production opened up new areas of application for this method. Besides being used in tall buildings, glulam girders were used in aeroplanes (Mosquitoes in England and Spruce Goose in the United States), sometimes with large wing spreads, and for mine casings because of their antimagnetic properties. Glulam girders are now used mainly for building in central and northern European countries and North America, where wood has traditionally been widely used for this purpose.

Glulam girders were traditionally produced in small, handcrafting workshops. These have been replaced in recent decades by more efficient industrial plants, which use machines and automatic production lines and can offer quality assurance.

Laminated wood is made mostly from spruce, which is the easiest wood to process and fulfils the specific requirements of strength and durability. Pine, larch, and Douglas fir are also used in special cases. Hardwoods are generally used only for building components that must support very heavy loads. For some hardwoods, the effect of wood components on the durability of adhesive compounds is not entirely clear.

In a normal indoor climate and for building components protected from weathering, glues based on urea resins are used. For building components that are exposed to strong climatic variations or weathering, weather-resistant adhesives based on phenol-resorcinol-formaldehyde (PRF) are employed. These PRF resins give a dark brown glue line, unlike adhesives based on amino resins. In the 1990s a special glue type for bonding laminates was developed based on using melamine-resorcinol-formaldehyde resins. These bondings are highly resistant to boiling water and give a pale glue line.

Lamellar boards are sawed from round wood trunks. The boards are pre-dried to ca. 10% residual moisture, graded, and then bonded in automatic plants to form continuous boards with dovetail joints. The moisture in the boards to be glued should be 2-3% below the expected equilibrium moisture level. Variations in the moisture content of the different boards should be not greater than 2%. The broad sides of these joined boards are planed until smooth and uniformly coated with adhesive. Constructing the beams, particular attention is paid to the positions of the annual rings in the cross section of the boards. The rifts (vertical) and the flats (horizontal) should be distributed regularly.

Depending on how exactly the surfaces fit together, 400-600 g of adhesive liquor is applied per square meter. Glued boards are stacked and stored under pressure at room temperature or slightly higher (40 °C) so that the adhesive resin cures after a few hours and a composite beam is formed.

Screwed laminated timber elements

Screwed laminated timber plates consist of a horizontal stack of single boards which are aligned on end and joined together using different techniques. Depending on the direction of loading and/or installation in reference to their plain, these construction parts serve as load bearing or non load bearing plates or panels. Most frequently, laminated timber plates are used for walls, roofs, ceilings and bridge decks.

Besides the quality of the timber used and its dimensions, the load bearing capacity of laminated timber structures depends on the way of joining the single boards. Common joining techniques for low to medium loaded parts are based on nails, screws and metal or wooden dowels which are arranged in a regular pattern along the boards. Glued "Glulam" or laterally pre-stressed laminated timber plates are preferably used for parts that bear medium to heavy loads, e.g. as building material for houses. Laminated timber plates are also used as basis for timber-concrete composite structures which provide increased load bearing capacities and, if used within house-building, positive effects on sound insulation.

For a couple of years, dowel laminated timber plates (using wooden dowels) have reached in Switzerland by far the highest volume of all techniques mentioned above. This is because they are regarded as environmentally benign (not containing any glue or metal parts), economic viable (they can be produced semi-automatically) and flexible for further processing (no limitation thanks to the absence of metal parts).

Three-layered laminated board (softwood)

The three layered laminated board is actually a special form of plywood featuring relatively thick layers. For details on the production see chapter 10.1.

The three-layered laminated board made of spruce is produced in a large wood processing company in the Canton of Fribourg, Switzerland: out of rough sawn timber ($u=70\%$) 22 mm lamellas are sawn, which are technically dried, planed, finger-jointed, glued with PVAc-adhesive at their ends to "never-ending" lamellas and finally glued to a bloc. This block is sawn to boards, which are planed. The spruce is mainly coming from the area around Fribourg. The thermal energy for the drying of the lamellas is mainly produced by industrial residual wood (8% of wood input) ((Werner et al. 1997), p. A7).

10.2.1 Laminated timber in ecoinvent database

The allocation of the burdens to the different outputs of this process is proposed according to the economic return of the products. Because both, the amount and the value of the industrial residue wood that is sold are small compared to those of glued laminated timber, the allocation factor gets zero or almost zero (for the softwood of the screw-laminated timber element).

The economic allocation can't respect the mass and energy balance of the products. Therefore, correcting modules are defined that can add or subtract the CO₂ uptake and the resource (wood and solar energy) consumption from nature. They are used as shown in the example in Fig. 5.2:

The background information to the processes are given in Tab. 10.2, flows are shown in:

- glued laminated timber, indoor use, production
Outputs:
 - glued laminated timber, indoor use, at plant
 - industrial residue wood, GLT production, indoor use, u=10%, at plantFig. 10.3
- glued laminated timber, outdoor use, production
Outputs:
 - glued laminated timber, outdoor use, at plant
 - industrial residue wood, GLT production, outdoor use, u=10%, at plantFig. 10.4
- laminated timber element, outdoor use, production
Outputs:
 - laminated timber element, transversally prestressed, for outdoor use, at plant
 - industrial residue wood, LTE production, softwood, u=20%, at plant
 - industrial residue wood, LTE production, hardwood, u=20%, at plantFig. 10.5
- three layered laminated board, production
Outputs:
 - three layered laminated board, at plant
 - industrial residue wood, 3-layered LB production, softwood, u=20%, at plantFig. 10.6

10.2.2 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Data for glued laminated timber are estimated based on data for three layered laminated board. Their overall quality however is as good as it gets for a class of products this wide. Data quality, however, is not sufficient to allow for comparing with other wood boards.

Tab. 10.2 Ecoinvent meta information for the laminated timber production processes

Name	glued laminated timber, indoor use, production	glued laminated timber, outdoor use, production
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Includes the inputs to the production processes, transports of those inputs and the process emissions as far as they are available.	Includes the inputs to the production processes, transports of those inputs and the process emissions as far as they are available.
Amount	1	1
Local Name	Brettschichtholz, verleimt, Innenanwendung, Produktion	Brettschichtholz, verleimt, Aussenanwendung, Produktion
Synonyms		
General Comment to reference function	The multi-output process "glued laminated timber, indoor use, production" delivers the two coproducts "glued laminated timber, indoor use, at plant" and "industrial residue wood, GLT production, indoor use, u=10%, at plant". Allocation is based on the over-all proceeds of the process.	The multi-output process "glued laminated timber, outdoor use, production" delivers the two coproducts "glued laminated timber, outdoor use, at plant" and "industrial residue wood, GLT production, outdoor use, u=10%, at plant". Allocation is based on the over-all proceeds of the process.
Start Date	1986	1986
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Germany used for central Europe	Data for Germany used for central Europe
Technology text	Estimate based on data for three layered laminated board	Estimate based on data for three layered laminated board
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that the technology is representative for the product and the region is considered in the uncertainties.	The assumption that the technology is representative for the product and the region is considered in the uncertainties.

Tab. 10.2 Ecoinvent meta information for the laminated timber production processes (cont.)

Name	laminated timber element, outdoor use, production	three layered laminated board, production
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Includes the inputs to the production processes and transports of those inputs. No process emission data are available.	Includes the inputs to the production processes and transports of those inputs. No process emission data are available.
Amount	1	1
Local Name	Brettstapelement, Aussenanwendung, Produktion	Dreischichtplatte, Produktion
Synonyms		
General Comment to reference function	Covers screwed-laminated timber element, transversally prestressed, for outdoor use, mainly for bridges. The multi-output process "laminated timber element, outdoor use, production" delivers the three coproducts "laminated timber element, transversally prestressed, for outdoor use, at plant", "industrial residue wood, LTE production, softwood, u=20%, at plant" and "industrial residue wood, LTE production, hardwood, u=20%, at plant". Allocation is based on the over-all proceeds of the process.	The multi-output process "three layered laminated board, production" delivers the two coproducts "three layered laminated board, at plant" and "industrial residue wood, 3-layered LB production, softwood, u=20%, at plant". Allocation is based on the over-all proceeds of the process.
Start Date	1986	1997
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Germany used for central Europe	Data for Germany used for central Europe
Technology text	Medium enterprise technology (1995)	Medium enterprised technology (1995)
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that the technology is representative for central Europe is considered in the uncertainties.	The assumption that the technology is representative for central Europe is considered in the uncertainties.

10 Wood boards based on round wood

General Flow information							Representation in ecoinvent								Uncertainty information			
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Sawn timber spruce rough, naturally dried, at sawmill (u=20%)	→	glued laminated timber, indoor use, production, inputs						wooden materials	extraction	No	RER	sawn timber, softwood, raw, air dried, u=20%, at plant	1.37E+00	m3	yield: Ressel 1986	1	2.01	(1,4,2,3,1,1,5)
Light fuel oil low sulfur burned in boiler 1 MW	→							oil	heating systems	No	RER	heat, light fuel oil, at industrial furnace 1MW	2.30E+01	MJ	Frühwald et al. 1996	1	1.12	(1,4,2,3,1,1,1)
Heat from wood furnace, fueled with industrial residual wood	→						Energy input; drying: 1610 MJ, hall heating 1072 MJ	wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 300kW	2.68E+03	MJ	Frühwald et al. 1996	1	1.12	(1,4,2,3,1,1,1)
Chips	→						Since the fuel for the furnace is included in the input, the corresponding amount inventoried with the energy module is subtracted. Since the volume of the chips is bulked, it is higher then the difference of input and output volume.	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40%, at plant	-8.48E-01	m3	calculated	1	1.00	calculated
Urea-formadehyde resin incl. hardener, at plant	→						without impregnation and surface treatment	paintings	production	No	RER	urea formaldehyde resin, at plant	1.20E+01	kg	Consumption of adhesives: Frühwald 1996	1	1.12	(1,4,2,3,1,1,3)
Diesel burned in building machine	→						internal transports	construction processes	machinery	No	GLO	diesel, burned in building machine	3.36E+01	MJ	Frühwald et al. 1996	1	2.01	(1,4,2,3,1,1,5)
Transport rail	→						Chemicals: 600 km, wood: 100 km	transport systems	train	No	RER	transport, freight, rail	8.12E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Transport lorry	→						Chemicals: 100 km, wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	3.82E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Electricity	→							electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.29E+02	kWh	Frühwald et al. 1996	1	1.12	(1,4,2,3,1,1,2)
plant	→							wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	3.33E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
Softwood, allocation correction	→				100% to GLT		adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-5.17E-02	m3	calculated	1	1.00	calculated correction term
Softwood, allocation correction	→				100% to industrial residual wood			wooden materials	extraction	No	RER	softwood, allocation correction, 2	5.17E-02	m3	calculated	1	1.00	calculated correction term

Fig. 10.3 Flows for "glued laminated timber, indoor use, production" and its representation in the ecoinvent database

10 Wood boards based on round wood

General Flow information							Representation in ecoinvent								Uncertainty information			
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
glued laminated timber, indoor use, production, outputs	→			Glued laminated timber, for indoor use, at plant	>800 CHF/m3; revenue process: >800 CHF; alloc. factor: 1.0	estimation based on price list from Hefti & Grob, Niederhasli		wooden materials	extraction	No	RER	glued laminated timber, indoor use, at plant	1.00E+00	m3				
	→			Industrial residual wood spruce, at GLT-plant I (u=10%)	39 CHF/m3; revenue: 1 CHF; revenue process: >800 CHF; alloc. factor: 0.0	estimation after Werner 2002, based on Gautschi 2001 (value increased by 1.61 based on wood with u=140% due to lower water content (after Vhe Nr. 407 (2002))	The remaining residual wood (0.344m3) is used for heating purposes and for technical drying	wooden materials	extraction	No	RER	industrial residue wood, GLT production, indoor use, u=10%, at plant	5.17E-02	m3	calculated by balance of dry wood mass			
	→			Adhesive residues in MWI				waste management	municipal incineration	No	CH	disposal, polyurethane, 0.2% water, to municipal incineration	9.74E-01	kg	Frühwald et al. 1996	1	1.12	(1,4,2,3,1,1,2)
	→			waste heat				air	unspecified			Heat, waste	4.63E+02	MJ	calculated	1	1.12	(1,4,2,3,1,1,13)
	→			Formaldehyde into air			0.1 mass% Emissionen	air	unspecified			Formaldehyde	1.20E-02	kg	according to Niemz 1997	1	2.81	(4,5,2,5,5,5,23)

Fig. 10.3 Flows for "glued laminated timber, indoor use, production" and its representation in the ecoinvent database (cont.)

10 Wood boards based on round wood

General Flow information							Representation in ecoinvent								Uncertainty information			
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Sawn timber spruce rough, naturally dried, at sawmill (u=20%)	→	glued laminated timber, outdoor use, production, inputs						wooden materials	extraction	No	RER	sawn timber, softwood, raw, air dried, u=20%, at plant	1.37E+00	m3	yield: Ressel 1986	1	1.12	(1,4,2,3,1,1,4)
Light fuel oil low sulfur burned in boiler 1 MW	→							oil	heating systems	No	RER	heat, light fuel oil, at industrial furnace 1MW	2.30E+01	MJ	Frühwald et al. 1996	1	1.12	(1,4,2,3,1,1,1)
Heat from wood furnace, fueled with industrial residual wood	→						Data per MJ energy carrier input (90% efficiency); contains emissions, waste and electricity consumption); this module is used for wood residues generated internally that are used for internal energy generation (so no additional wood input required)	wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 300kW	2.66E+03	MJ	Frühwald et al. 1996	1	1.12	(1,4,2,3,1,1,1)
Chips	→						Since the fuel for the furnace is included in the input, the corresponding amount inventoried with the energy module is subtracted. Since the volume of the chips is bulked, it is higher then the difference of input and output volume.	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40%, at plant	-8.41E-01	m3	calculated	1	1.00	calculated
Diesel burned in building machine	→						internal transports	construction processes	machinery	No	GLO	diesel, burned in building machine	3.36E+01	MJ	Frühwald et al. 1996	1	2.01	(1,4,2,3,1,1,5)
Melamin-formaldehyde resin incl. hardener, at plant	→						without impregnation and surface treatment	paintings	production	No	RER	melamine formaldehyde resin, at plant	1.20E+01	kg	Adhesive consumption: Frühwald 1996	1	1.12	(1,4,2,3,1,1,3)
Transport rail	→						Chemicals: 600 km, wood: 100 km	transport systems	train	No	RER	transport, freight, rail	8.12E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Transport lorry	→						Chemicals: 100 km, wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	3.82E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Electricity	→							electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.29E+02	kWh	Frühwald et al. 1996	1	1.12	(1,4,2,3,1,1,2)
plant	→							wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	3.33E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
Softwood, allocation correction	→				100% to GLT		adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-5.43E-02	m3	calculated	1	1.00	calculated correction term
Softwood, allocation correction	→				100% to industrial residual wood			wooden materials	extraction	No	RER	softwood, allocation correction, 2	5.43E-02	m3	calculated	1	1.00	calculated correction term

Fig. 10.4 Flows for "glued laminated timber, outdoor use, production" and its representation in the ecoinvent database

10 Wood boards based on round wood

General Flow information							Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
glued laminated timber, outdoor use, production, outputs	→			Glued laminated timber, for outdoor use, at plant	>800 CHF/m3; revenue process: >800 CHF; alloc. factor: 1.0	estimation based on price list from Hefti & Grob, Niederhasli		wooden materials	extraction	No	RER	glued laminated timber, outdoor use, at plant	1.00E+00	m3				
	→			Industrial residual wood spruce, at GLT-plant II (u=10%)	39 CHF/m3; revenue: 1 CHF; revenue process: >800 CHF; alloc. factor: 0.0	estimation after Werner 2002, based on Gautschi 2001 (value increased by 1.61 based on wood with u=140% due to lower water content (after Vhe Nr. 407 (2002))	The remaining residual wood (0.344m3) is used for heating purposes and for technical drying	wooden materials	extraction	No	RER	industrial residue wood, GLT production, outdoor use, u=10%, at plant	5.43E-02	m3	calculated			
	→			Waste heat into air				air	unspecified			Heat, waste	4.63E+02	MJ	calculated	1	1.12	(1,4,2,3,1,1,13)
	→			Formaldehyde into air				air	unspecified			Formaldehyde	1.20E-02	kg	according to Uhlmann 1997	1	2.81	(4,5,2,5,5,5,23)

Fig. 10.4 Flows for "glued laminated timber, outdoor use, production" and its representation in the ecoinvent database (cont.)

10 Wood boards based on round wood

General Flow information							Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Sawn timber spruce rough, naturally dried, at sawmill (u=20%)	→	laminated timber element, outdoor use, production, inputs					estimation: abatement according to glued laminated timber	wooden materials	extraction	No	RER	sawn timber, softwood, raw, air dried, u=20%, at plant	1.28E+00	m3	estimated	1	1.45	(4,5,1,3,3,5,3)
Sawn timber beech rough, naturally dried, at sawmill (u=20%)	→			allocation 100% to LTE		estimation: abatement according to glued laminated timber	wooden materials	extraction	No	RER	sawn timber, hardwood, raw, air dried, u=20%, at plant	9.20E-02	m3	estimated	1	1.45	(4,5,1,3,3,5,3)	
Electricity	→					mechanical treatment	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.00E+01	kWh	estimation based on Ressel 1986	1	1.45	(4,5,1,3,3,5,2)	
Steel low-alloyed	→				allocation to laminated timber element; allocation factor 1			metals	extraction	No	RER	steel, low-alloyed, at plant	5.23E+01	kg	calculation based on EGH/DBfH 1998	1	1.45	(4,5,1,3,3,5,3)
Transport rail	→				12.1% for steel; allocation fact. for laminated timber element: 97.5%; allocation fact. for industrial residual wood spruce: 2.5%	97.5% = (97% of 88%) + 12.1%	Wood: 100 km, steel: 200 km	transport systems	train	No	RER	transport, freight, rail	8.66E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Transport lorry	→						Wood: 50 km, steel: 100 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	4.33E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
plant	→							wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	3.33E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
Softwood, allocation correction	→				100% to LTE		adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-3.16E-01	m3	calculated	1	1.00	calculated correction term
Softwood, allocation correction	→				100% to industrial residual softwood			wooden materials	extraction	No	RER	softwood, allocation correction, 2	3.16E-01	m3	calculated	1	1.00	calculated correction term
Hardwood, allocation correction	→				100% to LTE		adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	hardwood, allocation correction, 1	-2.50E-02	m3	calculated	1	1.00	calculated correction term
Hardwood, allocation correction	→				100% to industrial residual hardwood			wooden materials	extraction	No	RER	hardwood, allocation correction, 2	2.50E-02	m3	calculated	1	1.00	calculated correction term

Fig. 10.5 Flows for "laminated timber element, outdoor use, production" and its representation in the ecoinvent database

10 Wood boards based on round wood

General Flow information							Representation in ecoinvent								Uncertainty information			
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
		laminated timber element, outdoor use, production, outputs	→	Laminated timber element, transversally prestressed, for outdoor use, at plant	allocation based on sawn timber planed, naturally dried: 450 CHF/m3; revenue: 450 CHF; revenue process: 465 CHF; alloc. factor (except for steel and transports: 0.97		u=20%	wooden materials	extraction	No	RER	laminated timber element, transversally prestressed, for outdoor use, at plant	1.00E+00	m3				
			→	Industrial residual wood spruce, at LTE-plant (u=20%)	39 CHF/m3; revenue: 14 CHF; revenue process: 465 CHF; alloc. factor: 0.03	estimation after Werner 2002, based on Gautschi 2001 (increased value due to lower water content)	estimation: abatement according to glued laminated timber	wooden materials	extraction	No	RER	industrial residue wood, LTE production, softwood, u=20%, at plant	3.54E-01	m3	calculated			
			→	Industrial residual wood beech, at LTE-plant (u=20%)	47 CHF/m3; revenue: 1 CHF; revenue process: 465 CHF; alloc. factor: 0.0	estimation after Werner 2002, based on Gautschi 2001 (value increased by 1.61 based on wood with u=140% due to lower water content (after Vhe Nr. 407 (2002))	estimation: abatement according to glued laminated timber	wooden materials	extraction	No	RER	industrial residue wood, LTE production, hardwood, u=20%, at plant	2.50E-02	m3	calculated			
			→	Waste heat into air				air	unspecified			Heat, waste	3.60E+01	MJ	calculated	1	1.45	(4,5,1,3,3,5,13)

Fig. 10.5 Flows for "laminated timber element, outdoor use, production" and its representation in the ecoinvent database (cont.)

10 Wood boards based on round wood

General Flow information							Representation in ecoinvent								Uncertainty information				
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Sawn timber spruce rough, at sawmill	→	three layered laminated board, production						wooden materials	extraction	No	RER	sawn timber, softwood, raw, air dried, u=20%, at plant	2.50E+00	m3	Werner 1997	1	1.13	(2,4,2,3,1,1,3)	
Lubricating oil	→							chemicals	organics	No	RER	lubricating oil, at plant	4.20E-03	kg	Werner 1997	1	1.13	(2,4,2,3,1,1,4)	
PVAc adhesive	→							plastics	monomers	No	RER	vinyl acetate, at plant	8.00E+00	kg	Werner 1997	1	1.13	(2,4,2,3,1,1,4)	
Heat from wood furnace, fueled with industrial residual wood	→						8% of wood input	wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 300kW	1.70E+03	MJ	Werner 1997	1	1.13	(2,4,2,3,1,1,1)	
Chips	→						Since the fuel for the furnace is included in the input, the corresponding amount inventoried with the energy module is subtracted. Since the volume of the chips is bulked, it is higher then the difference of input and output volume.	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40%, at plant	-5.36E-01	m3	calculated	1	1.00	calculated	
Diesel burned in building machine	→							construction processes	machinery	No	GLO	diesel, burned in building machine	1.80E+01	MJ	Werner 1997	1	2.02	(2,4,2,3,1,1,5)	
Electricity	→							electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.94E+02	kWh	Werner 1997	1	1.13	(2,4,2,3,1,1,2)	
Transport rail	→						Chemicals: 600 km, wood: 100 km	transport systems	train	No	RER	transport, freight, rail	1.40E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport lorry	→						Chemicals: 100 km, wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	6.83E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
plant	→							wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	3.33E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)	
Softwood, allocation correction	→				100% to 3-layered LB		adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-1.23E+00	m3	calculated	1	1.00	calculated correction term	
Softwood, allocation correction	→				100% to industrial residual wood			wooden materials	extraction	No	RER	softwood, allocation correction, 2	1.23E+00	m3	calculated	1	1.00	calculated correction term	
	→			Three-layered laminated board spruce, at plant	>1500 CHF/m3; revenue process: >1550 CHF; alloc. factor: 0.97		estimation based on price list from Hefti & Grob, Niederhasli	wooden materials	extraction	No	RER	three layered laminated board, at plant	1.00E+00	m3	Werner 1997				
	→			Industrial residual wood spruce, at LB-plant (u=20%)	39 CHF/m3; revenue: 50 CHF; revenue process: 1550 CHF; alloc. factor: 0.03		estimation after Werner 2002, based on Gautschi 2001 (value increased by 1.61 based on wood with u=140% due to lower water content (after Vhe Nr. 407 (2002))	52% of wood input	wooden materials	extraction	No	RER	industrial residue wood, 3-layered LB production, softwood, u=20%, at plant	1.30E+00	m3	Werner 1997			
	→			Waste heat into air					air	unspecified			Heat, waste	6.98E+02	MJ	calculated	1	1.13	(2,4,2,3,1,1,13)

Fig. 10.6 Flows for "three layered laminated board, production" and its representation in the ecoinvent database

10.3 Wood wool board

Wood wool is produced mainly from spruce by machines similar to planing machines. The shavings are 0.03 to 0.5 mm thick and usually 2 mm wide. The input wood has to be dry ($u < 20\text{-}30\%$) and its length usually is around 50 cm.

Wood wool is mainly used as packing material and as raw material for wood wool boards.

Wood wool boards are mineral bond low density ($360\text{-}570 \text{ kg/m}^3$) boards. They are used for light wall elements, as insulation material (thermal ($0.08 < \lambda < 0.14 \text{ W/m}$) and sound) for walls, floors and ceilings or as lost shuttering panels.

10.3.1 Wood wool board in ecoinvent database

The allocation of the burdens to the different outputs of the wood wool production is proposed according to the economic return of the products. Because both, the amount and the value of the industrial residue wood are small compared to those of wood wool, the allocation factor gets zero.

The economic allocation can't respect the mass and energy balance of the products. Therefore, correcting modules are defined that can add or subtract the CO_2 uptake and the resource (wood and solar energy) consumption from nature. They are used as shown in the example in Fig. 5.2:

The background information to the processes are given in Tab. 10.3, flows are shown in:

- wood wool production, $u=20\%$
Outputs:
 - wood wool, $u=20\%$, at plant
 - industrial residue wood, wood wool production, softwood, $u=20\%$, at plant Fig. 10.7
- wood wool boards, cement bonded, at plant Fig. 10.8

The energy consumption of the wood wool board production process is confidential and therefore not shown.

10.3.2 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

There might be considerable variations between different wood wool and wood wool board plants. The processes however are basically the same. Thus overall data quality is thought to be good.

Tab. 10.3 Ecoinvent meta information for the wood wool board production processes

Name	wood wool production, u=20%	wood wool boards, cement bonded, at plant
Location	RER	RER
Infrastructure Process	0	0
Unit	kg	m ³
Data Set Version	2.0	2.0
Included Processes	Includes all processes of wood wool production from industrial softwood (debarked), including air drying (from u=70 - 20%), external and internal transports, suction, air compression, sawmill production	Includes the inputs to the production processes, transports of those inputs. No process emission data are available.
Amount	1	1
Local Name	Holzwoleproduktion, u=20%	Holzwole-Leichtbauplatte, zementgebunden, ab Werk
Synonyms	Fichte, Verpackungsmaterial//Spruce, packaging material	Fichte, Duo-panel//Spruce
General Comment to reference function	Covers high quality wood wool without shives. The multi-output process "wood wool production, u=20%" delivers the two coproducts "wood wool, u=20%, at plant" and "industrial residue wood, wood wool production, softwood, u=20%, at plant". Allocation is based on the over-all proceeds of the process.	density of wood wool boards differs considerably (360-570 kg/m ³)
Start Date	2002	1989
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data from the only Swiss wood wool production plant used for central Europe	Swiss energy data. Production data based on composition from literature. Used for central Europe
Technology text	Medium enterprise technology (2000)	Medium enterprise technology (2000); data stemming from various sources
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	personal communication	Literature, personal communication
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that the technology is representative for central Europe is considered in the uncertainties.	The assumption that the technology is representative for central Europe is considered in the uncertainties.

10 Wood boards based on round wood

General Flow information								Representation in ecoinvent							Uncertainty information				
Input		Process Name		Output	Allocation	source allocation	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Industrial wood spruce debarked, at forest road (u=70%)	→	wood wool production, u=20%					Data from the only Swiss wood wool production plant. Shrinkage from drying is included.	wooden materials	extraction	No	RER	industrial residue wood, softwood, forest-debarked, u=70%, at plant	2.12E-03	m3	Lindner Produktions AG, Wattwil (2002); Hr. Wildnagel; www.lindner.ch ; Fax 071/987 61 59	1	1.12	(1,4,1,3,1,1,3)	
Electricity	→						contains all processes incl. Internal transports, suction, air compression, sawmill production; average years 2000 and 2001	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	5.92E-02	kWh	Lindner Produktions AG, Wattwil (2002); Taschenbuch der Holztechnologie , p. 443	1	1.12	(1,4,1,3,1,1,2)	
Transport rail	→						Wood: 100 km	transport systems	train	No	RER	transport, freight, rail	1.62E-01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport lorry	→						50 km transport distance	transport systems	road	No	RER	transport, lorry >16t, fleet average	8.11E-02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
plant	→						Sawmill used as proxy	wooden materials	extraction	Yes	RER	sawmill	8.48E-10	unit	estimated	1	3.36	(4,5,2,3,4,5,9)	
Softwood, allocation correction	→				100% to wood wool		adds / subtracts the amount of CO2 uptake, ressource consumption and embodied energy that is lacking / too much according to economic allocation	wooden materials	extraction	No	RER	softwood, allocation correction, 1	-9.26E-05	m3	calculated	1	1.00	calculated correction term	
Softwood, allocation correction	→				100% to saw dust			wooden materials	extraction	No	RER	softwood, allocation correction, 2	9.26E-05	m3	calculated	1	1.00	calculated correction term	
				→	Wood wool, at plant (u=20%)	2.2 CHF/kg; revenue: 2.2 CHF; revenue process: 2.2012 CHF; alloc. factor: 1.0	Personal communication, Mr. Wildhaber, Linder AG		wooden materials	extraction	No	RER	wood wool, u=20%, at plant	1.00E+00	kg	Lindner Produktions AG, Wattwil (2002)			
				→	Sawdust spruce, at woodwool-plant (u=20%)	13 CHF/m3; revenue: 0.0012 CHF; revenue process: 2.2012 CHF; alloc. factor: 0	estimation after Werner 2002, based on Gautschi 2001 (value increased by 1.61 based on wood with u=140% due to lower water content (after Vhe Nr. 407 (2002))	around 5% input material; used mainly by farmers	wooden materials	extraction	No	RER	industrial residue wood, wood wool production, softwood, u=20%, at plant	9.26E-05	m3	Lindner Produktions AG, Wattwil (2002)			
				→	Waste heat into air				air	unspecified			Heat, waste	5.92E-02	MJ	Lindner Produktions AG, Wattwil (2002)	1	1.12	(1,4,1,3,1,1,2)

Fig. 10.7 Flows for "wood wool production, u=20%" and its representation in the ecoinvent database

10 Wood boards based on round wood

General Flow information					Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Location	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Wood wool, at plant (u=20%)	➔	wood wool boards, cement bonded, at plant			120-180 kg	wooden materials	extraction	No	RER	wood wool, u=20%, at plant	1.50E+02	kg	Schniewind 1989, p. 197 ff.	1	1.24	(1,4,4,3,1,1,4)
Portland Cement	➔				180-250 kg	construction materials	binder	No	CH	cement, unspecified, at plant	2.15E+02	kg	Schniewind 1989, p. 197 ff.	1	1.24	(1,4,4,3,1,1,4)
Process and cooling water	➔				200-290 kg; evaporates partly during drying	resource	in water			Water, cooling, unspecified natural origin	2.45E-01	m3	Schniewind 1989, p. 197 ff.	1	1.24	(1,4,4,3,1,1,4)
Organic chemicals	➔				Additives: 6-8 kg; sometimes, no organic chemicals but waterglas is used	chemicals	organics	No	GLO	chemicals organic, at plant	7.00E+00	kg	Schniewind 1989, p. 197 ff.	1	1.24	(1,4,4,3,1,1,4)
Electricity	➔				confidential	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	confidential	kWh	CEWAG Düringen, Herr Kurzo, Tel. 026 492 94 50	1	1.12	(1,4,1,3,1,1,2)
Thermal heat from oil firing extra light	➔				confidential	oil	heating systems	No	RER	heat, light fuel oil, at industrial furnace 1MW	confidential	MJ	CEWAG Düringen, Herr Kurzo, Tel. 026 492 94 50	1	1.12	(1,4,1,3,1,1,1)
Transport rail	➔				Chemicals: 600 km, cement: 100 km, wood: 0 km	transport systems	train	No	RER	transport, freight, rail	2.57E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Transport lorry	➔				Chemicals & cement: 100 km, wood: 0 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.22E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
plant	➔					wooden materials	extraction	Yes	RER	wooden board manufacturing plant, cement bonded boards	4.00E-07	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
	➔			Woodwool boards, cement bonded, at plant		medium density 450 kg/m3 (360-570 kg/m3)	wooden materials	extraction	No	RER	wood wool boards, cement bonded, at plant	1.00E+00	m3			
	➔		Waste heat into air			air	unspecified			Heat, waste	3.29E+01	MJ	calculated	1	1.12	(1,4,1,3,1,1,13)

Fig. 10.8 Flows for "wood wool boards, cement bonded, at plant" and its representation in the ecoinvent database

11 Industrial residual wood

Wood is a naturally grown material; its shape and properties do not a priori suit industrial production processes. Consequently, many production processes throughout the wood processing chain generate industrial residual wood. Such industrial wood varies in shape, humidity, available quantities, etc.

If industrial residual wood is used as input in processes different from the one which has generated it, usually no information on the source of this wood is available. Therefore datasets for average industrial residual wood are calculated.

11.1.1 Industrial residue wood mixes in ecoinvent database

The wood chain considered in ecoinvent generates a variety of industrial residual wood (Tab. 11.1).

Tab. 11.1 Industrial residual wood generated in the wood chain in Ecoinvent 2000

Process generating industrial residual wood	Name of industrial residue wood
SAWMILL	
sawing / debarking, hardwood, at plant	industrial residue wood, hardwood, including bark, u=70%, at plant
sawing / debarking, softwood, at plant	industrial residue wood, softwood, plant-debarked, u=70%, at plant
sawing, softwood, forest-debarked, at plant	industrial residue wood, softwood, forest-debarked, u=70%, at plant
PLANING MILL	
planing, hardwood, air / kiln dried, u=10%	industrial residue wood, from planing, hard, air/kiln dried, u=10%, at plant
planing, hardwood, kiln dried, u=10%	industrial residue wood, from planing, hardwood, kiln dried, u=10%, at plant
planing, softwood, air dried, u=20%	industrial residue wood, from planing, softwood, air dried, u=20%, at plant
planing, softwood, kiln dried, u=10%	industrial residue wood, from planing, softwood, kiln dried, u=10%, at plant
WOOD BOARDS PRODUCTION BASED ON ROUNDWOOD	
plywood, indoor use, production	industrial residue wood, plywood prod., indoor use, hardwood, u=20%, at plant
plywood, outdoor use, production	industrial residue wood, plywood prod., outdoor use, hardwood, u=20%, at plant
glued laminated timber, indoor use, production	industrial residue wood, GLT production, indoor use, u=10%, at plant
glued laminated timber, outdoor use, production	industrial residue wood, GLT production, outdoor use, u=10%, at plant
laminated timber element, outdoor use, production	industrial residue wood, LTE production, softwood, u=20%, at plant
	industrial residue wood, LTE production, hardwood, u=20%, at plant
three layered laminated board, production	industrial residue wood, 3-layered LB production, softwood, u=20%, at plant
wood wool production, u=20%	industrial residue wood, wood wool production, softwood, u=20%, at plant

For quantities and allocation procedures applied to these particular processes, see the corresponding process descriptions.

The data used to calculate the average composition of industrial residual hardwood and softwood are given in Tab. 11.2 and Tab. 11.3 respectively. It is derived from production volume. If necessary, a split of 28% of hardwood and 72% of softwood is assumed. Data for Switzerland are taken. An average moisture content of 40% is assumed. Given the huge amount of industrial residual wood from sawmills, residual industrial wood from board production is neglected.

Tab. 11.2 Industrial residual hardwood generated: percentage to total industrial residual hardwood

	Average timber input m ³ /a	% residual hardwood on total output	Intermediate average residual quantities m ³ /a	Further specifications	Average quantities residual hardwood m ³ /a	Percentage to total industrial residual hardwood
Sawmill	644'000	37.5%			241'500	94.7%
Planing mill	32'760	15%	4914	50% air-dried/ technically dried 50% technically dried	2457 2457	0.95% 0.95%
Plywood	27'273	32%	8727	50% MUF-adhesive 50% UF-adhesive	4364 4364	1.7% 1.7%
Laminated timber element	neglected	neglected			0	0%

Tab. 11.3 Industrial residual softwood generated: percentage to total industrial residual softwood

	Average timber input m ³ /a	% residual softwood on total output	Intermediate average residual quantities m ³ /a	Further specifications	Average quantities residual softwood m ³ /a	Percentage to total industrial residual softwood
Sawmill	1'656'000	41.5%			687240	98.2%
Planing mill	32'760	15%	12636	50% air-dried/ technically dried 50% technically dried	6318 6318	0.9% 0.9%
Glued laminated timber	neglected	neglected			0	0%
Laminated timber element	neglected	neglected			0	0%
three layered laminated board	neglected	neglected			0	0%
Wood wool	neglected	neglected			0	0%

The background information to the mixes are given in Tab. 11.4, flows are shown in:

- industrial residue wood, mix, hardwood, u=40%, at plant Fig. 11.1
- industrial residue wood, mix, softwood, u=40%, at plant Fig. 11.2

11.1.2 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Assuming that the production mix for the different residual wood fractions in Switzerland is similar to that of central Europe, data quality is good.

Tab. 11.4 Ecoinvent meta information for industrial residue wood mixes

Name	industrial residue wood, mix, hardwood, u=40%, at plant	industrial residue wood, mix, softwood, u=40%, at plant
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Mix of the different industrial residual hardwood fractions in proportion to their production in Switzerland	Mix of the different industrial residual softwood fractions in proportion to their production in Switzerland
Amount	1	1
Local Name	Industrierestholz, Mix, Laubholz, u=40%, ab Werk	Industrierestholz, Mix, Nadelholz, u=40%, ab Werk
Synonyms	Abfallholz//Buche//Beech	Abfallholz//Fichte//Spruce
General Comment to reference function	refers to the actual volume of the wood (not bulked); does only include industrial residual wood from processes considered in ecoinvent; since this process is dominated by industrial residual wood from sawmills, this effect might not be to severe.	refers to the actual volume of the wood (not bulked); does only include industrial residual wood from processes considered in ecoinvent; since this process is dominated by industrial residual wood from sawmills, this effect might not be to severe.
Start Date	2002	2002
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Switzerland used for central Europe	Data for Switzerland used for central Europe
Technology text	Mix	Mix
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	survey	survey
Extrapolations	see Geography	see Geography
Uncertainty Adjustments	The assumption that the mix is representative for central Europe is considered in the uncertainties.	The assumption that the mix is representative for central Europe is considered in the uncertainties.

General Flow information					Representation in ecoinvent									Uncertainty information			
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Industrial residual wood beech incl. bark, at sawmill (u=70%)	→	industrial residue wood, mix, hardwood, u=40%, at plant															
Industrial residual wood beech planed, technically dried, at planing mill (u=10%)	→					wooden materials	extraction	No	RER	industrial residue wood, hardwood, including bark, u=70%, at plant	9.47E-01	m3	own calculation based on internal survey	1	1.31	(4,3,1,3,1,5,3)	
Industrial residual wood beech planed, naturally/technically dried, at planing mill (u=10%)	→					wooden materials	extraction	No	RER	industrial residue wood, from planing, hardwood, kiln dried, u=10%, at plant	9.63E-03	m3	own calculation based on internal survey	1	1.31	(4,3,1,3,1,5,3)	
Industrial residual wood beech, at LTE-plant (u=20%)	→					wooden materials	extraction	No	RER	industrial residue wood, from planing, hard, air/kiln dried, u=10%, at plant	9.63E-03	m3	own calculation based on internal survey	1	1.31	(4,3,1,3,1,5,3)	
Industrial residual wood beech, at plywood-plant I (u=20%)	→					neglected											
Industrial residual wood beech, at plywood plant II (u=20%)	→						wooden materials	extraction	No	RER	industrial residue wood, plywood prod., indoor use, hardwood, u=20%, at plant	1.71E-02	m3	own calculation based on internal survey	1	1.31	(4,3,1,3,1,5,3)
Industrial residual wood beech, at plywood plant II (u=20%)	→					wooden materials	extraction	No	RER	industrial residue wood, plywood prod., outdoor use, hardwood, u=20%, at plant	1.71E-02	m3	own calculation based on internal survey	1	1.31	(4,3,1,3,1,5,3)	
			→	Industrial residual wood hardwood (u=40%), at plant	Average mix calculated from the Swiss consumption mix of main products.	wooden materials	extraction	No	RER	industrial residue wood, mix, hardwood, u=40%, at plant	1.00E+00	m3					

Fig. 11.1 Flows for "industrial residue wood, mix, hardwood, u=40%, at plant" and its representation in the ecoinvent database

11 Industrial residual wood

General Flow information					Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Remarks	Category	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Industrial residual wood spruce, at sawmill (debarked in forest) (u=70%)	→	industrial residue wood, mix, softwood, u=40%, at plant			Average mix calculated from the Swiss consumption mix of main products	wooden materials	extraction	No	RER	industrial residue wood, softwood, forest-debarked, u=70%, at plant	9.82E-01	m3	own calculation based on internal survey	1	1.31	(4,3,1,3,1,5,3)
Industrial residual wood spruce, naturally dried, at planing mill (u=20%)	→					wooden materials	extraction	No	RER	industrial residue wood, from planing, softwood, air dried, u=20%, at plant	9.03E-03	m3	own calculation based on internal survey	1	1.31	(4,3,1,3,1,5,3)
Industrial residual wood spruce, technically dried, at planing mill (u=10%)	→					wooden materials	extraction	No	RER	industrial residue wood, from planing, softwood, kiln dried, u=10%, at plant	9.03E-03	m3	own calculation based on internal survey	1	1.31	(4,3,1,3,1,5,3)
Industrial residual wood spruce, at GLT-plant I (u=10%)	→				neglected											
Industrial residual wood spruce, at GLT-plant II (u=10%)	→				neglected											
Industrial residual wood spruce, at LTE-plant (u=20%)	→				neglected											
Industrial residual wood spruce, at woodwool-plant (u=20%)	→				neglected											
			→	Industrial residual wood softwood (u=40%), at plant	Average mix calculated from the Swiss consumption mix of main products	wooden materials	extraction	No	RER	industrial residue wood, mix, softwood, u=40%, at plant	1.00E+00	m3				

Fig. 11.2 Flows for "industrial residue wood, mix, softwood, u=40%, at plant" and its representation in the ecoinvent database

12 Boards based on industrial residual wood

This chapter is mainly based on (Nimz et al. 1997) and (Kollmann et al. 1975).

To overcome the pronounced anisotropy of solid wood, it must be cut up into small pieces; then the pieces can be oriented to a desired fibre direction. The following wood-based particles have achieved economic importance in the production of wood-based construction materials:

- Chips for defibration and grinding
- Shavings from cutting, peeling, and sawing
- Strands
- Wafers or flakes
- Fibres
- Wood wool

In industrial production, these particles are formed with a spectrum of dimensions and shapes. In practice, they must be sieved or classified before further utilisation. After drying and application of adhesive, the properties of the wood-based construction material can be influenced by targeted orientation of the particles during mat forming and by moulding.

12.1 Particle board

In 1902, Henry Watson of Indiana had the idea of making particle board from waste wood. Unfortunately, no adhesives available at that time could allow industrial realisation of this idea.

Only when the chemical industry made suitable adhesives available with the development of amino resins, could the industrial production of particle board start in Germany in 1941. After the end of World War II, the development of particle board technology made rapid advances in Germany and subsequently in the whole of Europe. Important contributions came from the Swiss F. Fahrni and the German W. Klauditz. In 1991, particle board was the most important industrially produced wood-based construction material, with an annual world-wide production of ca. 50 million m³.

Various types of particle board are available: flat pressed particle board, extruded particle board, and moulded particle board.

12.1.1 Types of particle board

Flat pressed particle board

In flat pressed particle board, the particles are arranged predominantly parallel to the board surface and bonded. Most of the particle board produced world-wide belongs to this group. The wafer boards, flake boards, strand boards and oriented structural boards (OSB) produced widely in North America are, in principle, flat pressed particle boards. Although wafer and flake boards are made up predominantly of (decorative) wood particles with large surfaces and a thickness up to 0.5 mm, strand boards are made from strands with a high ratio of slenderness having a thickness up to 0.5 mm and a length of 7.5-15 cm. The production, transport, gluing, and mat-forming of these particles require special machines.

Extruded particle board

These boards are produced by extrusion. The process was developed in the 1960s by O. Kreibbaum and is still termed Okal process. Particles are arranged predominantly at right angles to the board surface. Short particles or granulated wood are mainly used. Adhesive-coated particles arrive in portions in the heated press, are compressed by using a pressure piston driven by an eccentric flywheel, and are extruded through the hot zone of the press. A continuous band of board is formed as it leaves the press. Heating of the board is accelerated by heated rods that project into the press. The heated rods them-

selves form tube-shaped hollows in the board. Extruded particle board is generally used only in the planked or veneered state because of the low flexural strength resulting from the production process. The production of profiles constitutes a further variant of the extrusion process.

Moulded Objects

The production of moulded particle board was described in 1948 in a Swedish patent. In the same year, work began on the Thermodyne process, in which natural wood constituents are activated as binders at pressing temperatures up to 290 °C and pressures up to 300 bar. The process was not successful economically because of the extreme operating conditions.

For particle board, low-value wood and industrial waste wood (e.g., remainders, slabs, splinters, and shavings from planing and sawing) are used. In principle, all types of wood are suitable for making particle board. However, coniferous wood is used for most of the production in Europe and North America, which accounts for 75% of the total world-wide particle board production. A combination of hard and soft types of wood is frequently used; the fact that these types of wood affect the properties of the board because of their differing sorption behaviour is taken into account.

Wood bark used to be separated and incinerated or dumped, because the different structure of the bark generally required greater compression of the particles and the use of larger quantities of adhesive resins. Nowadays, however, the use of unpeeled wood has become common, even for the outer layer. Addition of 6-15% bark to middle-layer particles does not affect board quality significantly. Larger quantities of bark impair not only the appearance of the board, but also its strength and sorption properties. Excess bark in particle board factories is incinerated, and the waste gas is used for drying the particles.

Flat pressed particle board is now generally produced in several layers. The density of the individual layers and their construction determine the mechanical properties of the board. Light particle board has a density up to 500 kg/m³. Most of the board produced is, however, in the density range of 600-800 kg/m³. The thickness of the board ranges from 3 to 50 mm.

Several additives and adhesives are used in particle board production: Amino Resins (UF, MF, and MUF), Phenol-formaldehyde resins, Polymeric Methylene Diphenylene Isocyanate, Mineral Binders, etc.

12.1.2 Resins in particle boards

Amino Resins (UF, MF, and MUF)

Urea-formaldehyde adhesive resins are used as 55-72% aqueous solutions or as powders. For gluing particles, the resins used now predominantly have a basic U:F molar ratio of 1:1.0 to 1:1.3 to limit formaldehyde emissions. Catalysts based on ammonium salts or acids are used to accelerate the curing process. The use of resins with increasingly lower formaldehyde content finally led to a lowering of the cross-link density and thus to impaired quality of adhesion. This was corrected by reinforcing the UF resins with melamine. To adhere to the strict limiting values for formaldehyde emissions in many countries, a special process for low-emission boards was developed by using reactive formaldehyde-rich resins and formaldehyde scavengers. Urea-formaldehyde is used in quantities of 6-12% (w/w) solid resin, based on absolutely dry wood. UF-glued particle board is produced mainly for interior use and for furniture.

Higher-quality adhesions are achieved with melamine-formaldehyde, phenol-formaldehyde, or polyisocyanate resins. Because of their chemical similarity to UF resins, MF resins are mixed or co-condensed with them. Melamine-urea-formaldehyde and phenol-containing MUPF resins with 15-25% melamine and up to 5% phenol are used for particle board that is exposed to particularly high moisture levels. The dosage of resin is increased to 12-15%, based on the dry wood particles.

Phenol-formaldehyde resins

Phenol-formaldehyde resins are high-value products, particularly because of their resistance to hydrolysis. They are used as aqueous alkaline solutions with a P:F molar ratio of 1:1.5 to 1:2.5. The final alkali content is 3-10%, depending on the degree of condensation of the resin. Phenol-formaldehyde resins are generally less reactive than amino resins. This means that, in practice, longer pressing times must be used for processing PF resins compared with amino resins. Alkaline phenolic resins dye particle board light brown and increase its hygroscopic nature. PF-glued particle board produces no significant formaldehyde emissions. It is used for exterior applications. The usual dosage of resin is 8-12%.

Polymeric Methylene Diphenylene Isocyanate

Since 1980 adhesives based on PMDI have increasingly been used. They consist of a mixture of monomeric, polymeric, and oligomeric diisocyanates, which are capable of reacting with the cellulose, lignin, and moisture in wood. The quality of adhesions using PMDI is comparable to that using PF resins. A significant disadvantage of isocyanate adhesives is their tendency to stick to metals. PMDI can therefore not be used without release agents. Handling PMDI adhesives requires particular care because of their high reactivity with water and bases. However, they are very economical. Adequate board strengths are achieved with 3-5% adhesive.

Mineral Binders

Special boards for building purposes are also produced using inorganic binders. These include Portland cement, magnesia cement (Sorel cement), and gypsum.

Additives

For repelling water and delaying swelling during particle board production with synthetic resins, 0.1-1.5% paraffin, based on dry wood, is generally added. Paraffin is mixed with the adhesive liquor as dispersion or applied to the particles as a finely dispersed melt.

12.1.3 Production of particle board

This section describes the particle board production process.

The wood to be milled should have a moisture level of ca. 50% and be free of metallic or mineral substances. Low moisture levels increase the throughput of the size reduction machine, but also increase dust formation and fire hazards.

Chipping

Chipping machines of various types are used for size reduction. Machines with cutting tools fixed in a carrier disk or drum, together with spurs, are preferred to process lengthwood (round wood). By pressing the wood onto the rotating blade carriers, uniformly thick particles are formed, which can be cut up further in a subsequent mill if desired. The most powerful chipping machines of this type are knife shaft machines with an operating width of 2 m. For making larger particles, knife ring machines are used, which have combined cutting and striking tools. To cut up the particles further, hammer, wing beater, and toothed disk mills are used. The target of the size reduction process is the production of a mixture of particles that are as uniform as possible in shape and thickness. The geometry of the particles affects both the further processing (distribution and dispersion of glue) and the mechanical properties of the board. Since particle board for use in surface coating is produced mainly in three or five layers, the aim of particle preparation is to produce different mixtures of particles. Fine particles for the outer and coarse ones for the middle layer are obtained by air classifying and screening. For the production of particle board used in building, large proportions of fine particles and dust are undesirable.

Drying

To process the particles further, residual moisture levels of 1-3% are necessary. High moisture levels, together with the water applied with the adhesive resin, generally cause steam cracking during the pressing process. Drying the particles is therefore indispensable. Drying follows the rules of convection drying. In the first part of the drying process, wood moisture decreases almost linearly. The second part of the drying process is determined by vapour diffusion. With particles of varying thickness and size, uniform drying is difficult to achieve. Separate drying of the coarse middle-layer and fine outer-layer materials is therefore necessary. To dry the particles jet dryers, tubular dryers, stream dryers, and drum dryers are used.

Gluing

Glue blending of the particles occurs continuously in flow mixers to achieve as uniform a distribution of the adhesive on the surface of the particles as possible. Dried particles are fed to the mixer via a belt scale. The necessary quantity of adhesive is dosed volumetrically or gravimetrically with electronic control. In the mixer, adhesive is rubbed on the surface of the particles or applied by using spray jets. The efficiency of the mixer depends on the design of the mixing drum and the conveying and braking tools. Because of the friction heat produced, the mixing vessel must be cooled with water.

Forming Mats

Glue-blended particles are fed from the storage hopper into the dosing hopper of the forming station by using conveyor screws or belts.

By controlling the volume of the glued chips a uniform flow of particles is produced, which is deposited onto a moving forming belt via a toothed feed roller. The falling particles receive an impulse from a spreading roller, resulting in separation of the coarse and fine particles. Commonly, the mat is formed by using air classifying. In this process, air is blown against a curtain of falling particles so that the fine particles are separated from the coarse ones. Here also, classified particles fall onto a conveyor belt below. With two blowers set opposite one another, mats can be formed that have coarse particles in the centre and finer particles on the outside. For reducing thickness, the mats are pre-compressed in a cold press and then passed to a hot press for activation of the adhesive.

In hot pressing the blank is compressed to the desired final thickness and the adhesive is irreversibly cured. Both continuously operating and batch presses are used.

Batch Presses

The output of presses for batch processing is limited by the pressing and cycle times. With one-daylight presses, high pressing temperatures (200-230 °C) are used to shorten the setting time of the adhesive. Multi-daylight presses have up to 20 daylights and are operated at <200 °C. Automatic charging and emptying of a multi-daylight press require complex machinery. The tolerance in thickness of the boards produced by using the batch press is so great that calibration by sanding is indispensable.

Continuously Operating Presses

In larger particle board plants, continuously operating presses are mainly used. With this technique, boards are produced with significantly better dimensional stability in thickness than in batch production. Various continuously operating systems exist:

Roller Presses. In roller presses the application of pressure and transportation is effected by a bed of rods. The operating temperature is 220-250 °C. Continuously operating roller presses are usually designed for capacities of more than 400 m³ of board per day. The upper production costs are 8% less than those for batch presses because of the lower energy consumption.

Gliding Film Presses. In gliding film presses the pressure is transmitted by an oil film (Hydro Dyn process of the Bison company) or by air cushions (Hymmen company).

Calender Presses. In calender presses pressure is applied to the blank by pressing it with steel bands on a heated cylinder (calender). The system is particularly suitable for producing thin particle board.

Extrusion Presses. In extrusion presses the particles are extruded continuously between heated plates or rods. Profiles and tubular boards are produced by this method.

Steam Injection Presses. Recently, heating the mats has been accelerated by using steam injection presses. The steam is injected immediately before pressure is applied. The process is particularly suitable for thick boards, which are difficult to heat through. Since 1989, it is being developed further by Siempelkamp.

Presses are predominantly heated indirectly using heat-transfer oil as the energy carrier. Experience with the use of microwave energy is extensive. For cost reasons, it is now used only for warming the mats in pre-presses. The necessary pressing time is determined by the reactivity of the adhesive system and the pressing temperature. The specific pressure of 20-50 bar is decreased in steps as the spring back forces of the compressed mat decrease and the adhesive takes effect.

After treatment

After hot pressing, particle board is cooled and conditioned by storage for several days. This process leads to further setting of the adhesive so that the internal bonding of the board increases and the thickness swelling values decrease. The board is standardised by sanding.

Europe is the area with the highest particle board production. Here, efforts have been made to harmonise the standard requirements and tests. In EN 312, parts 1-7, the requirements are categorised according to the areas of use. In the United States, a differentiation is made between interior and exterior use in commercial standards CS 236. In the Japanese Industrial Standard JISA 5908, boards are classified according to their flexural strengths.

12.1.4 Particle board inecoinvent database

The background information to the processes are given in Tab. 12.1, flows are shown in:

- particle board, indoor use, at plant Fig. 12.1
- particle board, outdoor use, at plant Fig. 12.2

12.1.5 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Specific particle boards can contain different amounts of glue. The data given represent an average situation as good as possible. Specific products can differ considerably.

Tab. 12.1 Ecoinvent meta information for the particle board production processes

Name	particle board, indoor use, at plant	particle board, outdoor use, at plant
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Includes the inputs to the production processes, transports of those inputs and the process emissions as far as they are available.	Includes the inputs to the production processes, transports of those inputs and the process emissions as far as they are available.
Amount	1	1
Local Name	Spanplatte, Innenanwendung, ab Werk	Spanplatte, Aussenanwendung, ab Werk
Synonyms		
General Comment to reference function		
Start Date	1986	1986
End Date	2000	2000
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Germany used for central Europe	Data for Germany used for central Europe
Technology text	Average German Technology (1996)	Average German Technology (1996)
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that the technology is representative for central Europe is considered in the uncertainties.	The assumption that the technology is representative for central Europe is considered in the uncertainties.

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
Industrial wood beech, at forest road (u=80%)	particle board, indoor use, at plant, inputs		Amount reduced by 12% because bark is not included.	wooden materials	extraction	No	RER	industrial wood, hardwood, under bark, u=80%, at forest road	1.28E-01	m3	Wegener et al. (1994: 68ff); Ressel (1986)	1	1.13	(1,3,3,3,1,1,3)
Ammonia			Ammoniumchlorid 2 mass% on binding medium (accounted as ammonia and HCl)	chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	6.40E-01	kg	estimated	1	2.07	(3,3,1,3,5,5,3)
Paraffin			Paraffin 2 mass% of wood mass u=0%	chemicals	organics	No	RER	paraffin, at plant	1.10E+01	kg	estimated	1	2.07	(3,3,1,3,5,5,3)
Hydrochloric acid			Ammoniumchlorid 2 mass% on binding medium (accounted as ammonia and HCl)	chemicals	inorganics	No	RER	hydrochloric acid, 30% in H2O, at plant	1.36E+00	kg	estimated	1	2.07	(3,3,1,3,5,5,3)
Natural gas burned in industrial furnace >100 kW Euro				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.54E+02	MJ	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,1)
Industrial wood spruce, at forest road (u=140%)			Spruce (0.157 m3), pine (0.079 m3); Amount reduced by 10% because bark is not included.	wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=140%, at forest road	2.15E-01	m3	Wegener et al. (1994: 68ff); Ressel (1986)	1	1.13	(1,3,3,3,1,1,3)
Urea-formaldehyde 100%			Binding medium 9 mass% of wood mass u=0% (urea-formaldehyde)	paintings	production	No	RER	urea formaldehyde resin, at plant	5.10E+01	kg	estimated	1	2.07	(3,3,1,3,5,5,3)
Light fuel oil low sulfur burned in boiler 1 MW				oil	heating systems	No	RER	heat, light fuel oil, at industrial furnace 1MW	8.61E+01	MJ	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,1)
Heavy fuel oil high sulfur burned in boiler 1 MW				oil	heating systems	No	RER	heat, heavy fuel oil, at industrial furnace 1MW	8.61E+01	MJ	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,1)
Industrial residual wood hardwood (u=40%), at plant				wooden materials	extraction	No	RER	industrial residue wood, mix, hardwood, u=40%, at plant	2.17E-01	m3	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,3)
Industrial residual wood softwood (u=40%), at plant				wooden materials	extraction	No	RER	industrial residue wood, mix, softwood, u=40%, at plant	8.23E-01	m3	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,3)
Residual wood chips from process burned in 300kW furnace				wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 300kW	1.10E+03	MJ	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,1)
Chips			Since the fuel for the furnace is included in the input, the corresponding amount inventoried with the energy module is subtracted. Since the volume of the chips is bulked, it is higher than the difference of input and output volume.	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40%, at plant	-3.47E-01	m3	calculated	1	1.00	calculated
Electricity medium voltage - at grid UCTE				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.04E+02	kWh	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,2)
Transport lorry			Chemicals: 100 km, wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	6.13E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Transport rail			Chemicals: 600 km, wood: 100 km	transport systems	train	No	RER	transport, freight, rail	1.48E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Process and cooling water				resource	in water			Water, cooling, unspecified natural origin	3.04E-01	m3	Frühwald et al. 2000	1	1.08	(1,3,2,3,1,1,4)
plant				wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	3.33E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)

Fig. 12.1 Flows for "particle board, indoor use, at plant" and its representation in the ecoinvent database

General Flow information					Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
particle board, indoor use, at plant, outputs	→	Particle board, for dry applications, at plant	Density 680 kg/m ³	wooden materials	extraction	No	RER	particle board, indoor use, at plant	1.00E+00	m ³					
	→	Waste heat into air		air	unspecified			Heat, waste	3.75E+02	MJ	calculated	1	1.13	(1,3,3,3,1,1,13)	
	→	Formaldehyde into air	not inventoried: 50 g Formaldehyde during service life (according to Marutzky 1995, personal communication)	air	unspecified			Formaldehyde	3.00E-03	kg	Frühwald et al. 2000	1	2.01	(1,3,2,3,1,1,23)	
	→	Hydrocarbons (HC)		air	unspecified			NM VOC, non-methane volatile organic compounds, unspecified origin	1.66E-01	kg	Wegener et al. 1994	1	2.02	(1,3,3,3,1,1,23)	
	→	Dust	total dust: 0.078 kg	air	unspecified			Particulates, < 2.5 µm	3.90E-03	kg	total Dust: Frühwald et al. 2000	1	3.11	(3,3,1,3,3,5,27)	
	→	Dust		air	unspecified			Particulates, > 2.5 µm, and < 10 µm	3.51E-02	kg	Size fractions estimated	1	2.11	(3,3,1,3,3,5,26)	
	→	Dust		air	unspecified			Particulates, > 10 µm	3.90E-02	kg		1	1.64	(3,3,1,3,3,5,25)	
	→	Wastewater to treatment	0.196 kg BOD / m ³ waste water	waste management	wastewater treatment	No	CH	treatment, particle board production effluent, to wastewater treatment, class 3	3.60E-02	m ³	Frühwald et al. 2000	1	1.51	(1,3,2,3,1,1,32)	

Fig. 12.1 Flows for "particle board, indoor use, at plant" and its representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment	
Industrial wood beech, at forest road (u=80%)	particle board, outdoor use, at plant, inputs		Amount reduced by 12% because bark is not included.	wooden materials	extraction	No	RER	industrial wood, hardwood, under bark, u=80%, at forest road	1.28E-01	m3	Wegener et al. (1994: 68ff); Ressel (1986)	1	1.13	(1,3,3,3,1,1,3)	
Ammonia			Ammoniumchlorid 2 mass% on binding medium (accounted as ammonia and HCl)	chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	6.40E-01	kg	estimated	1	2.07	(3,3,1,3,5,5,3)	
Paraffin			2 mass% auf, atro	chemicals	organics	No	RER	paraffin, at plant	1.10E+01	kg	estimated	1	2.07	(3,3,1,3,5,5,3)	
Hydrochloric acid			Ammoniumchlorid 2 mass% on binding medium (accounted as ammonia and HCl)	chemicals	inorganics	No	RER	hydrochloric acid, 30% in H2O, at plant	1.36E+00	kg	estimated	1	2.07	(3,3,1,3,5,5,3)	
Natural gas burned in industrial furnace >100 kW Euro				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.54E+02	MJ	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,1)	
Industrial wood spruce, at forest road (u=140%)			Spruce (0.157 m3), pine (0.079 m3); Amount reduced by 10% because bark is not included.	wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=140%, at forest road	2.15E-01	m3	Wegener et al. (1994: 68ff); Ressel (1986)	1	1.13	(1,3,3,3,1,1,3)	
Phenole-formaldehyde resin 100%			Binding medium 9 mass% of wood u=0% (Phenole-formaldehyde); assumed quantities as UF-resin for particle board for dry applications	paintings	production	No	RER	phenolic resin, at plant	5.10E+01	kg	estimated	1	2.07	(3,3,1,3,5,5,3)	
Light fuel oil low sulfur burned in boiler 1 MW				oil	heating systems	No	RER	light fuel oil, burned in industrial furnace 1MW, non-modulating	8.61E+01	MJ	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,1)	
Heavy fuel oil high sulfur burned in boiler 1 MW				oil	heating systems	No	RER	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	8.61E+01	MJ	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,1)	
Industrial residual wood hardwood (u=40%), at plant				wooden materials	extraction	No	RER	industrial residue wood, mix, hardwood, u=40%, at plant	2.17E-01	m3	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,3)	
Industrial residual wood softwood (u=40%), at plant				wooden materials	extraction	No	RER	industrial residue wood, mix, softwood, u=40%, at plant	8.23E-01	m3	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,3)	
Residual wood chips from process burned in 300kW furnace				wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 300kW	1.10E+03	MJ	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,1)	
Chips			Since the fuel for the furnace is included in the input, the corresponding amount inventoried with the energy module is subtracted. Since the volume of the chips is bulked, it is higher then the difference of input and output volume.	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40%, at plant	-3.47E-01	m3	calculated	1	1.00	calculated	
Electricity medium voltage - at grid UCTE				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.04E+02	kWh	Wegener et al. 1994	1	1.13	(1,3,3,3,1,1,2)	
Transport lorry			Chemicals: 100 km, wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	6.13E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport rail			Chemicals: 600 km, wood: 100 km	transport systems	train	No	RER	transport, freight, rail	1.48E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Process and cooling water				resource	in water			Water, cooling, unspecified natural origin	3.04E-01	m3	Frühwald et al. 2000	1	1.08	(1,3,2,3,1,1,4)	
plant				wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	3.33E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)	

Fig. 12.2 Flows for "particle board, outdoor use, at plant" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Cate gory	Sub category	Infra struc ture	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
particle board, outdoor use, at plant, outputs		Particle board, for wet applications, at plant	Density 680 kg/m3	wooden materials	extraction	No	RER	particle board, outdoor use, at plant	1.00E+00	m3				
		Waste heat into air		air	unspecified			Heat, waste	3.75E+02	MJ	calculated	1	1.13	(1,3,3,3,1,1,13)
		Formaldehyde into air	Formaldehyde emission during service life is not included (50 g according to Marutzky 1995, personal communication)	air	unspecified			Formaldehyde	3.00E-03	kg	Frühwald et al. 2000	1	2.01	(1,3,2,3,1,1,23)
		Hydrocarbons (HC)		air	unspecified			NM/VOC, non-methane volatile organic compounds, unspecified origin	1.66E-01	kg	Wegener et al. 1994	1	2.02	(1,3,3,3,1,1,23)
		Dust	total dust: 0.078 kg (Frühwald et al. 2000)	air	unspecified			Particulates, < 2.5 µm	3.90E-03	kg	total Dust: Frühwald et al. 2000	1	3.11	(3,3,1,3,3,5,27)
		Dust		air	unspecified			Particulates, > 2.5 µm, and < 10µm	3.51E-02	kg	Size fractions estimated	1	2.11	(3,3,1,3,3,5,26)
		Dust		air	unspecified			Particulates, > 10 µm	3.90E-02	kg		1	1.64	(3,3,1,3,3,5,25)
		Wastewater to treatment	0.196 kg BOD / m3 waste water	waste management	wastewater treatment	No	CH	treatment, particle board production effluent, to wastewater treatment, class 3	1.90E-01	m3	Frühwald et al. 2000	1	1.51	(1,3,2,3,1,1,32)

Fig. 12.2 Flows for "particle board, outdoor use, at plant" and its representation in the ecoinvent database (cont.)

12.2 Oriented strand board (OSB)

Oriented strand board is basically a particle board made from strands with a high ratio of slenderness having a thickness up to 0.5 cm and a length of 7.6-15 cm. Because of the size of the strands, OSB production needs relatively large amounts of glue. In the board, the strands show a preferred orientation. For details see chapter 12.1.

12.2.1 Oriented strand board in ecoinvent database

The background information to the process are given in Tab. 12.2, flows are shown in Fig. 12.3.

12.2.2 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

The available input data are estimated of good quality. However, the corresponding emission data are not published due to confidentiality. This reduces the overall quality of this dataset. Data quality is not sufficient to allow for comparing with other wooden boards.

Tab. 12.2 Ecoinvent meta information for the oriented strand board production processes

Name	oriented strand board, at plant
Location	RER
Infrastructure Process	0
Unit	m ³
Data Set Version	2.0
Included Processes	Includes the inputs to the production processes and transports of those inputs. No process emission data are available. Formaldehyd emissions to air estimated.
Amount	1
Local Name	Oriented Strand Board, ab Werk
Synonyms	OSB
General Comment to reference function	Various OSB types exist; data depicts OSB Type 3 (DIN/EN 300): load-bearing, constructive OSB
Start Date	2000
End Date	2000
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data for Germany used for central Europe
Technology text	Medium enterprise technology (2000)
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	see Geography and Technology
Uncertainty Adjustments	The assumption that the technology is representative for central Europe is considered in the uncertainties.

General Flow information					Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment	
Industrial wood spruce, at forest road (u=140%)	oriented strand board, at plant		0.31 m3 used internally for heat generation; Amount reduced by 10% because bark is not included.	wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=140%, at forest road	1.19E+00	m3	Frühwald et al. 2000	1	1.08	(2,2,1,1,1,2,3)	
Residual wood spruce, naturally dried, at forest road (u=20%)			for heat generation; Amount reduced by 10% because bark is not included.	wooden materials	extraction	No	RER	residual wood, softwood, under bark, air dried, u=20%, at forest road	1.04E-01	m3	Frühwald et al. 2000	1	1.08	(2,2,1,1,1,2,3)	
Phenole-formaldehyde resin 100%			44.3 kg glue plus 0.4 kg hardener	paintings	production	No	RER	phenolic resin, at plant	4.47E+01	kg	Frühwald et al. 2000	1	1.08	(2,2,1,1,1,2,3)	
Paraffin				chemicals	organics	No	RER	paraffin, at plant	5.30E+00	kg	Frühwald et al. 2000	1	1.08	(2,2,1,1,1,2,3)	
Natural gas burned in industrial furnace				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	2.03E+02	MJ	Frühwald et al. 2000	1	1.08	(2,2,1,1,1,2,1)	
Electricity medium voltage - at grid UCTE				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.30E+02	kWh	Frühwald et al. 2000	1	1.08	(2,2,1,1,1,2,2)	
Diesel burned in building machine			Internal transports	construction processes	machinery	No	GLO	diesel, burned in building machine	1.50E+01	MJ	Frühwald et al. 2000	1	2.00	(2,2,1,1,1,2,5)	
Residual wood chips from process burned in 300kW furnace				wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 300kW	3.00E+03	MJ	Frühwald et al. 2000	1	1.08	(2,2,1,1,1,2,1)	
Chips			Since the fuel for the furnace is included in the input, the corresponding amount inventoried with the energy module is subtracted. Since the volume of the chips is bulked, it is higher than the difference of input and output volume.	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40%, at plant	-9.48E-01	m3	calculated	1	1.00	calculated	
Transport rail			Chemicals: 600 km, wood: 100 km	transport systems	train	No	RER	transport, freight, rail	1.64E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport lorry			Chemicals: 100 km, wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	7.20E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
plant				wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	3.33E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)	
		Oriented stranded board (OSB), at plant	OSB Typ 3; u=8.2%; no emission data available for confidentiality reason	wooden materials	extraction	No	RER	oriented strand board, at plant	1.00E+00	m3	Frühwald et al. 2000				
		Waste heat into air		air	unspecified			Heat, waste	4.68E+02	MJ	calculated	1	1.08	(2,2,1,1,1,2,13)	
		Formaldehyde into air		air	unspecified			Formaldehyde	2.63E-03	kg	estimated according to particleboard for use under humid conditions	1	2.08	(2,4,3,3,3,1,23)	

Fig. 12.3 Flows for "oriented strand board, at plant" and its representation in the ecoinvent database

12.3 Cement-bonded particleboard

Wood-based materials with mineral binders consist of 30-90% of a hardened inorganic binder and 10-70% of wood particles. Because of the high binder content, these materials are heavy and particularly suitable for use in building. The most important mineral binders are cement, gypsum, and magnesia (magnesite). Mineral-bonded boards are characterised by low thickness swelling through the action of water. Cement- and magnesia-based boards are highly resistant to mould and termites. Cement-bonded boards are used mainly for exterior applications because of their high resistance to weathering. All mineral-bonded wood-based construction materials are categorised as building materials of low flammability because of their high binder content.

Cement was the first mineral binder used in the industrial production of boards. The originally long hardening times of this hydraulic binder have recently been shortened greatly by technical measures. The setting time depends not only on the reactivity of a particular cement quality, but also on its compatibility with different types of wood. Wood components, such as free sugars, glucosides, and tannins, delay the hydration of cement and therefore impair the setting process. Cement-bonded wood-based boards are currently produced predominantly by the wet or semidry processes.

The original production process developed by Elmendorf is still used in different forms. In the process, wood particles, cement, additives, and water are mixed, moulded, and compressed. To shorten the hardening times, stacked boards are fed into a heated tunnel under pressure. The boards can generally be removed from their moulds after 6-8 h and then conditioned for 12-18 d in storage until the final strength is reached.

Finished boards consist of ca. 60% (w/w) cement and additives, ca. 20% (w/w) wood, and ca. 20% (w/w) water. The ratio of total water to cement determines board properties such as compressive and flexural strength.

12.3.1 Cement bonded particle board in ecoinvent database

The background information to the process are given in Tab. 12.3, flows are shown in Fig. 12.4.

The energy consumption data is partly based on confidential information and cannot be shown.

12.3.2 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

The available input data are estimated of good quality. No emission data are available. Especially water emissions are expected to have certain significance. Data quality is not sufficient to allow for comparing with other wooden boards.

Tab. 12.3 Ecoinvent meta information for the cement bonded particle board production processes

Name	particle board, cement bonded, at plant
Location	RER
Infrastructure Process	0
Unit	m ³
Data Set Version	2.0
Included Processes	Includes the inputs to the production processes and transports of those inputs. No process emission data are available.
Amount	1
Local Name	Holzspan-Leichtbauplatte, zementgebunden, ab Werk
Synonyms	
General Comment to reference function	
Start Date	1989
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Data for Switzerland used for central Europe
Technology text	Medium enterprise technology (2000)
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	Literature
Extrapolations	see Geography and Technology
Uncertainty Adjustments	The assumption that the technology is representative for central Europe is considered in the uncertainties.

General Flow information					Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment	
Industrial residual wood softwood (u=40%), at plant	particle board, cement bonded, at plant		Wood mix: 25% hardwood (beech), 75% softwood (pine and spruce); sources: 70% industrial residual wood, 30 % industrial wood; hardwood under bark and softwood under bark reduced by 12% and 10% respectively because bark is not included.	wooden materials	extraction	No	RER	industrial residue wood, mix, softwood, u=40%, at plant	2.33E-01	m3	Schniewind 1989, p. 197 ff.; Wegener et al. (1994: 68ff); Ressel (1986)	1	1.40	(3,3,4,3,3,5,3)	
Industrial residual wood hardwood (u=40%), at plant				wooden materials	extraction	No	RER	industrial residue wood, mix, hardwood, u=40%, at plant	2.31E-02	m3		1	1.40	(3,3,4,3,3,5,3)	
Industrial wood beech, at forest road (u=80%)				wooden materials	extraction	No	RER	industrial wood, hardwood, under bark, u=80%, at forest road	3.74E-02	m3		1	1.40	(3,3,4,3,3,5,3)	
Industrial wood spruce, at forest road (u=140%)				wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=140%, at forest road	5.30E-02	m3		1	1.40	(3,3,4,3,3,5,3)	
Portland Cement				construction materials	binder	No	CH	cement, unspecified, at plant	8.00E+02	kg	Schniewind 1989, p. 197 ff.	1	1.24	(1,4,4,3,1,1,4)	
Process and cooling water			evaporates partly during pressing and drying	resource	in water			Water, cooling, unspecified natural origin	3.00E-01	m3	Schniewind 1989, p. 197 ff.	1	1.24	(1,4,4,3,1,1,4)	
Organic chemicals			Additives	chemicals	organics	No	GLO	chemicals organic, at plant	4.50E+01	kg	Schniewind 1989, p. 197 ff.	1	1.24	(1,4,4,3,1,1,4)	
Electricity medium voltage - at grid UCTE			Total consumption estimated: chipping according to Wegner and Frühwald; board production according to data on cement-bonded woodwool board production	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	confidential	kWh	Wegener et al. 1994 for total wood mass into particleboard (586 kg/m3 normal particleboard); Frühwald et al. 2000 for part of chip production compared to total energy consumption (437 MJ of 1383 MJ Primary energy); CEWAG Düringen, Herr Kurzo, for mixing, molding, pressing	1	1.25	(3,3,2,3,1,5,2)	
Thermal heat from oil firing extra light			Total consumption estimated based on data on cement-bonded wood wool board production	oil	heating systems	No	RER	heat, light fuel oil, at industrial furnace 1MW	confidential	MJ	CEWAG Düringen, Herr Kurzo	1	1.22	(1,3,2,3,1,5,1)	
Transport rail			Chemicals: 600 km, wood & cement: 100 km	transport systems	train	No	RER	transport, freight, rail	1.34E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport lorry			Chemicals & cement: 100 km, wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	9.80E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
plant				wooden materials	extraction	Yes	RER	wooden board manufacturing plant, cement bonded boards	4.00E-07	unit	estimated	1	3.36	(4,5,2,3,4,5,9)	
		Particle board, cement bonded, at plant	Duripanel, medium-high density 1200 kg/m3 (1000-1350 kg/m3)	wooden materials	extraction	No	RER	particle board, cement bonded, at plant	1.00E+00	m3					
		Waste heat into air		air	unspecified			Heat, waste	8.77E+01	MJ	calculated	1	1.25	(3,3,2,3,1,5,13)	

Fig. 12.4 Flows for "particle board, cement bonded, at plant" and its representation in the ecoinvent database

12.4 Fibreboard, MDF

Fibreboard consists, like paper, of lignocellulose-containing fibres. Both materials are related with regard to production and construction, and the history of fibreboard is closely associated with that of pulp and paper.

The first fibreboard was softboard (an insulating board). It was produced in the United Kingdom around 1900 - and somewhat later in the United States - from waste fibres that were unsuitable for paper production because they were too short.

Hardboard was developed in the United States in the 1920s to utilise sawmill waste. The first hardboard was produced, like paper, by wet process from a thin aqueous fibre suspension.

12.4.1 Types of fibreboard

The classification and nomenclature of the different types of fibreboard on the market seem confusing because the names either have a historic origin or represent current types of boards or their properties. Two important criteria used in classification are the production process and the board density (Tab. 12.4).

Tab. 12.4 Classification of fibreboards

Name	low density fibreboard	medium density fibreboard	high density fibreboard
	Softboard	MDF	Hardboard
Density [kg/m ³]	< 350	350 - 800	>800

All three fibreboard types can be produced in wet or dry processes. Because of the sieve that is needed for draining the water, fibreboard from wet process usually shows the imprints of the sieve on the underside. These boards are called S1S (smooth one side). The products of the dry process, showing a smooth texture on both sides are called S2S (smooth two sides). It is also possible to produce S2S hardboard in the wet process by first producing a softboard and pressing it between two smooth plates.

Most commonly softboard and hardboard are produced using wet process and MDF is produced with dry process.

Plasterboard

Besides the classical wood-based fibreboard, fibreboard bonded with gypsum is becoming increasingly popular for use in interior furnishing for partitions and panels. Plasterboard can be produced from waste paper and gypsum from the desulphurisation of exhaust gases. The production process is similar to that used for wood particle board using mineral binders (see below).

12.4.2 Production of fibreboard

Raw Material

Fibreboard can be produced from all lignocellulose-containing materials with industrially utilisable fibres. However, for reasons of availability, wood is almost always used. To minimise variations in the process, only one type or a very limited number of types of wood are employed, the latter as a mixture of constant composition.

Deciduous and coniferous woods differ in their anatomical construction. Coniferous wood fibres are generally two to three times longer than those of deciduous wood. Unlike paper, in fibreboard the fibre length is relatively unimportant. However, to a certain extent it determines the orientation of fibres in

the board. Short fibres tend to arrange vertically and longer ones horizontally. Fibres with thin walls nestle together better in the press than those with thick walls. The density of the wood used determines the thickness of the fibre mat. The lower the density, the thicker the mat, and consequently the higher are the degree of compression, the contact between fibres, and the resulting strength of the board. Dry processes are generally less sensitive to the properties of wood than wet ones. In the wet process, deciduous woods give better hardboard and coniferous woods better softboard. Coniferous woods require longer steaming times and, therefore, more energy for fibre production than deciduous woods. Mixtures of deciduous woods are mostly available in groups classified according to density. The lower the density of the deciduous wood, the longer is the digestion time. For cost reasons, efforts are made to use waste wood instead of round wood to the extent possible. Debarked chips from sawmills give good fibreboard.

Sawdust is less suitable because of the irregular particle sizes and short fibres; it requires expensive preparation. Bark is not suitable because of its mineral components and dark colour. Of the lignocellulose-containing annuals, only sugarcane bagasse is important for softboard production.

Raw Material Processing

Raw material processing and fibre production in the wet and dry processes are very similar, frequently even identical.

The wood used is predominantly round wood, also partly debarked chips from sawmills or whole-tree chips. Pieces of bark are generally undesirable. They contain minerals that shorten the service life of refiner disks and processing tools, and impair surface quality.

Chips are most frequently produced by using disk chippers or cylinder chopping machines. The cutting disk of the disk chipper, arranged vertically or diagonally, is fitted with 4-16 blades, depending on the size. The positions of the blades determine the dimensions of the chips.

Fine chips (<4 mm sieve mesh) and coarse chips (>40 mm sieve mesh) are separated from normal chips by sieving on flat or drum sifters. The fine chips are incinerated, and the coarse ones are cut up further in hammer or wing beater mills. The optimum chip size is 20-25 × 20 × 5 mm.

To remove unwanted material and mineral components, chips are washed before further processing. This is particularly important with whole-tree chips. Washing is carried out in specially designed plants with pre-washers for separating coarse components and a screw drainer for separating fine mineral particles. The chips are then stored in reinforced concrete or metal silos. Storage acts as a buffer between chip and fibre production and at the same time allows the moisture to distribute more uniformly.

Fibre Production

The aim of defibration is to break up the previously chopped raw material as carefully as possible into individual fibres and bundles of fibres. This is carried out thermo-mechanically. The lignin-rich middle lamellae that bind the fibres are first softened by heat and steam. The fibres are then separated by mechanical abrasion. They are often broken in the process, giving fibre fragments. The anatomical nature of the fibres and their state after defibration are important for forming the fibre mat and for the properties of the board. This applies particularly to the wet process in which the freeness (draining behaviour) and the type of matting are very important.

Thermal treatment of the chips before and during defibration causes part of the hemicelluloses to go into solution. Increasing the temperature and the length of the treatment period leads to increased softening of the inter-fibre bonding. In the wet process, this improves the natural adhesion between fibres during board production. At the same time, the sugar content of the wastewater and the energy consumption increase, so that a compromise between environmentally friendly and economic process design and board properties is usually aimed at. Fibre production is energy intensive; it consumes more than half of the total process energy.

An important discovery was made by H. Mason, who developed the Mason cannon in 1926. In his process, the chips are first softened in a boiler under high pressure. This is followed by a sudden pressure release accompanied by a loud cracking sound, which indicates breakdown of the wood structure into fibres and bundles of fibres. These are pressed in a hot press to form hardboard without the addition of binders. This technique is used exclusively in the United States by the Masonite company (wet process).

In all other processes defibration is carried out by using grinding disks. Some defibrators operate at atmospheric, others at elevated pressure. Also, machines exist with only one rotating grinding disk (Sunds) or with two counterrotatory disks (Sprout-Bauer). Refiners operating at atmospheric pressure have the disadvantages of high energy consumption and moderate quality of the fibres produced.

The refiner now predominantly used, which operates under pressure, gives better results. The Asplund defibrator (Sunds) is a prototype. This makes use of the fact that from ca. 170 °C, separation of fibres in the middle lamella is considerably easier and is associated with a significant lowering of the energy consumption and an improvement in fibre quality.

In the defibration process, washed chips are fed into the pre-heater through a conical opening using a screw feeder. The pre-heater is usually a vertical, slightly conical pressure container, which is continuously charged with saturated steam at 150-180 °C. The steam heats the chips by condensation. Blowing steam out through the opening is hindered by the plug formed by the chips.

In 1-3 min the chips reach a conveyor screw (at the base of the pre-heater through which they are transferred to the grinding disks of the refiner, where defibration takes place under steam pressure. The necessary grinding power is ca. 200 kW per tonne of dry fibre.

The important parameters affecting the quality of the fibre material and the energy consumption are the residence time and vapour pressure in the pre-heater, throughput, defibrating pressure, distance between grinding disks (0.1-0.4 mm), number of revolutions per minute, state of the grinding disk, nature of the raw material, and moisture level.

Defibration is tested by determining the degree of grinding or by fractionating the fibre material. The degree of grinding is a measure of the freeness of the fibre material and determined by means of the dewatering speed of a fibre suspension (standard freeness tester, defibrator-seconds, DS). In the wet process, higher degrees of grinding usually result in an improvement in board properties.

After defibration fibres are blown out through a pressure-regulated exit valve. From this stage onward the wet and dry processes differ and are described separately below.

Wet Process

The fibre material is fed to silos for mixing and adjusting concentration (4-6%) or is further defibrated in disk refiners. For hard and semi-hard fibreboard a degree of grinding of 18-30 DS is aimed at, and for porous boards 42-70 DS.

A system of tanks and conveyors is used for mixing, storing, and adding adhesives and additives. A control panel connected to the system regulates concentration, temperature, and pH.

The fibre mat is formed on a sieve by drainage of water from the fibre suspension and subsequent sedimentation of the fibres. Mat formation was previously carried out as a batch process in tanks, but cylinder wet machines and conveyor sieve de-watering machines are predominantly used now. The main component of the latter is a continuously moving sieve. Fibre material is added through a headbox at a concentration of 1-2% and is subsequently drained and compressed by means of a forming table, suction box, and wet pressing section.

To improve the surface properties of the smooth side of the fibreboard, fibre material of higher quality can be applied to the mat on the forming table through a second headbox. After the wet press the thickness of the mat is 20-25 mm, which gives a final thickness of 3-5 mm. The mat is then separated into individual sections corresponding to the pressed board format of the hot press.

In the production of softboard or S2S-hardboard by the wet process, the drained fibre mats are dried in single- or multi-deck dryers at 150-170 °C from moisture levels of 100-120% to 1-4%.

To prevent dried boards from catching fire during subsequent storage, they are cooled immediately after drying by spraying with water and aspiration of ambient air.

S1S board is further de-watered in hydraulic multi-daylight presses by hot pressing using an established pressure–time program, compressed, and strengthened by curing the binders that are naturally present or have been added. For S1S, use of a draining sieve is necessary to remove the large quantities of water. For hardboard the pressing time is generally 2.0 – 3.5 min/mm at a pressing temperature of 180 – 200 °C. A typical pressing diagram consists of three phases, the compression and first high-pressure stage, drying, and the second high-pressure stage.

An improvement in the properties of fibreboard can be obtained by thermal after-treatment in chambers or continuously in tunnels or by impregnating with drying oils. Treatment results in a more uniform moisture distribution in the board as well as better dimensional stability.

Wet processes require enormous quantities of water. Pollutants, such as extractives, hemicelluloses, and lignin, are present in the wastewater. For environmental reasons the dry process was developed. Here, the fibres are dried, glue-blended, and pressed as in particle board production. This process gives an S2S board. For reasons of surface quality, gluing is currently carried out before drying. The semidry process lies between the wet and dry processes. Because fibre moisture is still present, sieves must be used and S1S board is produced.

Dry Process

After defibration the fibres pass down a blow line in which glue is added and then reach the fibre dryer. Alternatively, they can be blown directly into the fibre dryer by using of the vapour pressure present in the refiner. In this case they are subsequently glue-blended in high-speed centrifugal machines, as in particle board production. In almost all plants, glue blending occurs before drying, thus avoiding spots of adhesive on the board surface that impair its quality. Paraffin added to improve hydrophobic properties is generally charged in the molten state to the lower conveyor screw of the refiner.

Tube dryers are now used exclusively for drying the fibres. They are mostly two-stage dryers with tube lengths of 80-100 m and diameters of 1.3-1.6 m. The air temperature at the entry is 140-170 °C. Because of the high air speed (ca. 30 m/s) and the small dimensions of the fibres, the drying time is only a few seconds. The moisture level of the fibres is adjusted to 9-11% by regulating the exit temperature of the dryer.

The fibres reach the forming station via cyclones and fibre hoppers. The fibre mat is formed either mechanically by using a snowfall felter or pneumatically by using a pendistor on a vacuum screen conveyor.

The uniformity of distribution of the fibre material over the surface of the screen significantly affects fibreboard properties.

The mat coming out of the forming station has a very low bulk density and contains a large quantity of air, which must be removed before hot pressing. To decrease the size of the press opening, shorten the closing times of the hot press, and improve the transportability of the mat and the surface quality of the finished board, the mat is pre-compressed in continuously operating belt pre-presses. Mat density is thus doubled or trebled.

Hot pressing of the fibre mat occurs either batch wise on multi-daylight presses with preliminary separation into mat sheets or, as is now increasingly common, continuously on double-belt presses or calenders for thin boards (up to 8 mm thickness). The presses are the same as for particle board production. To heat the fibre mat more rapidly and shorten the pressing time, the fibre mat is often preheated

by microwaves before entering the hot press. Pressure control in MDF production differs from that in particle board production.

The bulk density profile (perpendicular to the board thickness) and thus the properties of the board can be widely varied through the type of compression and the temperature control. High compression in the outer zones and as uniform a density as possible in the inner zones are pursued.

12.4.3 Binders and additives in fibreboard

In the production of fibreboard by the wet process, no binders are required in principle because of the binding forces already present in the wood. However, adhesives are necessary for the dry process.

The main purpose of additives is to improve the mechanical properties and water resistance of fibreboard. In softboard the fibres are bonded predominantly by hydrogen bonds. For hydrophobing, molten paraffin (ca. 1% slack wax based on dry fibre) or asphalt emulsion is added. For hardboard, lignin, which is activated in the hot press, is the important binder.

Wet process softboard is produced mainly with the addition of highly condensed, alkaline PF resins (0.5-2.0% based on dry fibre), which are fixed on the fibre by the addition of $Al_2(SO_4)_3$. Paraffin (0.1-0.5% based on dry fibre) and drying oils, such as linseed oil, are also added.

For S2S board (i.e. hardboard) produced by the wet process, phenolic resins cannot be used because the severe drying conditions would result in premature curing. Resins such as rosin are therefore employed. They are fixed on the fibre by lowering the pH with $Al_2(SO_4)_3$.

Dry process hardboard is generally produced by using PF resins (ca. 6% based on dry fibre). For MDF produced by the dry process, UF resins, generally modified with melamine, are the most important binders (10-14% solid resin based on dry fibre). To a small extent, mainly for special boards, MUF, PF, and PMDI are used. As with S1S boards, paraffin is used for hydrophobing.

12.4.4 Use of fibreboard

The main areas of use for fibreboard are:

Insulation board (softboard) is used in interior and exterior applications, mostly as parts of walls, and for ceiling panelling, frequently in the form of acoustic panels.

Hardboard produced by the wet process generally consists of thin boards. It is employed mainly in buildings for inner and outer wall panelling. It is also used in the production of furniture or doors or moulded for use in motor vehicle interiors.

Because of its excellent edge and surface properties and its good workability, MDF is used mainly in furniture production.

12.4.5 Fibreboard in ecoinvent database

The background information to the processes are given in Tab. 12.5, flows are shown in:

- | | |
|---------------------------------------|-----------|
| - fibreboard soft, at plant | Fig. 12.5 |
| - medium density fibreboard, at plant | Fig. 12.6 |
| - fibreboard hard, at plant | Fig. 12.7 |

12.4.6 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Available input data are of good quality. The corresponding air emission data are only estimated. Water emissions for the wet process are measured.

Tab. 12.5 Ecoinvent meta information for the fibreboard production processes

Name	fibreboard soft, at plant	medium density fibreboard, at plant	fibreboard hard, at plant
Location	RER	RER	RER
Infrastructure Process	0	0	0
Unit	m ³	m ³	m ³
Data Set Version	2.0	2.0	2.0
Included Processes	Includes the inputs to the production processes, transports of those inputs and the process emissions as far as they are available.	Includes the inputs to the production processes, transports of those inputs and the process emissions as far as they are available.	Includes the inputs to the production processes, transports of those inputs and the process emissions as far as they are available.
Amount	1	1	1
Local Name	Faserplatte weich, ab Werk	Faserplatte mittlerer Dichte, ab Werk	Faserplatte hart, ab Werk
Synonyms		MDF	
General Comment to reference function			
Start Date	1996	1996	1996
End Date	2002	2002	2002
Data Valid For Entire Period	1	1	1
Other Period Text			
Geography text	Data for Switzerland used for central Europe	Data for Switzerland used for central Europe	Data for Switzerland used for central Europe
Technology text	Wet process, medium enterprise technology (1996)	Dry process, medium enterprise technology (1996)	Wet process, medium enterprise technology (1996)
Representativeness [%]			
Production Volume	unknown	unknown	unknown
Sampling Procedure	Literature	Literature	Literature
Extrapolations	see Geography and Technology	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	The assumption that the technology is representative for central Europe is considered in the uncertainties.	The assumption that the technology is representative for central Europe is considered in the uncertainties.	The assumption that the technology is representative for central Europe is considered in the uncertainties.

General Flow information					Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment	
Paraffin	fibreboard soft, at plant, part 1			chemicals	organics	No	RER	paraffin, at plant	1.38E+00	kg	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.16	(1,4,3,3,1,1,3)	
Aluminiumsulfat				chemicals	inorganics	No	RER	aluminium sulphate, powder, at plant	3.00E-01	kg	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.16	(1,4,3,3,1,1,3)	
Natural gas burned in industrial furnace >100 kW Euro				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.38E+03	MJ	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.16	(1,4,3,3,1,1,1)	
Industrial wood beech, at forest road (u=80%)			total input: 0.796 m3 u input material =44%; Wood mix: 25% hardwood (beech), 75% softwood (pine and spruce); sources: 70% industrial residual wood, 30 % industrial wood; hardwood under bark and softwood under bark reduced by 12% and 10% respectively because bark is not included.	wooden materials	extraction	No	RER	industrial wood, hardwood, under bark, u=80%, at forest road	5.33E-02	m3	based on mix used for particle board manufacturing (Wegener et al. 1994); based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.14	(2,3,3,3,1,1,3)	
Industrial wood spruce, at forest road (u=140%)				wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=140%, at forest road	1.63E-01	m3		1	1.14	(2,3,3,3,1,1,3)	
Industrial residual wood hardwood (u=40%), at plant				wooden materials	extraction	No	RER	industrial residue wood, mix, hardwood, u=40%, at plant	1.39E-01	m3		1	1.14	(2,3,3,3,1,1,3)	
Industrial residual wood softwood (u=40%), at plant				wooden materials	extraction	No	RER	industrial residue wood, mix, softwood, u=40%, at plant	4.18E-01	m3		1	1.14	(2,3,3,3,1,1,3)	
NaOH Sodium hydroxide				chemicals	inorganics	No	RER	sodium hydroxide, 50% in H2O, production mix, at plant	3.33E-02	kg	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.16	(1,4,3,3,1,1,3)	
Phenole-formaldehyde resin 100%				paintings	production	No	RER	phenolic resin, at plant	3.00E+00	kg		1	1.16	(1,4,3,3,1,1,3)	
Electricity medium voltage - at grid UCTE				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.36E+02	kWh		1	1.16	(1,4,3,3,1,1,2)	

Fig. 12.5 Flows for "fibreboard soft, at plant" and its representation in the ecoinvent database

General Flow information					Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Transport lorry	fibreboard soft, at plant, part 2		Chemicals: 100 km, wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	3.19E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport rail			Chemicals: 600 km, wood: 100 km	transport systems	train	No	RER	transport, freight, rail	6.57E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Process and cooling water				resource	in water			Water, cooling, unspecified natural origin	5.99E-02	m3	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.16	(1,4,3,3,1,1,4)	
plant				wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	3.33E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)	
		Fibreboard soft, wet processing, at plant	Density ca. 300 kg/m3	wooden materials	extraction	No	RER	fibreboard soft, at plant	1.00E+00	m3					
		Waste heat into air		air	unspecified			Heat, waste	4.89E+02	MJ	calculated	1	1.16	(1,4,3,3,1,1,13)	
		Formaldehyde to waste water	6.5% of the wood input and some of the glue is dissolved in water. Emissions are not to be balanced as water emissions. They are included in waste water to treatment						3.09E-03	kg	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976				
		Phenole to waste water							1.64E-03	kg					
		Further dissolved organic matter to waste water							2.27E+01	kg					
		Waste water to treatment	includes above mentioned emissions (as 100kg COD / m3 waste water)	waste management	wastewater treatment	No	CH	treatment, fibre board production effluent, to wastewater treatment, class 3	2.66E-01	m3		1	1.53	(1,4,3,3,1,1,32)	

Fig. 12.5 Flows for "fibreboard soft, at plant" and its representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment	
Paraffin	medium density fibreboard, at plant, part 1		Paraffin; total 12% of adhesives and additives; ratio of the adhesive and additive according to the fiberboard hard	chemicals	organics	No	RER	paraffin, at plant	2.28E+01	kg	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.16	(1,4,3,3,1,1,3)	
Aluminiumsulfat			Aluminiumsulfate; total 12% of adhesives and additives; ratio of the adhesive and additive according to the fiberboard hard	chemicals	inorganics	No	RER	aluminium sulphate, powder, at plant	4.36E+00	kg	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.16	(1,4,3,3,1,1,3)	
Natural gas burned in industrial furnace >100 kW Euro				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	1.67E+03	MJ	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.16	(1,4,3,3,1,1,1)	
Industrial wood beech, at forest road (u=80%)			total input: 1.90 m3 u input material =44%; Wood mix: 25% hardwood (beech), 75% softwood (pine and spruce); sources: 70% industrial residual wood, 30 % industrial wood; hardwood under bark and softwood under bark reduced by 12% and 10% respectively because bark is not included.	wooden materials	extraction	No	RER	industrial wood, hardwood, under bark, u=80%, at forest road	1.27E-01	m3	based on mix used for particle board manufacturing (Wegener et al. 1994); based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.14	(2,3,3,3,1,1,3)	
Industrial wood spruce, at forest road (u=140%)				wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=140%, at forest road	3.89E-01	m3		1	1.14	(2,3,3,3,1,1,3)	
Industrial residual wood hardwood (u=40%), at plant				wooden materials	extraction	No	RER	industrial residue wood, mix, hardwood, u=40%, at plant	3.33E-01	m3		1	1.14	(2,3,3,3,1,1,3)	
Industrial residual wood softwood (u=40%), at plant				wooden materials	extraction	No	RER	industrial residue wood, mix, softwood, u=40%, at plant	9.98E-01	m3		1	1.14	(2,3,3,3,1,1,3)	
Residual wood chips from process burned in 300kW furnace				wood energy	heating systems	No	CH	wood chips, from industry, softwood, burned in furnace 300kW	2.77E+03	MJ	calculated based on Frühwald 2000	1	1.16	(1,4,3,3,1,1,1)	
Chips			Since the fuel for the furnace is included in the input, the corresponding amount inventoried with the energy module is subtracted. Since the volume of the chips is bulked, it is higher then the difference of input and output volume.	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40%, at plant	-8.76E-01	m3	calculated	1	1.00	calculated	

Fig. 12.6 Flows for "medium density fibreboard, at plant" and its representation in the ecoinvent database

General Flow information					Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment	
Urea-formaldehyde 100%	medium density fibreboard, at plant, part 2		Aluminiumsulfate; total 12% of adhesives and additives; ratio of the adhesive and additive according to the fiberboard hard	paintings	production	No	RER	urea formaldehyde resin, at plant	4.96E+01	kg	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.16	(1,4,3,3,1,1,3)	
Electricity medium voltage - at grid UCTE				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.55E+02	kWh	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.16	(1,4,3,3,1,1,2)	
Transport lorry			Chemicals: 100 km, wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	8.27E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport rail			Chemicals: 600 km, wood: 100 km	transport systems	train	No	RER	transport, freight, rail	1.96E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Process and cooling water				resource	in water			Water, cooling, unspecified natural origin	1.80E-01	m3	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.16	(1,4,3,3,1,1,4)	
plant				wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	3.33E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)	
		Medium density fibreboard, dry processing, at plant	Density ca. 780 kg/m3 no water emission because of dry process	wooden materials	extraction	No	RER	medium density fibreboard, at plant	1.00E+00	m3					
		Waste heat into air		air	unspecified			Heat, waste	1.28E+03	MJ	calculated	1	1.16	(1,4,3,3,1,1,13)	
		Formaldehyde into air		air	unspecified			Formaldehyde	9.27E-03	kg	based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	2.03	(1,4,3,3,1,1,23)	

Fig. 12.6 Flows for "medium density fibreboard, at plant" and its representation in the ecoinvent database (cont.)

General Flow information					Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment	
Paraffin	fibreboard hard, at plant, inputs			chemicals	organics	No	RER	paraffin, at plant	4.14E+00	kg	Pavatex Cham, HR. Lörcher	1	1.16	(1,4,3,3,1,1,3)	
Aluminiumsulfat				chemicals	inorganics	No	RER	aluminium sulphate, powder, at plant	9.00E-01	kg	Pavatex Cham, HR. Lörcher	1	1.16	(1,4,3,3,1,1,3)	
Natural gas burned in industrial furnace >100 kW Euro				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	4.14E+03	MJ	Pavatex Cham, HR. Lörcher	1	1.16	(1,4,3,3,1,1,2)	
Industrial wood beech, at forest road (u=80%)			total input: 2.39 m3 u input material =44%; Wood mix: 25% hardwood (beech), 75% softwood (pine and spruce); sources: 70% industrial residual wood, 30 % industrial wood; hardwood under bark and softwood under bark reduced by 12% and 10% respectively because bark is not included.	wooden materials	extraction	No	RER	industrial wood, hardwood, under bark, u=80%, at forest road	1.60E-01	m3	based on mix used for particle board manufacturing (Wegener et al. 1994); based on data for fibreboard hard (Pavatex Cham, HR. Lörcher); Anonymus 1976	1	1.14	(2,3,3,3,1,1,3)	
Industrial wood spruce, at forest road (u=140%)				wooden materials	extraction	No	RER	industrial wood, softwood, under bark, u=140%, at forest road	4.89E-01	m3		1	1.14	(2,3,3,3,1,1,3)	
Industrial residual wood hardwood (u=40%), at plant				wooden materials	extraction	No	RER	industrial residue wood, mix, hardwood, u=40%, at plant	4.18E-01	m3		1	1.14	(2,3,3,3,1,1,3)	
Industrial residual wood softwood (u=40%), at plant				wooden materials	extraction	No	RER	industrial residue wood, mix, softwood, u=40%, at plant	1.25E+00	m3		1	1.14	(2,3,3,3,1,1,3)	
Sodium hydroxide				chemicals	inorganics	No	RER	sodium hydroxide, 50% in H2O, production mix, at plant	1.00E-01	kg	Pavatex Cham, HR. Lörcher	1	1.16	(1,4,3,3,1,1,3)	
Phenole-formaldehyde resin 100%			1 % phenole resin based on total mass of fibres	paintings	production	No	RER	phenolic resin, at plant	9.00E+00	kg	Pavatex Cham, HR. Lörcher	1	1.16	(1,4,3,3,1,1,3)	
Electricity medium voltage - at grid UCTE				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.08E+02	kWh	Pavatex Cham, HR. Lörcher	1	1.16	(1,4,3,3,1,1,2)	
Transport lorry			Chemicals: 100 km, wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	9.57E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport rail			Chemicals: 600 km, wood: 100 km	transport systems	train	No	RER	transport, freight, rail	1.97E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Process and cooling water				resource	in water			Water, cooling, unspecified natural origin	1.80E-01	m3	Pavatex Cham, HR. Lörcher	1	1.16	(1,4,3,3,1,1,4)	
plant				wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	3.33E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)	

Fig. 12.7 Flows for "fibreboard hard, at plant" and its representation in the ecoinvent database

General Flow information					Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment	
fibreboard hard, at plant, outputs	→	Fibreboard hard, wet processing, at plant	Density ca. 900 kg/m ³	wooden materials	extraction	No	RER	fibreboard hard, at plant	1.00E+00	m ³					
	→	Waste heat to air		air	unspecified			Heat, waste	1.47E+03	MJ	calculated	1	1.16	(1,4,3,3,1,1,13)	
	→	Formaldehyde to waste water	6.5% of the wood input and some of the glue is dissolved in water. Emissions are not to be balanced as water emissions. They are included in waste water to treatment						9.27E-03	kg	Pavatex Cham, HR. Lörcher				
	→	Phenole to waste water							4.93E-03	kg	Pavatex Cham, HR. Lörcher				
	→	Further dissolved organic matter to waste water							6.81E+01	kg	Pavatex Cham, HR. Lörcher				
	→	Waste water to treatment	includes above mentioned emissions (as 100kg COD / m ³ waste water)	waste management	wastewater treatment	No	CH	treatment, fibre board production effluent, to wastewater treatment, class 3	7.99E-01	m ³	Pavatex Cham, HR. Lörcher	1	1.53	(1,4,3,3,1,1,32)	

Fig. 12.7 Flows for "fibreboard hard, at plant" and its representation in the ecoinvent database (cont.)

13 Fuels based on industrial residual wood

13.1 Wood chips made of industrial residual wood

This chapter deals with wood chips from industrial residual wood as described in chapter 11, produced using a stationary chopper as described in section 16.5. For wood chips directly out of the forest see section 6.1.

13.2 Wood pellets made of industrial residual wood

Wood pellets are made from dry industrial residual wood, e.g. from planing. They are pressed in pellet mills as described in section 16.6 without binders or additives. Their length varies between 5 and 30 mm and their diameter is 6 or 10 mm.

The main advantage of pellets over chips is their constant properties (dimension, moisture and calorific value) that make pellet boilers less service intensive than chips boilers.

13.3 Wood chips and pellets in ecoinvent database

The background information to the processes are given in Tab. 13.1, flows are shown in:

- | | |
|--|-----------|
| - wood chips, hardwood, from industry, u=40%, at plant | Fig. 13.1 |
| - wood chips, softwood, from industry, u=40%, at plant | Fig. 13.2 |
| - wood chips, mixed, from industry, u=40%, at plant | Fig. 13.3 |
| - wood pellets, u=10%, at storehouse | Fig. 13.4 |

13.3.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Although the maintenance of the electric chopper is neglected, the overall data quality is thought to be good.

Tab. 13.1 Ecoinvent meta information for the wood chips and pellets production processes

Name	wood chips, hardwood, from industry, u=40%, at plant	wood chips, softwood, from industry, u=40%, at plant
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Includes the chopping of residual hardwood with a stationary chopper in the sawmill. No transports for the inputs are assumed.	Includes the chopping of residual softwood with a stationary chopper in the sawmill. No transports for the inputs are assumed.
Amount	1	1
Local Name	Holzschnitzel, Laubholz, aus Industrie, u=40%, ab Werk	Holzschnitzel, Nadelholz, aus Industrie, u=40%, ab Werk
Synonyms	Buche//Beech//Häcksel	Fichte//Spruce//Häcksel
General Comment to reference function	The volume refers to the bulked volume: dried matter content = 239 kg/m ³ dried matter (u=0%), lower heating value = 4220 MJ/m ³	The volume refers to the bulked volume: dried matter content = 169 kg/m ³ dried matter (u=0%), lower heating value = 3169 MJ/m ³
Start Date	1996	1996
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Austria used for central Europe	Data for Austria used for central Europe
Technology text	Chopping with a large stationary chopper.	Chopping with a large stationary chopper.
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography	see Geography
Uncertainty Adjustments	The assumption that Austria is representative for central Europe is considered in the uncertainties.	The assumption that Austria is representative for central Europe is considered in the uncertainties.

Tab. 13.1 Ecoinvent meta information for the wood chips and pellets production processes (cont.)

Name	wood chips, mixed, from industry, u=40%, at plant	wood pellets, u=10%, at storehouse
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Includes the chopping of the average mix of 72% residual softwood and 28% residual hardwood (reflecting Swiss average wood consumption) with a stationary chopper in the sawmill. No transports are included.	Contains the pressing of pellets out of dried industrial residual wood from planing mill (u=10%). Transports of the input material is not considered because pellet mill and planing mill are on the same site.
Amount	1	1
Local Name	Holz schnitzel, Mix, aus Industrie, u=40%, ab Werk	Holz pellets, u=10%, ab Regionallager
Synonyms	Häcksel	
General Comment to reference function	The volume refers to the bulked volume: dried matter content = 188.6 kg/m ³ dried matter (u=0%), lower heating value = 3463 MJ/m ³	The volume refers to the bulked volume: dried matter content = 650 kg/m ³ , lower heating value = 12'164 MJ/m ³ . Various technologies for pellet production are available. The ratio of beech and spruce reflects current consumption of the species
Start Date	2002	1996
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Austria (process) and Switzerland (mix) used for central Europe	Data for Austria used for central Europe
Technology text	Mix	Average production in Austria 1996
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography	see Geography
Uncertainty Adjustments	The assumption that Switzerland/Austria is representative for central Europe is considered in the uncertainties.	The assumption that Austria is representative for central Europe is considered in the uncertainties.

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Industrial residual wood hardwood, at plant	wood chips, hardwood, from industry, u=40%, at plant		volume under bark	wooden materials	extraction	No	RER	industrial residue wood, mix, hardwood, u=40%, at plant	3.68E-01	m3	calculated	1	1.35	(3,3,3,3,3,5,3)
Stationary electric chopper for industrial residual wood				wooden materials	extraction	No	RER	Industrial residual wood chopping, stationary electric chopper, at plant	2.39E+02	kg	calculated	1	1.35	(3,3,3,3,3,5,2)
		Wood chips hardwood from industry, at plant (u=40%)	bulk volume: dried matter content = 239 kg/m3 dried matter (u=0%) --> heating value = 4220 MJ/m3	wood energy	fuels	No	RER	wood chips, hardwood, from industry, u=40%, at plant	1.00E+00	m3	Frischknecht et al. 1996			

Fig. 13.1 Flows for "wood chips, hardwood, from industry, u=40%, at plant" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Industrial residual wood softwood, at plant	wood chips, softwood, from industry, u=40%, at plant		volume under bark	wooden materials	extraction	No	RER	industrial residue wood, mix, softwood, u=40%, at plant	3.75E-01	m3	calculated	1	1.35	(3,3,3,3,3,5,3)
Stationary electric chopper for industrial residual wood				wooden materials	extraction	No	RER	Industrial residual wood chopping, stationary electric chopper, at plant	1.69E+02	kg	calculated	1	1.35	(3,3,3,3,3,5,2)
		Wood chips softwood from industry, at plant (u=40%)	bulk volume: dried matter content = 169 kg/m3 dried matter (u=0%), lower heating value = 3169 MJ/m3	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40%, at plant	1.00E+00	m3	Frischknecht et al. 1996			

Fig. 13.2 Flows for "wood chips, softwood, from industry, u=40%, at plant" and its representation in the ecoinvent database

13 Fuels based on industrial residual wood

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Wood chips softwood from industry, at plant (u=40%)	wood chips, mixed, from industry, u=40%, at plant		bulk volume: dried matter content = 169 kg/m ³ dried matter (u=0%) (Frischknecht et al. 1996:X-14)	wood energy	fuels	No	RER	wood chips, softwood, from industry, u=40%, at plant	7.20E-01	m ³	BFS/BUWAL 2000	1	1.35	(3,3,3,3,3,5,3)
Wood chips hardwood from industry, at plant (u=40%)			bulk volume: dried matter content = 239 kg/m ³ dried matter (u=0%) (Frischknecht et al. 1996:X-14)	wood energy	fuels	No	RER	wood chips, hardwood, from industry, u=40%, at plant	2.80E-01	m ³	BFS/BUWAL 2000	1	1.35	(3,3,3,3,3,5,3)
		Wood chips mix from industry, at plant (u=40%)	bulk volume: dried matter content = 188.6 kg/m ³ dried matter (u=0%), lower heating value = 3463 MJ/m ³	wood energy	fuels	No	RER	wood chips, mixed, from industry, u=40%, at plant	1.00E+00	m ³				

Fig. 13.3 Flows for "wood chips, mixed, from industry, u=40%, at plant" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Industrial residual wood spruce, technically dried, at planing mill (u=10%)	wood pellets, u=10%, at storehouse		72% of total volume according to current use ratio in Switzerland	wooden materials	extraction	No	RER	industrial residue wood, from planing, softwood, kiln dried, u=10%, at plant	9.25E-01	m ³	according to Anonymous 1998; BFS/BUWAL 2000	1	1.35	(3,3,3,3,3,5,3)
Industrial residual wood beech planed, naturally/technically dried (u=10%)			28% of total volume according to current use ratio in Switzerland	wooden materials	extraction	No	RER	industrial residue wood, from planing, hard, air/kiln dried, u=10%, at plant	#BEZUG!	m ³	according to Anonymous 1998; BFS/BUWAL 2000	1	1.35	(3,3,3,3,3,5,3)
Electricity medium voltage - at grid UCTE			20 kW/t dried matter chopper; 77 kWh/t dried matter press	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.64E+02	kWh	Bergmair 1996 according to Anonymous 1998 (GEMIS)	1	1.33	(1,3,3,3,3,5,2)
Transport rail			Wood 100 km	transport systems	train	No	RER	transport, freight, rail	7.15E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,5)
Transport lorry			Wood: 50 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	3.58E+01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,5)
				wood energy	fuels	Yes	RER	wood pellet manufacturing, infrastructure	1.00E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
plant		Wood pellets, at plant (u=10%)	bulk volume: dried matter content = 650 kg/m ³ , heating value = 12'164 MJ/m ³	wood energy	fuels	No	RER	wood pellets, u=10%, at storehouse	1.00E+00	m ³				
		Waste heat into air		air	unspecified			Heat, waste	5.91E+02	MJ	calculated	1	1.33	(1,3,3,3,3,5,13)

Fig. 13.4 Flows for "wood pellets, u=10%, at storehouse" and its representation in the ecoinvent database

14 Impregnation: Agents and processes

This chapter is mainly based on (Pommer 1997) and (Kollmann et al. 1975).

Wood can undergo physical, thermal, and chemical changes, and can be attacked and even destroyed by fungi, insects, and marine animals. Wood with a moisture level of <18-20% is not attacked by fungi, but can be attacked by insects - in certain climates, often predominantly by termites. If constructive wood preservation measures are not possible, or durable and weather-resistant types of wood are not available, chemical preservatives must be used. Depending on the hazard posed, different preservatives that can include flame retardant are used.

14.1 Types of wood preservatives

Wood preservatives can be divided into:

- creosote and creosote-coal-containing preparations
- water-soluble inorganic salt preservatives and other water-soluble, predominantly organic compounds
- solvent-based preservatives (solvent-containing preparations)

The last group, in particular, includes various combinations of active substances with different types of additives and solvents.

14.1.1 Creosote and creosote-coal containing preparations

Creosotes, in particular coal-tar distillates with boiling points between 200 and 400 °C belong to the most important wood preservatives with a broad spectrum of applications. These tar oil fractions are also used in mixture with the original coal-tar or mineral oils in some countries (e.g. the United States). For more than 150 years, creosotes have been used to protect railway ties, masts, poles, and wood for cooling towers and for marine timbers. The estimated period of effectiveness of the preservation is between 20 and 50 years, depending on conditions. A large number of chemical compounds in combination are responsible for long-term biological protection. The most important aromatic hydrocarbons contained are acenaphthene, fluorene, phenanthrene, and anthracene. Under unfavourable conditions the addition of termiticides is necessary. In coastal waters, boring isopods (*Limnoria tripunctata*) can destroy wood in contact with water if insufficient preservative is used.

A large number of specifications and standards for coal-tar oils and their use exist. In Europe the Western European Institute for Wood Preservation (WEI) is in charge of the specification of tar oils. Fractions with a higher boiling range are used for wood in contact with soil or water because of the better long-term protection. Middle distillate fractions are suitable for wood subjected to less demanding conditions. The creosote is usually applied to wood by using the pressure process. Fence, tree, and vine posts are also treated using butt-end treatment.

Benzo[a]pyrene (BaP) present in coal tar-creosote oil is potentially carcinogenic. It is regarded as the key substance from which the content of carcinogenic, mutagenic, and teratogenic polycyclic aromatic hydrocarbons (PAHs) is derived. Polycyclic PAHs that can be present in coal-tar oil are naphthalene, methylnaphthalenes, dimethylnaphthalenes, acenaphthene, fluorene, methylfluorenes, phenanthrene, methylphenanthrenes, anthracene, methylanthracenes, fluoranthene, pyrene, and benzo[a]pyrene.

Because of their strong odour, colour, and tendency to sweat, particularly in impregnated pine wood, creosotes can be employed only for outdoor use. The combustibility of wood impregnated with creosote is not increased but actually decreased.

14.1.2 Water-soluble inorganic salt preservatives

The most important water-borne wood preservatives are those that are rendered insoluble by reaction of the components with one another in the wood and those that become tightly bonded to the wood fibres, so that they are resistant to being leached out. The first group comprises inorganic salt mixtures that are fixed in the wood by chromium(VI) compounds [alkali-metal chromates and dichromates, and chromium(VI) oxide]. Chromium is responsible for fixation. It is reduced during the fixation process to almost insoluble chromium(III). The second group consists of organic compounds that are applied to wood in the aqueous phase and become bonded to the fibres or are precipitated to form insoluble compounds in the wood. Waterborne preservatives are usually applied by the pressure process. However, tank and butt-end treatment processes can also be used. Certain fluorine and boron compounds, which are suitable wood preservatives, are not fixed and can therefore be used to protect only wood that is not exposed to moisture. Fluorine is partially fixed in the presence of chromium(VI) compounds.

14.1.3 Solvent-based preservatives

Solvent-borne wood preservatives are formulations comprising non-polar, organic solvents and fungicides or insecticides. The largest group consists of products with volatile (light) organic solvents. They have the advantage that a second coat can be applied to the protected wood, which is not possible for wood treated with heavy-oil-containing preparations. Depending on the purpose envisaged, solvent-borne wood preservatives frequently contain mixtures of various active substances to broaden the spectrum of activity, together with solubilisers, binders, fixing agents, agents that counteract efflorescence of the active substance, pigments, dyes, light stabilisers, and water repellents. Solvent-borne wood preservatives penetrate wood without altering its structure (dimensions). These preservatives can be applied by brushing, injecting, spraying, dipping, or the double vacuum process with or without pressure. Most of the active substances used in solvent-borne wood preservatives originated in other areas of use or in plant protection.

14.2 Use of wood preservatives

Wood preservatives can be used all over the wood chain: in forestry, sawmills, structural engineering and wood-based materials. The processes presented below suite wood protection in structural engineering.

Building timber, wooden building components, and wood-based construction materials can be attacked and destroyed by fungi and insects, and also by termites in tropical and subtropical regions, depending on the type of wood and the environmental conditions. The wood preservation measures needed depend on the stress and the endangerment of the wood, for example, through continuous contact with soil, continuous or frequent wetting by rain, or use indoors in wet areas. Water-soluble salt preservatives are most common. These are applied to the wood by vacuum, pressure, tank, impregnation, dipping, brushing, injection, spraying tunnel, or flow coating processes. In addition, solvent-containing, low-solvent, and recently solvent-free formulations are used to protect non-load-bearing building components, such as outer doors and windows. In countries, such as Japan, the United States, and Canada, where wood is used widely in building, flame retardant are important, either in the form of water-soluble salts or flame retardant paints or as coatings that form an insulating layer. Particular demands are made on wood preservatives and impregnating processes if, for example, highly stressed poles, railway ties, mine props, marine timber, cooling tower timbering, and posts need to be protected. Apart from a broad spectrum of activity, good penetration capacity and high resistance to leaching are required.

14.2.1 Application processes for wood preservatives for structural engineering

Pressure processes have been used for more than 150 years for impregnating wood with preservatives. The distribution of preservative should be as deep and uniform as possible in the parts of the wood that can be impregnated. Heartwood generally absorbs very little preservative. Different types of wood vary in their ease of impregnation. Treatment using the pressure process is absolutely necessary for wood in constant contact with water or soil. The process is often employed also for treating wood used outdoors in frequent contact with water. The choice of pressure process depends on the intended use, the nature and quantity of the preservative and the moisture level of the wood at the time of treatment.

Reduced pressure (vacuum) processes are used for treating dried construction wood, in particular, with solvent-containing preservatives (double vacuum). Water-soluble preservatives (simple vacuum) are also suitable for this process. Higher penetration depths are achieved than with the dipping process, and less preservative is used than with the pressure process. Vacuum impregnation is not suitable for wood that is in continuous contact with water or soil. Green wood (e.g. poles or round wood) can be impregnated with salt preservative solutions by sap displacement.

In pressure-free processes, no plants with pressure vessels and pressure or vacuum pumps are necessary for applying the preservative. The wood absorbs preservative solutions through capillary forces and diffusion. Penetration of the preservative can also be promoted by the hydrostatic pressure of the liquid covering the wood in the impregnating container and by temperature differences in the impregnating liquids (hot-cold).

Short-term processes are wood treatment methods that are easily carried out and generally do not require expensive equipment. The preservative solution is applied to the wood surface only once or twice. Since preservatives penetrate the wood to a maximum of a few millimetres, only surface protection is achieved. These processes are therefore suitable only for wood that is not exposed to any particular hazard. The simplest treatment methods are brushing and spraying. Spraying should be carried out only if precautionary measures are taken with regard to environmental safety and health. Solvent-containing preparations and oily formulations that can be diluted with water are mainly used. The quantity of preservative absorbed essentially depends on the type of wood and the nature of its surface. Two treatments are generally necessary. If water-soluble preservatives are used, wet-in-wet processing is generally employed to achieve greater depths of penetration.

14.3 Preservatives in ecoinvent database

The background information to the processes are given in Tab. 14.1, flows are shown in:

- wood preservative, creosote, at plant Fig. 14.1
- wood preservative, inorganic salt, containing Cr, at plant Fig. 14.2
- wood preservative, organic salt, Cr-free, at plant Fig. 14.3

The material input data for the salt preservatives are confidential. Thus only the energy inputs are visible in the figures.

14.3.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Data quality for the preservative salts is good for the specific products. How good this data fit other products is not known. Creosote data are rather poor.

Tab. 14.1 Ecoinvent meta information for the preservative production processes

Name	wood preservative, creosote, at plant	wood preservative, inorganic salt, containing Cr, at plant	wood preservative, organic salt, Cr-free, at plant
Location	RER	RER	RER
Infrastructure Process	0	0	0
Unit	kg	kg	kg
Data Set Version	2.0	2.0	2.0
Included Processes	Includes the inputs to the production processes, transports of those inputs and the process emissions as far as they are available.	Includes the inputs to the production processes and transports of those inputs. No process emission data are available.	Includes the inputs to the production processes and transports of those inputs. No process emission data are available.
Amount	1	1	1
Local Name	Holzschutzmittel, Teeröl, ab Werk	Holzschutzmittel, Salz anorganisch, Cr-haltig, ab Werk	Holzschutzmittel, Salz organisch, Cr-frei, ab Werk
Synonyms			
General Comment to reference function		Data for one specific product. Material Input data are confidential.	Data for one specific product. Material Input data are confidential.
Start Date	1992	2000	2000
End Date	2002	2002	2002
Data Valid For Entire Period	1	1	1
Other Period Text			
Geography text	Data for UK used for central Europe	Data for Germany used for central Europe	Data for Germany used for central Europe
Technology text	Distillation of the coal tar to creosote	Mixing of ingredients	Mixing of ingredients
Representativeness [%]			
Production Volume	unknown	unknown	unknown
Sampling Procedure	Literature	Literature	Literature
Extrapolations	see Geography	see Geography	see Geography
Uncertainty Adjustments	The assumption that the technology is representative for central Europe is considered in the uncertainties.	The assumption that the technology is representative for central Europe is considered in the uncertainties.	The assumption that the technology is representative for central Europe is considered in the uncertainties.

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Fuel oil high sulfur content Euro, in boiler 1 MW	wood preservative, creosote, at plant		For the distillation of the coal tar to creosote, data is used from 1992 stemming from an english company (LCA "Creosote treated distribution poles case study" , Hillier 1997); it is estimated that 35% of the raw tar can be used for the production of creosote; allocation is made by mass; estimate of airborne emissions 2% HC; emissions into water are taken from an Anonymous 1990	oil	heating systems	No	RER	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	4.20E+00	MJ	Hillier 1997	1	1.16	(1,4,3,3,1,1,1)
Coal tar coke			Starting material coal tar is inventoried as coal tar coke (source: ESU)	hard coal	fuels	No	RER	hard coal coke, at plant	2.92E+01	MJ	Hillier 1997	1	1.16	(1,4,3,3,1,1,3)
Electricity medium voltage - at grid UCTE				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.11E+00	kWh	Hillier 1997	1	1.16	(1,4,3,3,1,1,2)
plant			organic plant as proxy	chemicals	organics	Yes	RER	chemical plant, organics	4.00E-10	unit	estimated	1	3.79	(5,5,5,5,4,5,9)
		Wood preservative, creosote, at plant		paintings	production	No	RER	wood preservative, creosote, at plant	1.00E+00	kg				
		Waste heat into air		air	unspecified			Heat, waste	3.99E+00	MJ	calculated	1	1.16	(1,4,3,3,1,1,13)
		NMVOC		air	unspecified			NMVOC, non-methane volatile organic compounds, unspecified origin	2.05E-02	kg	estimated	1	2.11	(4,3,3,3,1,5,23)
		COD	calculated from BOD, assuming C6H12O6	water	unspecified			COD, Chemical Oxygen Demand	1.00E-03	kg	calculated	1	1.75	(3,5,4,5,3,5,32)
		DOC	calculated from BOD, assuming C6H12O6	water	unspecified			DOC, Dissolved Organic Carbon	3.92E-04	kg	calculated	1	1.75	(3,5,4,5,3,5,32)
		TOC	calculated from BOD, assuming C6H12O6	water	unspecified			TOC, Total Organic Carbon	3.92E-04	kg	calculated	1	1.75	(3,5,4,5,3,5,32)
		BOD		water	unspecified			BOD5, Biological Oxygen Demand	1.00E-03	kg	Anonymous 1990	1	1.75	(3,5,4,5,3,5,32)

Fig. 14.1 Flows for "wood preservative, creosote, at plant" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent									Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment	
Boric acid (H ₃ BO ₃)	wood preservative, inorganic salt, containing Cr, at plant		Quantities (Application in Switzerland): hazard class 4 => 12 kg wood preservative for 1m ³ wood; hazard class 3 => 7.5 kg kg wood preservative for 1m ³ wood	chemicals	inorganics	No	RER	boric acid, anhydrous, powder, at plant	confidential	kg	Künniger et al. 2000	1	1.32	(4,4,1,3,1,5,4)	
Chromium acid				chemicals	inorganics	No	RER	chromium oxide, flakes, at plant	confidential	kg	Künniger et al. 2000	1	1.32	(4,4,1,3,1,5,4)	
Copper(II)oxide (CuO)				chemicals	inorganics	No	RER	copper oxide, at plant	confidential	kg	Künniger et al. 2000	1	1.32	(4,4,1,3,1,5,4)	
Electricity medium voltage - at grid UCTE				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	confidential	kWh	Künniger et al. 2000	1	1.24	(1,4,1,3,1,5,2)	
Transport rail			Chemicals: 600 km	transport systems	train	No	RER	transport, freight, rail	2.94E-01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport lorry			Chemicals: 100 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	4.90E-02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Tap water				water supply	production	No	RER	tap water, at user	confidential	kg	Künniger et al. 2000	1	1.24	(1,4,1,3,1,5,4)	
plant			organic plant as proxy	chemicals	organics	Yes	RER	chemical plant, organics	4.00E-10	unit	estimated	1	3.79	(5,5,5,5,4,5,9)	
		Wood preservative containing chrome, at plant		paintings	production	No	RER	wood preservative, inorganic salt, containing Cr, at plant	1.00E+00	kg					
		Waste heat into air		air	unspecified			Heat, waste	1.30E-01	MJ	calculated	1	1.24	(1,4,1,3,1,5,13)	

Fig. 14.2 Flows for "wood preservative, inorganic salt, containing Cr, at plant" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent									Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment	
Boric acid (H3BO3)	wood preservative, organic salt, Cr-free, at plant		Quantities (Application in Switzerland): hazard class 4 => 3 kg wood preservative for 1m3 wood; hazard class 3 => 2 kg wood preservative for 1m3 wood	chemicals	inorganics	No	RER	boric acid, anhydrous, powder, at plant	confidential	kg	Künniger et al. 2000	1	1.32	(4,4,1,3,1,5,4)	
Ethylenediamine				chemicals	organics	No	RER	ethylenediamine, at plant	confidential	kg	Künniger et al. 2000	1	1.32	(4,4,1,3,1,5,4)	
Copper (II) carbonate				chemicals	inorganics	No	RER	copper carbonate, at plant	confidential	kg	Künniger et al. 2000	1	1.32	(4,4,1,3,1,5,4)	
Copper-HDO			Total Copper-HDO 0.061 kg. Balanced as ethanolamine and copperoxide	chemicals	organics	No	RER	triethanolamine, at plant	confidential	kg	Künniger et al. 2000	1	1.32	(4,4,1,3,1,5,4)	
Copper-HDO				chemicals	inorganics	No	RER	copper oxide, at plant	confidential	kg	Künniger et al. 2000	1	1.32	(4,4,1,3,1,5,4)	
Electricity medium voltage - at grid UCTE				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	confidential	kWh	Künniger et al. 2000	1	1.24	(1,4,1,3,1,5,2)	
Transport rail			Chemicals: 600 km	transport systems	train	No	RER	transport, freight, rail	3.84E-01	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport lorry			Chemicals: 100 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	6.39E-02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Tap water				water supply	production	No	RER	tap water, at user	confidential	kg	Künniger et al. 2000	1	1.24	(1,4,1,3,1,5,4)	
plant			organic plant as proxy	chemicals	organics	Yes	RER	chemical plant, organics	4.00E-10	unit	estimated	1	3.79	(5,5,5,4,5,9)	
		Wood preservative chrome-free, at plant		paintings	production	No	RER	wood preservative, organic salt, Cr-free, at plant	1.00E+00	kg					
		Waste heat into air		air	unspecified			Heat, waste	7.20E-02	MJ	calculated	1	1.24	(1,4,1,3,1,5,13)	

Fig. 14.3 Flows for "wood preservative, organic salt, Cr-free, at plant" and its representation in the ecoinvent database

14.4 Preservative treatment in ecoinvent database

The background information to the processes are given in Tab. 14.2, flows are shown in:

- preservative treatment, logs, pressure vessel Fig. 14.4
- preservative treatment, sawn timber, pressure vessel Fig. 14.5

14.4.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Data are reliable for the process described and despite their age of good quality.

Tab. 14.2 Ecoinvent meta information for the preservative treatment processes

Name	preservative treatment, logs, pressure vessel	preservative treatment, sawn timber, pressure vessel
Location	RER	RER
Infrastructure Process	0	0
Unit	m ³	m ³
Data Set Version	2.0	2.0
Included Processes	Impregnation process only. No wood nor preservative included. No transports included. No data on waste water available.	Impregnation process only. No wood nor preservative included. No transports included. No data on waste water available.
Amount	1	1
Local Name	Kesseldruckimprägnierung, Rundholz	Kesseldruckimprägnierung, Schnittholz
Synonyms	mineralize//creosote	mineralize//creosote
General Comment to reference function	For wood in permanent soil contact 6-12 kg/m ³ of Cr-containing, 3-4 kg/m ³ of Cr-free preservative or 115 kg/m ³ creosote is needed. For weather exposed wood 4-7.5 of Cr-containing or 2-3 kg/m ³ of Cr-free preservative is required.	For wood in permanent soil contact 9-18 kg/m ³ of Cr-containing or 4.5-6 kg/m ³ of Cr-free preservative is needed. For weather exposed wood 6-11.25 of Cr-containing or 3-4.5 kg/m ³ of Cr-free preservative is required.
Start Date	2000	2000
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Germany used for central Europe	Data for Germany used for central Europe
Technology text	Pressure vessel treatment	Pressure vessel treatment
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	Literature
Extrapolations	see Geography	see Geography
Uncertainty Adjustments	The assumption that the technology is representative for central Europe is considered in the uncertainties.	The assumption that the technology is representative for central Europe is considered in the uncertainties.

General Flow information					Representation in ecoinvent									Uncertainty information		
Input		Process Name		Output	Remarks	Cate gory	Sub category	Infra struc ture	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Electricity medium voltage - at grid UCTE	→	preservative treatment, logs, pressure vessel				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.17E+00	kWh	Künniger et al. 2000	1	1.24	(1,4,1,3,1,5,2)
Process and cooling water	→					resource	in water			Water, cooling, unspecified natural origin	3.00E-01	m3	Künniger et al. 2000	1	1.24	(1,4,1,3,1,5,4)
plant	→					wooden materials	extraction	Yes	RER	preservative treatment, infrastructure	3.33E-06	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
				Impregnation pressure vessel (dried logs)	without wood and wood preservatives	wooden materials	refinement	No	RER	preservative treatment, logs, pressure vessel	1.00E+00	m3				
				Waste heat into air		air	unspecified			Heat, waste	1.50E+01	MJ	calculated	1	1.24	(1,4,1,3,1,5,13)

Fig. 14.4 Flows for "preservative treatment, logs, pressure vessel" and its representation in the ecoinvent database

General Flow information					Representation in ecoinvent								Uncertainty information			
Input		Process Name		Output	Remarks	Cate gory	Sub category	Infra struc ture	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Electricity medium voltage - at grid UCTE	→	preservative treatment, sawn timber, pressure vessel				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	1.02E+01	kWh	Künniger et al. 2000	1	1.24	(1,4,1,3,1,5,2)
Process and cooling water	→					resource	in water			Water, cooling, unspecified natural origin	1.50E-01	m3	Künniger et al. 2000	1	1.24	(1,4,1,3,1,5,4)
plant	→					wooden materials	extraction	Yes	RER	preservative treatment, infrastructure	3.33E-06	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
				Impregnation pressure vessel (dried sawn timber)	without wood and wood preservatives	wooden materials	refinement	No	RER	preservative treatment, sawn timber, pressure vessel	1.00E+00	m3				
				Waste heat into air		air	unspecified			Heat, waste	3.65E+01	MJ	calculated	1	1.24	(1,4,1,3,1,5,13)

Fig. 14.5 Flows for "preservative treatment, sawn timber, pressure vessel" and its representation in the ecoinvent database

15 Charcoal

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Summary

This section investigates the production of charcoal as a pre-product for technical processes. The investigation is based on literature information. An important aspect of charcoal production is the use of wood from native forests in developing countries. The inventory of charcoal production can be used for the assessment of this material used for technical applications.

15.1 Introduction

This chapter investigates the production of charcoal from wood as a pre-product for technical processes. Charcoal is produced mainly in small-scale production plants in the Southern Hemisphere. Large quantities are used in metal processing.

15.2 Reserves and Resources

Wood is the basic resource for the production of charcoal. For environmental assessments the type of wood growing is important. Only a sustainable forest management with a organized planting of new trees after the cutting can ensure that the CO₂ uptake during growing equals the releases during use and combustion or degradation. A large portion of today production comes from South America and Asia from countries with an unsustainable use of rain forests ((Eikeland et al. 2001). In Brazil only 39% of the charcoal is produced from forest plantations while the majority comes from native forests ((Rosillo-Calle et al. 1996)).

15.3 Production and Use of Charcoal

Charcoal has been produced since prehistoric times. First it was used as a smokeless, odourless fuel for heating and cooking. In industry, charcoal is used metallurgically, e.g. as a reduction agent for the production of magnesium or silicon.

Tab. 15.1 shows some specifications for the investigated charcoal. The heating value of charcoal ranges between 29 and 33MJ/kg. The carbon content of the charcoal produced ranges between 78 and 90-wt% and is assumed here to be 80% ((Brocksiepe 1997)). Thus 2.93kg of CO₂ are formed during the combustion of 1kg charcoal.

Tab. 15.1 Specifications of charcoal in this study (partly based on (Rosillo-Calle et al. 1996))

Parameter	Range	Value
Lower heating value	29-33MJ/kg	30 MJ/kg
Carbon content	70-80%	80%
Ash content	3-5%	4%
Volatiles	20-25%	-
Moisture	5-35%	-
CO ₂ emission during combustion		2.93kg/kg
Density	200-280kg/m ³	250kg/m ³
	(Rosillo-Calle et al. 1996)	

The worldwide production was estimated to be 18 million tonnes in 1984 (Brocksiepe 1997). Charcoal production is an integral part of the informal economy in many developing countries. The largest producer is Brazil with 7.3 million tonnes in 1992 (Rosillo-Calle et al. 1996). The usage varies from country to country. In the USA 95% of production are used for barbecuing, while in South America nearly all of the charcoal production is consumed by iron works (Brocksiepe 1997). In developing countries charcoal is an important energy source for cooking and heating (Rosillo-Calle et al. 1996). In Norway and other countries charcoal is also used for the production of silicon (Eikeland et al. 2001).

In Switzerland about 7000t of charcoal are used per year. Most of it is imported. Only small amounts mainly for barbecuing are produced from Swiss wood. It is difficult to extend this production due to the labour intensive process and thus comparable high costs for the product (Duss 2002).

15.4 System Characterisation

Different industrial and small-scale processes are known for the production of charcoal (Briane et al. 1985; Brocksiepe 1997; Emrich 1985; Rosillo-Calle et al. 1996). There are many techniques, both simple and sophisticated, for producing charcoal. Necessary is a restricted supply of air during carbonisation. Two main methods are used in Brazil (Rosillo-Calle et al. 1996):

- stacking the timber on the ground and covering it with soil, often referred as an earth-clamp or earth-mound kiln
- digging a hole into which the timber is placed and which is then covered with soil, commonly referred to as a pit kiln

A series of decomposition reactions take place when wood is heated in the absence of atmospheric oxygen above 100°C. Macromolecular chains of the wood components depolymerise to a variety of smaller molecules. From 100-170°C, the free and bound water is evaporated. At the same temperature small amounts of carbon dioxide, carbon monoxide, formic acid, and acetic acid are formed. With an increase of the temperature to 200 to 350°C increasing amounts of reaction water, methanol, and acetic acid are released. To obtain the desired carbon content, which depends on the final temperature, post heating is necessary (Brocksiepe 1997).

15.5 Life Cycle Inventory

Literature data are used in order to investigate the production process. The inventory for charcoal production is shown in Tab. 15.2. The efficiency of charcoal making shows large variations from 9 to 35% depending on the state of technology applied (Rosillo-Calle et al. 1996).

Here it is assumed that the by-products of the pyrolysis (methanol, acetic acid, etc.) can be used (0.23 kg/kg wood), but all environmental impacts of the production are allocated to the main product charcoal. Here we assume the production of 0.3 kg charcoal/kg wood (Briane et al. 1985 :28-29). The wood consumption is allocated to the charcoal. For the production of 1 kg charcoal, including the drying of the wood, 2.5 MJ of heat and 0.075 kWh of electricity are needed, as well as 50 litres of cooling water. About 1.6 MJ/kg of heat is released during the carbonisation. It is assumed that the remaining heat of 0.9 MJ/kg necessary for production is provided by wood boilers (Brocksiepe 1997).

The composition of flue gas from the pyrolysis has been investigated in a laboratory (Briane et al. 1985 :29). The figures for leafy wood given as percentage of wood input have been used for this inventory. All gaseous emissions are allocated to the charcoal production. A waste gas treatment is not assumed due to the normally small capacity of the plants. Data for particle emissions were not available. They have been assumed with data from carbon black production (Berdowski et al. 2002).

Data for the infrastructure of production plants were not available. Often there are no fixed plants, but production places near the forest with no fixed infrastructure parts. Construction work is mainly hu-

man labour. Land use can be considered neglect able in comparison to the use for the forest. Thus infrastructure is not assessed.

The inventory is calculated with wood from forest plantation. Thus the full CO₂ uptake has been considered. For inventories where charcoal from Asia are South America is an important input, it is recommended to calculate a part of the CO₂ emissions as fossil emissions in a sensitivity analysis.

Tab. 15.2 Unit process raw data and data quality indicators for the production of charcoal. References and assumptions for the inventory are presented in the text

	Name	Location	Category	Subcategory	Infrastructure	Unit	charcoal, at plant	Uncertainty	GeneralComment
	Location InfrastructureProcess Unit						GLO 0 kg	Deviation max.	
output	charcoal, at plant	GLO	-	-	0	kg	1.00E+0		
resource, in water	Water, cooling, unspecified natural origin	-	resou in water	-	-	m3	5.00E-2	1 1.40	(3,5,1,1,3,5); Literature
technosphere	logs, hardwood, at forest	RER	-	-	0	m3	5.1282E-3	1 1.35	(3,4,2,3,3,5); Literature, 0.3kg charcoal per kg of wood, allocation to charcoal production
	electricity, medium voltage, production UCTE, at grid	UCTE	-	-	0	kWh	7.50E-2	1 1.26	(3,4,2,3,1,5); Literature
	wood chips, from forest, hardwood, burned in furnace 50kW	CH	-	-	0	MJ	9.00E-1	1 1.26	(3,4,2,3,1,5); Literature
	transport, lorry >16t, fleet average	RER	-	-	0	tkm	3.94E-7	1 2.09	(4,5,na,na,na,na); Standard distance 50km
emission air, low population density	Heat, waste	-	air	low population	-	MJ	1.16E+1	1 1.26	(3,4,2,3,1,5); Literature
	Carbon dioxide, biogenic	-	air	low population	-	kg	2.695E+0	1 1.26	(3,4,2,3,1,5); Calculated
	Carbon monoxide, biogenic	-	air	low population	-	kg	1.90E-1	1 5.08	(3,4,2,3,1,5); Laboratory measurement 5.7% of wood used in process as emission released
	Methane, biogenic	-	air	low population	-	kg	4.03E-2	1 1.59	(3,4,2,3,1,5); Laboratory measurement 1.21% of wood used in process as emission released
	Ethene	-	air	low population	-	kg	2.33E-3	1 2.93	(5,5,1,3,5,na); Laboratory measurement 0.07% of wood used in process as emission released
	Ethane	-	air	low population	-	kg	7.00E-3	1 2.08	(3,4,2,3,1,5); Laboratory measurement 0.21% of wood used in process as emission released
	Particulates, < 2.5 um	-	air	low population	-	kg	4.50E-4	1 3.91	(5,na,2,5,5,na); Estimation
	Particulates, > 2.5 um, and < 10um	-	air	low population	-	kg	5.00E-5	1 2.90	(5,na,2,5,5,na); Estimation
	Particulates, > 10 um	-	air	low population	-	kg	5.50E-5	1 2.47	(5,na,2,5,5,na); Estimation

15.6 Data Quality Considerations

Tab. 15.2 shows the data quality indicators for the inventory of charcoal production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

A large uncertainty exists concerning the CO₂ uptake and release in the life cycle which is not directly quantified here. For charcoal from developing countries comes from native forest. Thus this charcoal use cannot be considered to be CO₂ neutral.

15.7 Cumulative Results and Interpretation

Results of the cumulative inventory can be downloaded from the database.

15.8 Conclusions

The inventory of charcoal production can be used for the assessment of this material used for technical applications.

15.9 EcoSpold Meta Information

ReferenceFunction	401	Name	charcoal, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	41
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of charcoal from forest wood including emissions. Infrastructure can be considered of minor importance and is neglected.
	404	Amount	1
	490	LocalName	Holzkohle, ab Werk
	491	Synonyms	carbon, Aktivkohle
	492	GeneralComment	A series of decomposition reactions take place when wood is heated in the absence of atmospheric oxygen above 100°C. Macromolecular chains of the wood components depolymerise to a variety of smaller molecules.
	494	InfrastructureIncluded	
	495	Category	wood energy
	496	SubCategory	fuels
	497	LocalCategory	Holzenergie
	498	LocalSubCategory	Brenn- und Treibstoffe
	499	Formula	C
	501	StatisticalClassification	
	502	CASNumber	7440-44-0
TimePeriod	601	StartDate	1985
	602	EndDate	1996
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	Most of global production takes place in South America and Asia. Background data were available only for Europe.
Technology	692	Text	Pyrolysis of wood in a small scale production plant (kiln). No waste gas or effluent treatment. Production of charcoal with 80% carbon content.
Representativeness	722	Percent	
	724	ProductionVolume	18 Mio. Tonnes worldwide in 1984
	725	SamplingProcedure	Literature survey
	726	Extrapolations	From single data.
	727	UncertaintyAdjustments	none
DataGenerator	751	Person	41
AndPublication	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	9
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	chapter charcoal

16 Auxiliary modules: processes and infrastructure

16.1 Sawmill and planing mill

The land and machinery use in a sawmill is estimated for two different sizes of plants. From these estimates the figures for a medium sized plant with 50'000 m³/a sawn timber output are calculated.

Planing mills usually are located on the sites of sawmills. Thus only the built-up area is considered as land occupied. The hall includes space for storing the dry wood to be planed and the planed wood to be delivered. The figures have been estimated for a medium sized plant with 25'000 m³/a planed timber output.

The plants are assumed to operate 50 years.

The background information to the processes are given in Tab. 16.1, flows are shown in:

- sawmill Fig. 16.1
- planing mill Fig. 16.2

16.1.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Although data quality is poor, they are regarded good enough for the purpose of inventory of the infrastructure of wood products.

Tab. 16.1 Ecoinvent meta information for the sawmill and planing mill infrastructure

Name	sawmill	planing mill
Location	RER	RER
Infrastructure Process	1	1
Unit	unit	unit
Data Set Version	2.0	2.0
Included Processes	The land area and the material and their transports needed for building and disposal of a sawmill with a yearly output of 50'000 m ³ sawn timber is included. Land used for air drying of wood is not included.	The land area and the material and their transports needed for building and disposal of a planing mill with a yearly output of 25'000 m ³ timber is included. The space for storing dried wood before and after planing and for packageing etc. is included.
Amount	1	1
Local Name	Sägerei	Hobelwerk
Synonyms	Sägewerk//sawing mill	
General Comment to reference function	The service life is assumed to be 50 years. Thus the life time output is 2'500'000 m ³ sawn timber. This data must only be used to consider the infrastructure needed for sawing wood if this is not relevant.	The service life is assumed to be 50 years. Thus the life time output is 1'250'000 m ³ planed timber. This data must only be used to consider the infrastructure needed for planing wood if this is not relevant.
Start Date	2002	2002
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geograhpy text	Data are derived by averaging rough estimates of a small sawmill in Switzerland and a large sawmill in Germany. Estimates are so rough that the Geograhpy is not relevant.	Estimate is so rough that the Geograhpy is not relevant.
Technology text	Sawmill with it's input wood storage place, a lop station, a hall with the saw and a conveyer belt.	Medium sized planing mill, located with sawmill
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	estimation	estimation
Extrapolations	none	none
Uncertainty Adjustments	none	none

General Flow information					Representation in ecoinvent								Uncertainty information				
Input		Process Name		Output	Remarks	Cate gory	Sub category	Infra struc ture	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Land transformation from unknown	→	sawmill				resource	land			Transformation, from unknown	7.88E+04	m2	estimated		1	2.16 (3,5,1,3,3,5,8)	
Land transformation to industrial built up	→					resource	land			Transformation, to industrial area, built up	7.78E+03	m2	estimated		1	2.16 (3,5,1,3,3,5,8)	
Land transformation industrial vegetation	→					resource	land			Transformation, to industrial area, vegetation	5.06E+04	m2	estimated		1	2.16 (3,5,1,3,3,5,8)	
Land transformation industrial unspecific	→					resource	land			Transformation, to industrial area	2.04E+04	m2	estimated		1	2.16 (3,5,1,3,3,5,8)	
Land occupation as industrial built up	→					resource	land			Occupation, industrial area, built up	3.89E+05	m2a	estimated		1	1.69 (3,5,1,3,3,5,7)	
Land occupation as industrial vegetation	→					resource	land			Occupation, industrial area, vegetation	2.53E+06	m2a	estimated		1	1.69 (3,5,1,3,3,5,7)	
Land occupation as industrial unspecific	→					resource	land			Occupation, industrial area	1.02E+06	m2a	estimated		1	1.69 (3,5,1,3,3,5,7)	
Hall	→					construction processes	buildings	Yes	CH	building, hall	7.78E+03	m2	estimated		1	3.15 (3,5,1,3,3,5,9)	
Additional foundation of hall	→					Only special fundation for heavy machines	construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	7.60E+02	m3	estimated		1	3.15 (3,5,1,3,3,5,9)
Conveyer belt	→						construction processes	machinery	Yes	RER	conveyor belt, at plant	2.24E+03	m	estimated		1	3.15 (3,5,1,3,3,5,9)
Machines (cast iron)	→						metals	extraction	No	RER	cast iron, at plant	8.17E+05	kg	estimated		1	3.15 (3,5,1,3,3,5,9)
Machines (steel)	→						metals	extraction	No	RER	steel, low-alloyed, at plant	8.55E+05	kg	estimated		1	3.15 (3,5,1,3,3,5,9)
Steel sheet rolling	→						metals	processing	No	RER	sheet rolling, steel	8.55E+05	kg	estimated		1	3.15 (3,5,1,3,3,5,9)
Transport rail	→					Steel / cast iron: 200 km,	transport systems	road	No	RER	transport, lorry >16t, fleet average	3.35E+05	tkm	estimated		1	2.09 (4,5,nA,nA,nA,nA,5)
Transport lorry	→					Concrete: 50 km, Steel / cast iron: 100 km	transport systems	train	No	RER	transport, freight, rail	2.51E+05	tkm	estimated		1	2.09 (4,5,nA,nA,nA,nA,5)
	→		Sawmill infrastructure	relates to 2.5E6 m3 sawn timber output in 50 years		wooden materials	extraction	Yes	RER	sawmill	1.00E+00	unit					
	→		Steel to recycling	not balanced							2.24E+03	kg					
	→		Disposal of foundation			waste management	building demolition	No	CH	disposal, building, reinforced concrete, to final disposal	1.67E+06	kg	calculated		1	3.15 (3,5,1,3,3,5,9)	

Fig. 16.1 Flows for "sawmill" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent									Uncertainty information		
Input		Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Land transformation from unknown	→	planing mill			resource	land			Transformation, from unknown	1.17E+04	m2	estimated	1	2.16	(3,5,1,3,3,5,8)
Land transformation to industrial built up	→				resource	land			Transformation, to industrial area, built up	1.17E+04	m2	estimated	1	2.16	(3,5,1,3,3,5,8)
Land occupation as industrial built up	→				resource	land			Occupation, industrial area, built up	5.84E+05	m2a	estimated	1	1.69	(3,5,1,3,3,5,7)
Hall	→				construction processes	buildings	Yes	CH	building, hall	1.17E+04	m2	estimated	1	3.15	(3,5,1,3,3,5,9)
Additional foundation of hall	→			Only special foundation for heavy machines	construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	1.14E+03	m3	estimated	1	3.15	(3,5,1,3,3,5,9)
Machines (steel)	→				metals	extraction	No	RER	steel, low-alloyed, at plant	7.78E+05	kg	estimated	1	3.15	(3,5,1,3,3,5,9)
Steel sheet rolling	→				metals	processing	No	RER	sheet rolling, steel	7.78E+05	kg	estimated	1	3.15	(3,5,1,3,3,5,9)
Transport rail	→			Steel: 200 km,	transport systems	road	No	RER	transport, lorry >16t, fleet average	1.56E+05	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Transport lorry	→			Concrete: 50 km, Steel: 100 km	transport systems	train	No	RER	transport, freight, rail	2.03E+05	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
	→		Planning mill infrastructure	relates to 1.25E6 m3 planed wood output in 50 years	wooden materials	extraction	Yes	RER	planing mill	1.00E+00	unit				
	→		Steel to recycling	not balanced						1.17E+04	kg				
	→		Disposal of foundation		waste management	building demolition	No	CH	disposal, building, reinforced concrete, to final disposal	2.51E+06	kg	calculated	1	3.15	(3,5,1,3,3,5,9)

Fig. 16.2 Flows for "planing mill" and its representation in the ecoinvent database

16.2 Technical wood drying, infrastructure

Kilns for wood drying are usually located where sawmills and planing mills are. Thus only the built-up area is considered as land occupied. The Kiln itself is a shed of 5 m by 3 m, 4 m in height. Thus a wood volume of 30 m³ can be dried per charge. Its outer skin is made of polystyrene foam insulated aluminium panels. It is assumed that the kiln is operated for 50 years and that it is in use for 90% of the time. The drying time and the resulting demand of the infrastructure for the different processes are given in Tab. 16.2.

Tab. 16.2 Drying times and use of kiln for drying processes

Drying time ¹	70% --> 20%	70% --> 10%	20% --> 10%	Unit
Hardwood, 50 mm thickness		30	10	Days
Softwood, 50 mm thickness	3	9		Days
Fraction of Kiln needed for drying 1 m ³				
Hardwood		6.09E-05	2.03E-05	Unit/m ³
Softwood	6.09E-06	1.83E-05		Unit/m ³

The background information to the process "technical wood drying, infrastructure" are given in Tab. 16.3, flows are shown in Fig. 16.3.

16.2.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Although data quality is poor, they are good enough to inventory the impacts of the infrastructure.

¹ Drying times from: <http://www.brunner-hildebrand.com/de/main/home.shtml> (accessed Oct. 2002)

Tab. 16.3 Ecoinvent meta information for the technical wood drying infrastructure

Name	technical wood drying, infrastructure
Location	RER
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	The land area and the materials and their transports needed for building and disposal of a kiln with a per charge loading capacity of 30 m ³ wood (under bark) are included. Not included is the land used for storage etc.
Amount	1
Local Name	Technische Holzrocknung, Infrastruktur
Synonyms	kiln//Trocknungskammer
General Comment to reference function	The service life is assumed to be 50 years. The kiln has a capacity of 30 m ³ wood per charge. Drying time depends on wood type and in- and output moisture content. This data must only be used to consider the infrastructure needed for kiln drying wood if this is not relevant.
Start Date	2002
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geograhpy text	Estimate is so rough that the Geograhpy is not relevant.
Technology text	Medium sized kiln of polystyren foam insulated aluminium panels, located with sawmill
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	none
Uncertainty Adjustments	none

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Land transformation from unknown	technical wood drying, infrastructure			resource	land			Transformation, from unknown	1.50E+01	m2	estimated	1	2.16	(3,5,1,3,3,5,8)
Land transformation to industrial built up				resource	land			Transformation, to industrial area, built up	1.50E+01	m2	estimated	1	2.16	(3,5,1,3,3,5,8)
Land occupation as industrial built up				resource	land			Occupation, industrial area, built up	7.50E+02	m2a	estimated	1	1.69	(3,5,1,3,3,5,7)
Foundation of kiln			concrete, thickness 0.3 m	construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	4.50E+00	m3	estimated	1	3.15	(3,5,1,3,3,5,9)
Insulation material				insulation materials	production	No	RER	polystyrene foam slab, at plant	4.51E+02	kg	estimated	1	3.15	(3,5,1,3,3,5,9)
Aluminium				metals	extraction	No	RER	aluminium, production mix, wrought alloy, at plant	6.40E+02	kg	estimated	1	3.15	(3,5,1,3,3,5,9)
Aluminium sheet rolling				metals	processing	No	RER	sheet rolling, aluminium	6.40E+02	kg	estimated	1	3.15	(3,5,1,3,3,5,9)
Transport rail			Aluminium, insulation material: 200 km, Concrete: 50 km, aluminium, insulation material: 100 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.18E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Transport lorry				transport systems	train	No	RER	transport, freight, rail	6.04E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
		Kiln	relates to Kiln for 30 m3 of wood per charge and 50 years	wooden materials	extraction	Yes	RER	technical wood drying, infrastructure	1.00E+00	unit				
		Aluminium to recycling	not balanced						4.51E+02	kg				
		Disposal of foundation		waste management	building demolition	No	CH	disposal, building, reinforced concrete, to final disposal	9.90E+03	kg	calculated	1	3.15	(3,5,1,3,3,5,9)
		Disposal of insulation material		waste management	building demolition	No	CH	disposal, building, polystyrene isolation, flame-retardant, to final disposal	4.51E+02	kg	calculated	1	3.15	(3,5,1,3,3,5,9)

Fig. 16.3 Flows for "technical wood drying, infrastructure" and its representation in the ecoinvent database

16.3 Wooden board manufacturing plant

Two types of plants are here separately modelled, one fabricating organic bonded boards and one manufacturing mineral bonded boards because they usually have different sizes.

The plant for organically bonded boards in this project is assumed to produce 600'000 m³/year. The plant for cement bonded boards is assumed much smaller with only 50'000 m³/year output. The machines used (chopper, press and others) are balanced as steel. The additional foundation of the hall is needed for the heavy machines.

The background information to the processes are given in Tab. 16.4, flows are shown in:

- wooden board manufacturing plant, organic bonded boards Fig. 16.4
- wooden board manufacturing plant, cement bonded boards Fig. 16.5

16.3.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Although data quality is poor, they are good enough to inventory the impacts of the infrastructure.

Tab. 16.4 Ecoinvent meta information for the wooden board manufacturing plants

Name	wooden board manufacturing plant, organic bonded boards	wooden board manufacturing plant, cement bonded boards
Location	RER	RER
Infrastructure Process	1	1
Unit	unit	unit
Data Set Version	2.0	2.0
Included Processes	The land area and the material and their transports needed for building and disposal of a plant for organic bonded wooden boards with a yearly output of 600'000 m ³ is included.	The land area and the material and their transports needed for building and disposal of a plant for cement bonded wooden boards with a yearly output of 50'000 m ³ is included.
Amount	1	1
Local Name	Holzplattenwerk, organisch gebundene Platten	Holzplattenwerk, zementgebundene Platten
Synonyms	Spanplattenwerk//MDF-Werk//OSB-Werk//Sperrholzwerk//Faserplattenwerk	
General Comment to reference function	The service life is assumed to be 50 years. Thus the life time output is 30'000'000 m ³ board. This data must only be used to consider the infrastructure needed for wood board production if this is not relevant.	The service life is assumed to be 50 years. Thus the life time output is 2'500'000 m ³ board. This data must only be used to consider the infrastructure needed for wood board production if this is not relevant.
Start Date	2002	2002
End Date	2002	2002
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Estimate is so rough that the Geography is not relevant.	Estimate is so rough that the Geography is not relevant.
Technology text	Relates to a medium to large production plant for wooden boards using one or several Conti-presses.	Relates to a medium sized production plant for cement bonded wooden boards using one or several stack-presses.
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	estimation	estimation
Extrapolations	none	none
Uncertainty Adjustments	none	none

General Flow information				Representation in ecoinvent									Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment	
Land transformation from unknown	wooden board manufacturing plant, organic bonded boards			resource	land			Transformation, from unknown	5.20E+05	m2	estimated	1	2.16	(3,5,1,3,3,5,8)	
Land transformation to industrial built up				resource	land			Transformation, to industrial area, built up	1.20E+05	m2	estimated	1	2.16	(3,5,1,3,3,5,8)	
Land transformation industrial vegetation				resource	land			Transformation, to industrial area, vegetation	1.00E+05	m2	estimated	1	2.16	(3,5,1,3,3,5,8)	
Land transformation industrial unspecific				resource	land			Transformation, to industrial area	3.00E+05	m2	estimated	1	2.16	(3,5,1,3,3,5,8)	
Land occupation as industrial built up				resource	land			Occupation, industrial area, built up	6.00E+06	m2a	estimated	1	1.69	(3,5,1,3,3,5,7)	
Land occupation as industrial vegetation				resource	land			Occupation, industrial area, vegetation	5.00E+06	m2a	estimated	1	1.69	(3,5,1,3,3,5,7)	
Land occupation as industrial unspecific				resource	land			Occupation, industrial area	1.50E+07	m2a	estimated	1	1.69	(3,5,1,3,3,5,7)	
Hall				construction processes	buildings	Yes	CH	building, hall	1.20E+05	m2	estimated	1	3.15	(3,5,1,3,3,5,9)	
Additional foundation of hall			Only special foundation for heavy machines	construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	6.36E+03	m3	estimated	1	3.15	(3,5,1,3,3,5,9)	
Conveyer belt				construction processes	machinery	Yes	RER	conveyor belt, at plant	2.00E+03	m	estimated	1	3.15	(3,5,1,3,3,5,9)	
Machines (steel)				metals	extraction	No	RER	steel, low-alloyed, at plant	1.40E+07	kg	estimated	1	3.15	(3,5,1,3,3,5,9)	
Steel sheet rolling				metals	processing	No	RER	sheet rolling, steel	1.40E+07	kg	estimated	1	3.15	(3,5,1,3,3,5,9)	
Transport rail			Steel: 200 km,	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.80E+06	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport lorry			Concrete: 50 km, Steel: 100 km	transport systems	train	No	RER	transport, freight, rail	2.10E+06	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
		Organic bonded board manufacturing infrastructure	relates to 30 mio m3 board output in 50 years	wooden materials	extraction	Yes	RER	wooden board manufacturing plant, organic bonded boards	1.00E+00	unit					
		Steel to recycling	not balanced						2.00E+03	kg					
		Disposal of foundation		waste management	building demolition	No	CH	disposal, building, reinforced concrete, to final disposal	1.40E+07	kg	calculated	1	3.15	(3,5,1,3,3,5,9)	

Fig. 16.4 Flows for "wooden board manufacturing plant, organic bonded boards" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent									Uncertainty information		
Input		Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment
Land transformation from unknown	→	wooden board manufacturing plant, cement bonded boards			resource	land			Transformation, from unknown	1.00E+04	m2	estimated	1	2.16	(3,5,1,3,3,5,8)
Land transformation to industrial built up	→				resource	land			Transformation, to industrial area, built up	5.00E+03	m2	estimated	1	2.16	(3,5,1,3,3,5,8)
Land transformation industrial vegetation	→				resource	land			Transformation, to industrial area, vegetation	1.50E+03	m2	estimated	1	2.16	(3,5,1,3,3,5,8)
Land transformation industrial unspecific	→				resource	land			Transformation, to industrial area	3.50E+03	m2	estimated	1	2.16	(3,5,1,3,3,5,8)
Land occupation as industrial built up	→				resource	land			Occupation, industrial area, built up	2.50E+05	m2a	estimated	1	1.69	(3,5,1,3,3,5,7)
Land occupation as industrial vegetation	→				resource	land			Occupation, industrial area, vegetation	7.50E+04	m2a	estimated	1	1.69	(3,5,1,3,3,5,7)
Land occupation as industrial unspecific	→				resource	land			Occupation, industrial area	1.75E+05	m2a	estimated	1	1.69	(3,5,1,3,3,5,7)
Hall	→				construction processes	buildings	Yes	CH	building, hall	5.00E+03	m2	estimated	1	3.15	(3,5,1,3,3,5,9)
Additional foundation of hall	→			Only special fundation for heavy machines	construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	1.83E+02	m3	estimated	1	3.15	(3,5,1,3,3,5,9)
Conveyer belt	→				construction processes	machinery	Yes	RER	conveyor belt, at plant	1.00E+02	m	estimated	1	3.15	(3,5,1,3,3,5,9)
Machines (steel)	→				metals	extraction	No	RER	steel, low-alloyed, at plant	4.03E+05	kg	estimated	1	3.15	(3,5,1,3,3,5,9)
Steel sheet rolling	→				metals	processing	No	RER	sheet rolling, steel	4.03E+05	kg	estimated	1	3.15	(3,5,1,3,3,5,9)
Transport rail	→			Steel: 200 km,	transport systems	road	No	RER	transport, lorry >16t, fleet average	8.07E+04	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Transport lorry	→			Concrete: 50 km, Steel: 100 km	transport systems	train	No	RER	transport, freight, rail	6.05E+04	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
	→	Cement bonded board manufacturing infrastructure		relates to 2.5E6 m3 board output in 50 years	wooden materials	extraction	Yes	RER	wooden board manufacturing plant, cement bonded boards	1.00E+00	unit				
	→	Steel to recycling		not balanced						1.00E+02	kg				
	→	Disposal of foundation			waste management	building demolition	No	CH	disposal, building, reinforced concrete, to final disposal	4.03E+05	kg	calculated	1	3.15	(3,5,1,3,3,5,9)

Fig. 16.5 Flows for "wooden board manufacturing plant, cement bonded boards" and its representation in the ecoinvent database

16.4 Mobile choppers in ecoinvent database

This section is mainly based on (Bollens & Grasser 1996).

Mobile choppers are used for the production of wood chips from residual wood and/or industrial wood in the forest. This machine consists of a carrier vehicle of 20 t and of the chopper itself of 4 t. The whole machine consumes 1.4 kg diesel fuel per piled m^3 including approach and moving within the forest. The chopper is assumed to have an average service life of 10 years, the carrier vehicle of 20 years. Total weight includes 6 rubber-tyres of 300 kg each and cutting knives made of low-alloyed steel with a total weight of 60 kg. The rubber-tyres have a service life of 5 years; the steel content is about 16.8 percent of total weight. The cutting knives are replaced every 4'500 piled m^3 . The average production of such a machine lies around 20'000 piled m^3 . During service, one litre lubricant oil is used for chopping 44.5 m^3 piled wood (Kym 1993).

The emissions stemming from fuel and lubricant oil consumption are inventoried according to the module "Diesel burned in building machine".

The background information to the processes are given in Tab. 16.5, flows are shown in:

- | | |
|--|-----------|
| - wood chopping, mobile chopper, in forest | Fig. 16.6 |
| - diesel, burned in chopper | Fig. 16.7 |
| - chopper, mobile, diesel | Fig. 16.8 |

16.4.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Several types of choppers exist. Here only one model is developed. Therefore, the uncertainty especially for energy consumption data is rather high.

Tab. 16.5 Ecoinvent meta information for the mobile wood chopping processes

Name	wood chopping, mobile chopper, in forest	diesel, burned in chopper	chopper, mobile, diesel
Location	RER	RER	RER
Infrastructure Process	0	0	1
Unit	kg	MJ	unit
Data Set Version	2.0	2.0	2.0
Included Processes	Chopping of wood in mobile chopper and transport of the chopper to the place of use, per kg chopped wood (dry mass); inkl. fuel lubricating oil and spare parts	Chopping of wood in mobile chopper, per MJ diesel required for driving and chopping. inkl. fuel lubricating oil and spare parts	The material (production and disposal) and their transports needed for a mobile chopper
Amount	1	1	1
Local Name	Holz häckseln, mobiler Häcksler, im Wald	Diesel, in Grosshacker	Hacker, mobil, Diesel
Synonyms			
General Comment to reference function			This machine consists of a carrier vehicle of 20 t and of the chopper itself of 4 t. The chopper has an average service life of 10 years, the carrier vehicle of 20 years. Total weight includes 6 rubber-tyres of 300 kg each and cutting knives made of low-alloyed steel with a total weight of 60 kg. The rubber-tyres have a service life of 5 years; the steel content is about 16.8 % of total weight. The cutting knives are replaced every 4'500 piled m ³ . The average production of such a machine lies around 20'000 piled m ³ .
Start Date	1996	1996	1996
End Date	2002	2002	2002
Data Valid For Entire Period	1	1	1
Other Period Text			
Geography text	Data for Switzerland used for central Europe	Data for Switzerland used for central Europe	Data for Switzerland used for central Europe
Technology text	carrier with chopper	carrier with chopper	carrier with chopper
Representativeness [%]			
Production Volume	unknown	unknown	unknown
Sampling Procedure	Literature	Literature	Literature
Extrapolations	none	none	none
Uncertainty Adjustments	none	none	none

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Cate gory	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Diesel burned in building machine	wood chopping, mobile chopper, in forest		includes chopping and driving	construction processes	machinery	No	GLO	diesel, burned in building machine	1.41E-01	MJ	Bergmair 1996 according to Anonymous 1998 (GEMIS)	1	2.12	(3,3,3,3,3,5,5)
Building machine			specific infrastructure --> unspecific infrastructure is subtracted	construction processes	machinery	Yes	RER	building machine	-1.89E-08	unit	calculated	1	1.00	calculated
Steel for tyres and cleaver				metals	extraction	No	RER	steel, low-alloyed, at plant	8.67E-05	kg	Frischknecht et al. 1996	1	1.35	(3,3,3,3,3,5,4)
Rubber for tyres				plastics	polymers	No	RER	synthetic rubber, at plant	7.94E-05	kg	Frischknecht et al. 1996	1	1.35	(3,3,3,3,3,5,4)
Lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	5.12E-05	kg	Frischknecht et al. 1996	1	1.35	(3,3,3,3,3,5,4)
Mobile chopper				wooden materials	extraction	Yes	RER	chopper, mobile, diesel	1.33E-08	unit	Frischknecht et al. 1996	1	3.36	(4,5,2,3,4,5,9)
		wood chopping, mobile chopper, in forest	reference to dry wood mass	wooden materials	extraction	No	RER	wood chopping, mobile chopper, in forest	1.00E+00	kg	Bergmair 1996 according to Anonymous 1998 (GEMIS)			

Fig. 16.6 Flows for "wood chopping, mobile chopper, in forest" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent								Uncertainty information		
Input	Process Name	Output	Remarks	Cate gory	Sub category	Infra structure	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Diesel burned in building machine	diesel, burned in chopper		includes chopping and driving	construction processes	machinery	No	GLO	diesel, burned in building machine	1.00E+00	MJ	Bergmair 1996 according to Anonymous 1998 (GEMIS)	1	2.12	(3,3,3,3,3,5,5)
Building machine			specific infrastructure --> unspecific infrastructure is subtracted	construction processes	machinery	Yes	RER	building machine	-1.34E-07	unit	calculated	1	1.00	calculated
Steel for tyres and cleaver				metals	extraction	No	RER	steel, low-alloyed, at plant	6.16E-04	kg	Frischknecht et al. 1996	1	1.35	(3,3,3,3,3,5,4)
Rubber for tyres				plastics	polymers	No	RER	synthetic rubber, at plant	5.64E-04	kg	Frischknecht et al. 1996	1	1.35	(3,3,3,3,3,5,4)
Lubricating oil				chemicals	organics	No	RER	lubricating oil, at plant	3.64E-04	kg	Frischknecht et al. 1996	1	1.35	(3,3,3,3,3,5,4)
Mobile chopper				wooden materials	extraction	Yes	RER	chopper, mobile, diesel	9.41E-08	unit	Frischknecht et al. 1996	1	3.36	(4,5,2,3,4,5,9)
		Mobile chopper for wet wood in forest		wooden materials	extraction	No	RER	diesel, burned in chopper	1.00E+00	MJ	Bergmair 1996 according to Anonymous 1998 (GEMIS)			

Fig. 16.7 Flows for "diesel, burned in chopper" and its representation in the ecoinvent database

General Flow information					Representation in ecoinvent								Uncertainty information			
Input		Process Name		Output	Remarks	Cate gory	Sub category	Infra struc ture	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment
Steel	→	chopper, mobile, diesel				metals	extraction	No	RER	steel, low-alloyed, at plant	2.61E+04	kg	Frischknecht et al. 1996	1	1.35	(3,3,3,3,3,5,4)
Transport lorry 32t	→			Steel: 100 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.61E+06	tkm	estimated	1	2.09	} (4,5,nA,nA,nA,nA,5) (4,5,nA,nA,nA,nA,5)	
Transport rail	→			Steel: 200 km	transport systems	train	No	RER	transport, freight, rail	5.22E+06	tkm	estimated	1	2.09		
Natural gas burned in industrial furnace	→				natural gas	heating systems	No	RER	natural gas, burned in industrial furnace >100kW	2.62E+05	MJ	Frischknecht et al. 1996	1	1.35	(3,3,3,3,3,5,1)	
Electricity medium voltage - at grid UCTE	→				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.40E+04	kWh	Frischknecht et al. 1996	1	1.35	(3,3,3,3,3,5,2)	
				→ Mobile chopper (infrastructure)	The chopper has a life expectancy of 400'000 m3 bulked chips (≈7.544E7 kg dried matter)	wooden materials	extraction	Yes	RER	chopper, mobile, diesel	1.00E+00	unit				
				→ Waste heat into air		air	unspecified			Heat, waste	8.65E+04	MJ	Frischknecht et al. 1996	1	1.35	(3,3,3,3,3,5,13)

Fig. 16.8 Flows for "chopper, mobile, diesel" and its representation in the ecoinvent database

16.5 Stationary electric chopper

In sawmills, sidings and squaring are transported directly to electric choppers by band conveyors. The following data are provided by a company (Etienne 1993) and describe a sawmill with an annual output of 5'000 m³; compared to sizes of plants elsewhere, this is a very small plant.

The chopper is powered by a motor of 37 kW; additionally a 5 kW-motor is used for the feeding of the wood. The chopper mainly consists of unalloyed steel plates and has a total weight of 2.5 t. Its service life is around 30'000 service hours or 100'000 piled m³. Energy consumption is 27 MJ/piled m³ including no-load operating time. The chopper contains 5 litre of gear lubricant oil, which is exchanged every 4'000 service hours or 13'000 piled m³. This gear lubricant oil is inventoried as "Fuel oil petro, at storehouse CH". The cutting knives weight 5 kg, a service life of 2'000 h or 6'500 piled m³ and consist of low-alloyed steel. For the production of the chopper it is assumed that 25 MJ/kg primary energy for unalloyed steel and 35 MJ/kg primary energy for low-alloyed steel must be used. Primary energy resources are 50% electricity and 50% gas. Material input and energy consumption for the band conveyors are not inventoried.

Direct emissions into air are not caused by the electric chopper, except wood dust. Unfortunately, data on wood dust emissions are currently not available.

The background information to the processes are given in Tab. 16.6, flows are shown in:

- industrial residual wood chopping, stationary electric chopper, at plant Fig. 16.9
- chopper, stationary, electric Fig. 16.10

16.5.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Several types of choppers exist. Here only one model is developed. Therefore, the uncertainty especially for energy consumption data is rather high.

Tab. 16.6 Ecoinvent meta information for the stationary wood chopping processes

Name	Industrial residual wood chopping, stationary electric chopper, at plant	chopper, stationary, electric
Location	RER	RER
Infrastructure Process	0	1
Unit	kg	unit
Data Set Version	2.0	2.0
Included Processes	Chopping of industrial residual wood in stationary electric chopper, per kg chopped wood (dry mass)	The material (production and disposal) and their transports needed for a stationary chopper in a sawmill. Land used is not included as it is already included in the sawmill.
Amount	1	1
Local Name	Industrierestholz häckseln, stationärer Häcksler elektrisch, ab Werk	Grosshacker elektrisch
Synonyms		Häckseler
General Comment to reference function		Chopper with an hourly output of 3.3 m ³ /h bulked chips and a life time output of 100'000 m ³ (=1.886E7 kg dried matter) bulked chips. This data must only be used to consider the infrastructure needed for chopping industrial residue wood if this is not relevant.
Start Date	1996	1996
End Date	2002	1996
Data Valid For Entire Period	1	1
Other Period Text		
Geography text	Data for Switzerland used for central Europe	Estimate is so rough that the Geography is not relevant.
Technology text	Stationary electric chopper	Stationary electric chopper
Representativeness [%]		
Production Volume	unknown	unknown
Sampling Procedure	Literature	estimation
Extrapolations	none	none
Uncertainty Adjustments	none	none

General Flow information				Representation in ecoinvent									Uncertainty information		
Input		Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
Electricity medium voltage - at grid UCTE	→	Industrial residual wood chopping, stationary electric chopper, at plant			electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.00E-02	kWh	Bergmair 1996 according to Anonymous 1998 (GEMIS)	1	1.35	(3,3,3,3,3,5,2)
Cleaver	→				metals	extraction	No	RER	steel, low-alloyed, at plant	4.08E-06	kg	Frischknecht et al. 1996	1	1.35	(3,3,3,3,3,5,4)
Lubricating oil	→				chemicals	organics	No	RER	lubricating oil, at plant	2.04E-06	kg	Frischknecht et al. 1996	1	1.35	(3,3,3,3,3,5,4)
Stationary chopper	→				wooden materials	extraction	Yes	RER	chopper, stationary, electric	5.30E-08	unit	estimated	1	3.36	(4,5,2,3,4,5,9)
			Industrial residual wood chopping, stationary electric chopper, at plant	reference to dry wood mass	wooden materials	extraction	No	RER	Industrial residual wood chopping, stationary electric chopper, at plant	1.00E+00	kg	Bergmair 1996 according to Anonymous 1998 (GEMIS)			
			Waste heat into air		air	unspecified			Heat, waste	7.20E-02	MJ	calculated	1	1.35	(3,3,3,3,3,5,13)

Fig. 16.9 Flows for "industrial residual wood chopping, stationary electric chopper, at plant" and its representation in the ecoinvent database

General Flow information				Representation in ecoinvent									Uncertainty information		
Input		Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	StdV 95%	General Comment
Steel unalloyed	→	chopper, stationary, electric			metals	extraction	No	RER	reinforcing steel, at plant	2.50E+03	kg	Frischknecht et al. 1996	1	3.15	(3,5,1,3,3,5,9)
steel rolling	→				metals	processing	No	RER	sheet rolling, steel	2.50E+03	kg	Frischknecht et al. 1996	1	3.15	(3,5,1,3,3,5,9)
Land use	→			included in sawmill											
Foundation	→			concrete, thickness 0.3 m	construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	6.00E+00	m3	estimated	1	3.15	(3,5,1,3,3,5,9)
Transport lorry	→			Steel: 100 km	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.50E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
Transport rail	→			Steel: 200 km	transport systems	train	No	RER	transport, freight, rail	5.00E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)
			Electric chopper (infrastructure)	The chopper has a life expectancy of 100'000 m3 bulked chips (=1.886E7 kg dried matter)	wooden materials	extraction	Yes	RER	chopper, stationary, electric	1.00E+00	unit				
			Steel to recycling	not balanced						2.50E+03	kg				
			Disposal of foundation		waste management	building demolition	No	CH	disposal, building, reinforced concrete, to final disposal	1.32E+04	kg	calculated	1	3.15	(3,5,1,3,3,5,9)

Fig. 16.10 Flows for "chopper, stationary, electric" and its representation in the ecoinvent database

16.6 Pellet mill

Pellet production usually takes place in sawmills or other plants where dry industrial residual wood is generated. Thus only a small production and storage hall and the machine (as steel) are balanced. The production volume is assumed as 2'000'000 kg/year.

The background information to the process "wood pellet manufacturing, infrastructure" are given in Tab. 16.7, flows are shown in Fig. 16.11.

16.6.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Although data quality is poor, they are good enough to inventory the impacts of the infrastructure.

Tab. 16.7 Ecoinvent meta information for the pellet mill

Name	wood pellet manufacturing, infrastructure
Location	RER
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	The material (production and disposal) and their transports needed for a wood pellet manufacturing machine in a sawmill. Land used is only included for the hall in which the process is run and the pellets are stored as the rest is already included in the sawmill.
Amount	1
Local Name	Holzpelletherstellung, Infrastruktur
Synonyms	Pelletierpresse//Kolbenstrangpresse//extruder
General Comment to reference function	The capacity of the pellet extruder (output = 1250 kg/h) fits the need better than edge mills which would have higher output per machine weight. The life time of the infrastructure is assumed as 50 years (machines are replaced once in this time). The life time output is 1E8 kg pellets. This data must only be used to consider the infrastructure needed for pelletising wood if this is not relevant.
Start Date	2002
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Estimate is so rough that the Geography is not relevant.
Technology text	Pellet extruder
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	none
Uncertainty Adjustments	none

General Flow information				Representation in ecoinvent									Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub category	Infrastructure	Location	Module name in ecoinvent	Mean value	Unit	Source mean value	Type	Stdv 95%	General Comment	
Land transformation from unknown	wood pellet manufacturing, infrastructure			resource	land			Transformation, from unknown	2.00E+02	m2	estimated	1	2.16	(3,5,1,3,3,5,8)	
Land transformation to industrial built up				resource	land			Transformation, to industrial area, built up	2.00E+02	m2	estimated	1	2.16	(3,5,1,3,3,5,8)	
Land occupation as industrial built up				resource	land			Occupation, industrial area, built up	2.00E+02	m2a	estimated	1	1.69	(3,5,1,3,3,5,7)	
Hall				construction processes	buildings	Yes	CH	building, hall	2.00E+02	m2	estimated	1	3.15	(3,5,1,3,3,5,9)	
Additional foundation of hall			Only special foundation for heavy machines	construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	4.55E-01	m3	estimated	1	3.15	(3,5,1,3,3,5,9)	
Machines (steel)				metals	extraction	No	RER	steel, low-alloyed, at plant	2.00E+03	kg	estimated	1	3.15	(3,5,1,3,3,5,9)	
Steel sheet rolling				metals	processing	No	RER	sheet rolling, steel	2.00E+03	kg	estimated	1	3.15	(3,5,1,3,3,5,9)	
Transport rail			Steel: 200 km,	transport systems	road	No	RER	transport, lorry >16t, fleet average	4.00E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
Transport lorry			Concrete: 50 km, Steel: 100 km	transport systems	train	No	RER	transport, freight, rail	2.50E+02	tkm	estimated	1	2.09	(4,5,nA,nA,nA,nA,5)	
		Pellet manufacturing infrastructure	relates to 1E8 kg pellet output in 50 years	wood energy	fuels	Yes	RER	wood pellet manufacturing, infrastructure	1.00E+00	unit					
		Steel to recycling	not balanced						2.00E+02	kg					
		Disposal of foundation		waste management	building demolition	No	CH	disposal, building, reinforced concrete, to final disposal	1.00E+03	kg	calculated	1	3.15	(3,5,1,3,3,5,9)	

Fig. 16.11 Flows for "wood pellet manufacturing, infrastructure" and its representation in the ecoinvent database

16.7 Preservative treatment, infrastructure

Preservative treatment infrastructure is estimated according to pictures and general information of a treatment facility. Because the preservative treatment is usually done in sawmills the wood storage area and the infrastructure for loping, debarking and cranes have not to be considered here. The vessels are balanced as chromium steel.

The background information to the process "preservative treatment, infrastructure" are given in Tab. 16.8, flows are shown in Fig. 16.12.

16.7.1 Data quality

Data quality is assessed using the pedigree matrix as described in (Frischknecht et al. 2007)

Although data quality is poor, they are good enough to inventory the impacts of the infrastructure.

Tab. 16.8 Ecoinvent meta information for the preservative treatment infrastructure

Name	preservative treatment, infrastructure
Location	RER
Infrastructure Process	1
Unit	unit
Data Set Version	2.0
Included Processes	The land area and the material (production and disposal) and their transports needed for a preservative treatment plant is included. Land used for the storing of incoming wood is not included because most preservative treatment is done in the sawmills. The space used for drying after the treatment is included.
Amount	1
Local Name	Kesseldruckimprägnierung, Infrastruktur
Synonyms	
General Comment to reference function	The plant has a yearly throughput of 6'000 m ³ timber, a life time of 50 years (machines are replaced once in this time) and a life time output of 3E5 m ³ treated timber. This data must only be used to consider the infrastructure needed for preservative treatment of wood if this is not relevant.
Start Date	2002
End Date	2002
Data Valid For Entire Period	1
Other Period Text	
Geography text	Estimate is so rough that the Geography is not relevant.
Technology text	Infrastructure for pressurised impregnation. Dip treatment would require less steel but about the same land use.
Representativeness [%]	
Production Volume	unknown
Sampling Procedure	estimation
Extrapolations	none
Uncertainty Adjustments	none

General Flow information					Representation in ecoinvent									Uncertainty information			
Input		Process Name		Output	Remarks	Cate gory	Sub category	Infra struc ture	Loca tion	Modul name in ecoinvent	Mean value	Unit	Source mean value	Type	StDv 95%	General Comment	
Land transformation from unknown	→	preservative treatment, infrastructure				resource	land			Transformation, from unknown	1.20E+03	m2	estimated		1	2.16 (3,5,1,3,3,5,8)	
Land transformation to industrial built up	→					resource	land			Transformation, to industrial area, built up	1.20E+03	m2	estimated		1	2.16 (3,5,1,3,3,5,8)	
Land occupation as industrial built up	→					resource	land			Occupation, industrial area, built up	6.00E+04	m2a	estimated		1	1.69 (3,5,1,3,3,5,7)	
Hall	→					construction processes	buildings	Yes	CH	building, hall	1.20E+03	m2	estimated		1	3.15 (3,5,1,3,3,5,9)	
Additional foundation of hall	→				Only special fundation for heavy machines	construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	3.00E+00	m3	estimated		1	3.15 (3,5,1,3,3,5,9)	
Vessel (steel)	→					metals	extraction	No	RER	chromium steel 18/8, at plant	1.32E+04	kg	estimated		1	3.15 (3,5,1,3,3,5,9)	
Steel sheet rolling	→					metals	processing	No	RER	sheet rolling, chromium steel	1.32E+04	kg	estimated		1	3.15 (3,5,1,3,3,5,9)	
Transport rail	→				Steel: 200 km,	transport systems	road	No	RER	transport, lorry >16t, fleet average	2.64E+03	tkm	estimated		1	2.09 (4,5,nA,nA,nA,nA,5)	
Transport lorry	→				Concrete: 50 km, Steel: 100 km	transport systems	train	No	RER	transport, freight, rail	1.65E+03	tkm	estimated		1	2.09 (4,5,nA,nA,nA,nA,5)	
	→			Preservative treatment infrastructure	relates to 3E5 m3 treated timber output in 50 years	wooden materials	extraction	Yes	RER	preservative treatment, infrastructure	1.00E+00	unit					
	→			Steel to recycling	not balanced							1.20E+03	kg				
	→			Disposal of foundation		waste managemnet	building demolition	No	CH	disposal, building, reinforced concrete, to final disposal	6.61E+03	kg	calculated		1	3.15 (3,5,1,3,3,5,9)	

Fig. 16.12 Flows for "preservative treatment, infrastructure" and its representation in the ecoinvent database

17 References

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